

# A Large-Scale Evaluation of Merger Simulations

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**ABSTRACT.** Prospective merger simulations are a commonly used tool in industrial organization and antitrust, but evidence about their accuracy and predictive ability is limited. We study 101 mergers in consumer packaged goods and compare the realizations of price changes with predictions from merger simulations. In our sample of consummated mergers, predicted price changes from merger simulations are typically larger than realized ones. Despite the overprediction, we find that full merger simulations are more effective than both structural presumptions and approximations of the merger effect at identifying mergers with large price changes.

**KEYWORDS.** Antitrust, Merger Simulation.

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## I. Introduction

Antitrust agencies frequently make decisions about whether to challenge proposed mergers and thus must make prospective predictions of their effects. A commonly used tool to guide these decisions is merger simulation. Merger simulations have become a workhorse tool in industrial organization and are prominent in both the classroom and the courtroom (Werden and Froeb, 2006). However, despite repeated calls for it (Whinston, 2007; Angrist and Pischke, 2010; Nevo and Whinston, 2010), there is limited work comparing the predictions from these simulations to realized price changes. Thus, the literature lacks a systematic evaluation of the accuracy of the predictions derived from the tool or of its ability to screen price-increasing mergers.

Prior work that has evaluated merger simulations has typically focused on a handful of mergers at a time. This paper does so in the context of 101 mergers in US consumer packaged goods (CPG) from 2006 to 2017. These mergers comprise a large subset of all transactions with a deal size larger than \$280 million involving CPG products sold through retail outlets. For each merger in our sample, we run every step of a textbook merger simulation (Hausman et al., 1994; Werden and Froeb, 1994; Nevo, 2000) using state-of-the-art methods (Conlon and Gortmaker, 2020, 2025). We first collect product characteristics and cost shifters for each market and estimate a mixed logit demand system (Berry et al., 1995) using pre-merger data on prices and shares. We recover implied marginal costs and project them onto cost shifters using nonparametric methods. We finally forecast demand and costs into future periods and solve the first-order conditions of the Bertrand-Nash game with the post-merger ownership structure. This procedure gives us predicted prices for about 1,000 markets (geography-month pairs) for each merger.

This scale enables us to identify patterns in the comparisons between merger simulations and realizations of post-merger prices. We first do so through the lens of accuracy: do predictions from merger simulations align with realizations of price changes on average, or are they systematically biased? We then analyze merger simulations from the perspective of an agency trying to screen mergers and thus values predictive power rather than accuracy itself (Shapiro, 2010). The 2010 Horizontal Merger Guidelines write that the agencies “place more weight on whether their merger simulations consistently predict substantial price increases than on the precise prediction of any single simulation” (p. 21). We compare the predictive ability of the merger simulation toolkit vis-à-vis the structural presumptions, approximations to a merger’s effects, and simpler models. The first set of analyses teaches us about systematic mistakes made by the model, shedding light on potential avenues for improvement. The second evaluates the value-added of the model relative to other tools at policymakers’ disposal.

We find that merger simulations systematically overpredict price increases of completed mergers. The median overprediction is 2.0% overall, and 3.8% when considering merging party price changes.

Simulations overpredict price changes in about 70% of mergers. This overprediction increases over time and is thus not due to a delay in adjusting to the next equilibrium. It is also not due to a continuation of pre-merger trends: we find zero average prediction error from placebo simulations of the year before the merger.

We investigate the extent to which deviations from standard modeling assumptions could explain this prediction error. We first consider demand misspecification and find that allowing for a wide range of perturbations of our estimates can explain at most a fifth of average prediction error for merging parties. We then explore alternative explanations: changes to demand and product assortment following the mergers, departures from Bertrand-Nash conduct in either the pre- or post-merger periods, and accounting for retail markups. In all cases, our finding of overprediction persists and therefore appears robust to these modeling assumptions.

We next consider the possibility of post-merger marginal cost changes. Assuming the specified supply-and-demand system is correct, merging party costs must decline by 2.1% of prices on average to rationalize prices after the merger. While we lack cost data that would allow us to validate these “implied synergies” directly, we analyze auxiliary patterns that one might expect them to follow if they were indeed merger-induced cost efficiencies. We find that implied synergies grow over time, are low on average for non-merging parties, and are correlated within product in a manner consistent with production efficiencies. However, we also find that implied synergies realize immediately following the merger and are correlated with unilateral effects, suggesting that these prediction errors may also be consistent with model misspecification. While this correlation is muted by allowing for deviations from Bertrand-Nash conduct before and (especially) after the merger, we ultimately find that post-merger synergies are needed to fully explain overprediction in each alternative model we consider. We caution against interpreting these results as *ex ante* evidence of synergies for any hypothetical merger, as our sample is a selected set of mergers that were not blocked due to antitrust scrutiny.

Finally, we turn our attention to the predictive ability of the toolkit. We find that merger simulations are directionally calibrated: a regression of a measure of the causal price change due to the merger on the unilateral effect predicted by the merger simulation yields a slope of 0.77 (s.e. 0.40) across mergers, with similar results within-merger across geographies. Despite this positive slope, the predictive power of the toolkit is low for the actual price effect of a merger: unilateral effects only explain about 8%–14% of the variation in causal effects. Of course, agencies may not be interested in predicting the exact price effect of mergers that are likely to have small price effects, and they may instead place particular attention on the ability of the toolkit to identify mergers that will lead to a substantial lessening of competition. Using a series of metrics for classification, we find that simulations are better at discriminating mergers with large price changes than predictions from structural presumptions and approximations such as GUPPI. We conclude by

analyzing the implications of these results for merger policy through a statistical decision theory framework, and we find that merger simulations allow decision-makers to target a challenge rate closer to the optimal one. Although merger simulation achieves the lowest error rate out of all methods we consider, its performance is far from the oracle, suggesting that there is room for both the evidentiary process followed by antitrust agencies and future methodological innovations to bridge this gap.

Taken together, our findings paint a nuanced picture of merger simulation as an antitrust tool. In consummated CPG mergers, textbook simulations tend to overpredict price increases under standard assumptions, and understanding that overprediction sheds light on the relative importance of efficiencies, conduct, and other supply-side forces. At the same time, the simulations contain useful information for enforcement, particularly when the goal is to screen for mergers likely to generate substantial price increases rather than to forecast exact magnitudes.

*Related Literature.* The literature comparing predictions from merger simulations to realizations is sparse, and each paper in this literature studies a handful of mergers. Nevo (2000) studies two ready-to-eat cereal mergers, Peters (2006) studies five airline mergers, Weinberg (2011) and Weinberg and Hosken (2013) together study three mergers of consumer goods, Houde (2012) studies one merger in retail gasoline, Björnerstedt and Verboven (2016) study one merger of pharmaceuticals, and Miller and Weinberg (2017) study one joint venture in beer.<sup>1</sup> With the exception of Nevo (2000) and one merger in Weinberg and Hosken (2013), most simulations depart significantly from the realizations. Reasons for the departure are varied and include unexplained supply-side issues (Peters, 2006), potential issues in demand estimation (Weinberg and Hosken, 2013; Björnerstedt and Verboven, 2016), and post-merger coordination (Björnerstedt and Verboven, 2016; Miller and Weinberg, 2017). Relative to this literature, our main contribution is scale: by studying a large set of mergers—and incorporating both cross-merger and within-merger heterogeneity in results—we can find systematic patterns.

The literature on synergies attributable to mergers is also sparse: Asker and Nocke (2021, p. 221) note that “[e]vidence as to the typical size of merger efficiencies is also extremely limited.” The extant literature can be organized into three groups. Merger retrospectives (Kim and Singal, 1993; Focarelli and Panetta, 2003; Bhattacharya et al., 2026) often attribute decreases in prices to synergies but do not provide a quantification of them. A second set of papers retrospectively estimates direct proxies for synergies, such as distribution distance in beer (Ashenfelter et al., 2015), heat rate in

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<sup>1</sup>Garmon (2017) is distinct in both the focus on hospital mergers and in the quantity: he studies 28 mergers and evaluates a number of screening tools, although he assesses “reduced-form” merger simulations. Similar papers include Fournier and Gai (2007) and May and Noether (2014), which evaluate other reduced-form screens used in merger analysis, and Dranove and Ody (2016), which evaluates the effectiveness of these screens in predicting cross-sectional variation in prices. Finally, though it does not evaluate merger simulation directly, Panhans and Taragin (2023) emphasizes the effects that a potentially misspecified demand model can have on the outcome of merger simulation and merger review.

power plants (Demirer and Karaduman, 2025), inventory and capacity utilization in manufacturing (Braguinsky et al., 2015), or costs in hospitals (Schmitt, 2017). Finally, some of the aforementioned papers evaluating merger simulations provide implied synergies required to rationalize the price effect as an auxiliary result.<sup>2</sup> We take a similar approach and are able to provide such estimates on a much broader scale than previous papers in the literature, and we evaluate them next to other potential supply-side explanations. We view validating these implied synergies with direct measures of cost as a fruitful avenue for future work.

Moving beyond merger simulations, despite the rapidly growing importance of structural modeling in empirical economics, there are relatively fewer tests of the accuracy of counterfactual predictions. A limited set of examples include an analysis of microfinance (Kaboski and Townsend, 2011), predictions of auctions models (Bajari and Hortacsu, 2005; Athey et al., 2011), discrete choice models (Pathak and Shi, 2021; Raval et al., 2022), dynamic games (Salz and Vespa, 2020), spatial economics (Dingel and Tintelnot, 2026), and trade (Kehoe et al., 2017; Adão et al., 2025). We believe more work is warranted, and we view this paper as a contribution to this literature.

## II. Data and Sample Selection

### II.A. Data

Our main source of data is from NielsenIQ (2025). NielsenIQ provides a Retail Scanner Dataset, which contains prices and quantities at the UPC-store-week level for about 2.6–4.5 million UPCs, depending on the sample year. The dataset covers food, non-food grocery items, health and beauty aids, and select general merchandise. It also has fairly comprehensive coverage of retail outlets in the United States.<sup>3</sup> NielsenIQ provides an additional dataset called the Consumer Panel, which tracks shopping behavior by individual households over time and provides properties of these households. We have access to this sample from 2006 to 2018.

We collect external sources of data to supplement the NielsenIQ dataset on three dimensions relevant for demand estimation. First, we obtain ownership information for each product from a combination of Euromonitor Passport (Euromonitor International, 2022) and manual searches. Second, we collect cost shifters for each merger by obtaining commodity price indices for relevant inputs, typically from Federal Reserve Economic Data (FRED) from the Federal Reserve Bank of

<sup>2</sup>Some papers cannot be classified into these bins. Grieco et al. (2018) estimates returns to scale from a production function, makes predictions of these efficiencies based on expected changes to quantities, and verifies these predictions with post-merger data in the context of beer. Jeziorski (2014) uses the decision to merge as a way to estimate expected synergies. Fan (2013) estimates a cost function and computes synergies that would be realized from a hypothetical merger from economies of scale and scope.

<sup>3</sup>NielsenIQ describes this dataset as providing “scanner data from 35,000 to 50,000 grocery, drug, mass merchandise, and other stores, covering more than half the total sales volume of US grocery and drug stores and more than 30 percent of all US mass merchandiser sales volume.”

St. Louis (2025). Third, the NielsenIQ dataset has limited information about product characteristics, with variables that differ at the product module level. We supplement this dataset with additional product characteristics. Some of these characteristics are from unstructured searches, but many are from decoding the abbreviations and information embedded in the NielsenIQ UPC description. Table C.1 includes a list of the instruments and product characteristics used in each merger.

## II.B. Merger Selection, Market Definition, and Product Definition

NielsenIQ’s data structure provides additional information that is useful for demand estimation. First, Nielsen defines a unit of geography called a designated market area (DMA), which is a collection of counties that is subject to the same local television stations. We follow other papers that conduct demand estimation (Nevo, 2000; Miller and Weinberg, 2017) and use the DMA to construct a geographic market. Second, NielsenIQ assigns products to “modules.” We typically use modules as our definition of a product market, a decision that is aligned with other papers that do demand estimation at scale (Atalay et al., 2025; Döpper et al., 2025). However, we have found through manual inspection that it is sometimes reasonable to combine modules together: for instance, we combine the “Wine-Domestic Dry Table” and “Wine-Imported Dry Table” modules. In a handful of cases, we also found it reasonable to split modules based on characteristics when we think that products in each split are unlikely to be substitutes.<sup>4</sup> Table C.1 provides the full list of product markets used in our analysis. As discussed in Bhattacharya et al. (2026), these market definitions seem consistent with both the so-called *Brown Shoe* factors cited in the 2023 Merger Guidelines and with definitions used historically by the DOJ and FTC. We also believe that our definitions of markets are likely to contain most relevant substitutes for the products of interest.

Our goal is to obtain a relatively large sample of mergers that are representative of typical mergers in consumer packaged goods. To do so, we follow a procedure very similar to the one in Bhattacharya et al. (2026). We start with a set of deals tracked by SDC Platinum from Thomson Reuters (2021), which provides comprehensive information on mergers, acquisitions, and joint ventures, and we restrict to deals that involve mergers of manufacturers of products sold in groceries and mass merchandisers. We consider every such deal with a transaction size larger than \$280 million, and we define a merger to be a combination of a deal and a product market. We only keep mergers where both the target and the acquirer sold at least one UPC in the same market with non-negligible market share—which ensures that we retain national, seasonal, and important regional products.<sup>5</sup> Finally, for computational reasons, we filter each merger to the top 50 DMAs by

<sup>4</sup>For instance, we split “Baby Milk and Milk Flavoring” by whether the formula is for infants. The “Seasoning-Dry” module contains many different types of spices, such as garlic powder and cinnamon, which likely do not compete with each other. Dog and cat treats likely do not compete with each other in the “Dog and Cat Treats” module.

<sup>5</sup>Specifically, we only include a product in a DMA if it ever achieves an inside share of at least 1% in the four years surrounding the merger. Appendix D of Bhattacharya et al. (2026) shows that a similar sample selection procedure

sales volume. This leads to 115 mergers over 45 deals.

We must also take a stance on the definition of a product in our sample. Papers differ in whether the definition of a product is the UPC (Atalay et al., 2025), the brand (Döppler et al., 2025), or some level of aggregation between the two (Backus et al., 2021). The granularity of brands differs significantly across modules in the Nielsen data, so we also adopt an intermediate approach: in most markets our definition of a product is at the unique brand-characteristics level, but we sometimes define a product as a UPC when aggregating is not useful. For instance, the “Chili-Shelf Stable” module has 343 unique UPCs, only 215 of which have unique brand-characteristics, while the “Stew - Beef - Shelf Stable,” “Stew - Chicken - Shelf Stable,” and “Stew - Remaining - Shelf Stable” modules (which we group into one product market) have 98 UPCs, 86 of which have unique brand-characteristics, despite the ex-ante similarity of the characteristics we define on each.<sup>6</sup> We therefore aggregate within brand-characteristics for chili but keep stew at the UPC level.

## II.C. Descriptive Statistics

Panel A of Table 1 reports summary statistics for our initial sample. The markets in our sample are fairly concentrated, with the median Herfindahl-Hirschman Index (HHI) around 2,700.<sup>7</sup> Many mergers have limited impacts on concentration: the median merger has a change in HHI (DHHI) around 15, and about a third of mergers in our sample have a DHHI larger than 100. We also have a large variety of different mergers in the dataset. Variation in the market size of the mergers is large, and there is a fair amount of variation in the number of products per merger. Finally, the most common NielsenIQ “department” for our mergers is dry groceries.

We are unable to obtain reasonable demand estimates for some of the mergers in our dataset: Section III.B provides the criteria. Panel B shows the same summary statistics for the sample for which we have estimates, and the patterns are similar. The share of large-DHHI mergers is slightly smaller, and we have a smaller share of mergers in health and beauty products.

Table C.1 shows the characteristics we collect. Since they are varied across mergers, we are unable to provide detailed distributions for each of the potential characteristics. However, given the characteristics are used to capture substitution between products, we compute distance between these characteristics. To compute this distance in a uniform way across mergers, we conduct a principal component analysis of each set of non-price characteristics, take the first two principal

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leads of high coverage of UPCs within each product market.

<sup>6</sup>In addition to package size and store brand, which are used for every product market, we define “meat type,” “is hot,” “has beans,” and “chili con carne” characteristics for the chili product market, and “Brunswick,” “contains beef,” “contains chicken,” “microwave,” and “regular flavor” characteristics for the stew product market.

<sup>7</sup>The HHI is the sum of the squared market shares of each firm, and we compute this using the inside shares. We compute the change in the HHI (which we refer to as the DHHI) as the naive, or pro-forma, DHHI, assuming that the market shares of merged entities would be the sum of the pre-merger market shares. Without divestitures, this is simply twice the product of the pre-merger market shares of the merging parties. We scale both these from 0 to 10,000.

|                                    |       |          | Quantiles |       |       |       |       |
|------------------------------------|-------|----------|-----------|-------|-------|-------|-------|
|                                    | Mean  | St. Dev. | 10%       | 25%   | 50%   | 75%   | 90%   |
| A. Initial Sample ( $N = 115$ )    |       |          |           |       |       |       |       |
| HHI                                | 2,990 | 1,313    | 1,522     | 2,076 | 2,715 | 3,748 | 4,602 |
| DHHI                               | 133.0 | 290.7    | 0.0       | 1.5   | 15.9  | 161.4 | 308.9 |
| $\mathbb{1}[\text{DHHI} > 100]$    | 0.31  | 0.47     | 0         | 0     | 0     | 1     | 1     |
| Merging Party Share                | 0.20  | 0.18     | 0.03      | 0.05  | 0.14  | 0.31  | 0.49  |
| Avg. Market Size (\$1,000s)        | 631   | 891      | 29        | 79    | 304   | 713   | 1,703 |
| Number of Products                 | 80    | 47       | 22        | 42    | 80    | 113   | 146   |
| Share in Department                |       |          |           |       |       |       |       |
| Dry Grocery                        | 0.62  |          |           |       |       |       |       |
| Health & Beauty Care               | 0.10  |          |           |       |       |       |       |
| Alcoholic Beverages                | 0.13  |          |           |       |       |       |       |
| B. Estimation Sample ( $N = 101$ ) |       |          |           |       |       |       |       |
| HHI                                | 2,959 | 1,254    | 1,527     | 2,030 | 2,686 | 3,781 | 4,588 |
| DHHI                               | 113.4 | 221.1    | 0.0       | 1.2   | 15.3  | 163.8 | 293.6 |
| $\mathbb{1}[\text{DHHI} > 100]$    | 0.3   | 0.46     | 0         | 0     | 0     | 1     | 1     |
| Merging Party Share                | 0.19  | 0.17     | 0.03      | 0.05  | 0.14  | 0.30  | 0.49  |
| Avg. Market Size (\$1,000s)        | 610   | 908      | 29        | 65    | 242   | 655   | 1,622 |
| Number of Products                 | 77    | 46       | 21        | 41    | 76    | 111   | 138   |
| Share in Department                |       |          |           |       |       |       |       |
| Dry Grocery                        | 0.62  |          |           |       |       |       |       |
| Health & Beauty Care               | 0.08  |          |           |       |       |       |       |
| Alcoholic Beverages                | 0.15  |          |           |       |       |       |       |

Table 1: Summary statistics. HHI, DHHI, merging party share, average market size, and number of products are all averaged at the market (DMA-month) level within merger. Other departments include Frozen Food, General Merchandise, Non-Food Grocery, and Packaged Meat.

|             | All Characteristics |        |             | Price    |        |             | Non-Price Characteristics |        |             |
|-------------|---------------------|--------|-------------|----------|--------|-------------|---------------------------|--------|-------------|
|             | Acquirer            | Target | Non-Merging | Acquirer | Target | Non-Merging | Acquirer                  | Target | Non-Merging |
| Acquirer    | 1.50                | —      | —           | 0.52     | —      | —           | 1.21                      | —      | —           |
| Target      | 1.77                | 1.47   | —           | 0.81     | 0.63   | —           | 1.35                      | 1.05   | —           |
| Non-Merging | 2.00                | 2.14   | 1.95        | 0.88     | 1.02   | 0.82        | 1.59                      | 1.65   | 1.57        |

Table 2: Average Euclidean distance in normalized characteristics space between competing products. All characteristics are normalized into  $z$ -scores before computing distances. Averages are weighted by shares within markets and then unweighted across markets and mergers.

components, and normalize these to  $z$ -scores. This procedure maps to the characteristics used for demand estimation in Section III.A. We also normalize prices to  $z$ -scores to make units comparable across product markets. We then compute the average Euclidean distance between competing products in normalized characteristics space. Each set of three columns of Table 2 shows this distance between different sets of products. Our first observation is that the diagonal of each matrix



is smaller than the elements in corresponding rows and columns: a given firm's products are more similar to other products they own than to other firms'. Second, the distance between the acquirer and the target products is smaller than that between the non-merging parties' products and either the acquirers' or targets'. This provides some evidence that firms with similar products are more likely to merge, an observation that has not been made systematically in the literature. Moreover, to the extent that these characteristics are valued by consumers, they will feed into larger diversion between merging parties than between merging and non-merging parties.

### III. Demand Estimation

#### III.A. Model

For each merger, we estimate a differentiated products demand system following Berry et al. (1995), which has become the standard model in industrial organization for demand for consumer packaged goods. We define a market to be a DMA-month pair and aggregate price and quantities to the product level across all retailers and weeks within a market, weighing prices by the relative share in each retailer-week. For each product  $j$  in DMA  $d$  at time  $t$ , we define the indirect utility to consumer  $i$  as

$$u_{ijdt} = \beta_i^0 + \alpha_i p_{jdt} + \beta'_i \mathbf{x}_j + \xi_j + \xi_d + \xi_{\text{quarter}(t)} + \xi_{jdt} + \epsilon_{ijdt}, \quad (1)$$

where  $p_{jdt}$  is the price of the product,  $\mathbf{x}_j$  is a vector of product characteristics,  $(\xi_j, \xi_d, \xi_{\text{quarter}(t)})$  are product, DMA, and quarter-of-year fixed effects,  $\xi_{jdt}$  is an unobserved shifter of utility common across all individuals, and  $\epsilon_{ijdt}$  is a Type I extreme value shock that is iid across products. The utility of the outside option is  $u_{i0dt} = \epsilon_{i0dt}$ , another independent draw from a Type I extreme value distribution. We parameterize the vector of random coefficients as

$$\begin{aligned} \beta_i^0 &= \Pi^{\beta^0} D_i + \sigma^0 v_i^0 \\ \alpha_i &= \alpha + \Pi^\alpha D_i + \sigma^\alpha v_i^\alpha \\ \beta_{ki} &= \Pi^{\beta^k} D_i + \sigma^k v_i^k, \end{aligned} \quad (2)$$

where the  $v_i^\times$  are independent standard normal draws, and  $D_i$  is a vector of demographics. We use  $\theta$  to refer to all the structural parameters to be estimated; this consists of  $\alpha$ , all  $\Pi^\times$ , and all  $\sigma^\times$ .

The parameterization in (1) and (2) allows for both vertical differentiation across products and horizontal differentiation through the random coefficients. The random coefficients also allow for a departure from share-based substitution and allow for substitution towards products that are closer in characteristics space or in price. The demographic and random coefficient on the constant  $\beta_i^0$  breaks share-based substitution to the outside option. Throughout our analysis, we focus on an indicator for whether household income is above \$100,000 as the demographic of interest, although

in some specifications we set some  $\Pi^\times$  to 0, and sometimes  $\sigma^0$  to 0.

Papers that estimate demand at scale (Atalay et al., 2025; Döpper et al., 2025) do not use product characteristics to model substitution, likely due to the difficulty both in collecting these data and in choosing characteristics to use in the demand system. As mentioned in Section II.C, we tackle the second problem by using the first two principal components of the set of characteristics we collected by hand, an approach used by Backus et al. (2021). A benefit of this approach in our setting is that it allows us to avoid arguably arbitrary decisions about the choice of characteristics for each market and limits researcher degrees of freedom. It also allows us to account for discretely coded characteristics in a sensible manner. This approach allows us to incorporate the most variation possible from the characteristics while respecting computational limits. There are some drawbacks. First, the first two principal components do not always capture a large share of the variation in the characteristics (see Table C.1). Second, this approach also limits the interpretability of random coefficients themselves. Perhaps more importantly, it is not clear that distance in the space of hand-coded characteristics is the variation that is perceived by consumers when they assess whether products are substitutes: an approach such as in Magnolfi et al. (2025) that determines characteristics from embeddings of consumer surveys would be promising, but is infeasible for historical data.

Finally, we must compute market sizes (or, equivalently, shares of the outside option). Some prior papers studying individual markets take market sizes as proportional to population, where the proportion is based on an upper bound for inside purchases devised from facts about the product market (Nevo, 2000; Backus et al., 2021). The diversity of product markets in our sample makes this approach difficult to implement. Instead, we take a simpler approach by calibrating the market sizes in each DMA as 1.5 times the largest sales quantity in that DMA throughout the sample, such that the minimum share of the outside good is 1/3. This is consistent with the approach in Miller and Weinberg (2017) and Atalay et al. (2025).

### III.B. Estimation

Demand estimation follows the algorithm of Berry et al. (1995). Given a guess of the nonlinear parameters,  $\Pi^\times$  and  $\sigma^\times$ , we solve for values of mean utilities  $\delta_{jdt}$  such that observed and predicted shares match. We then concentrate out the linear parameters  $(\alpha, \xi_j, \xi_d, \xi_{\text{quarter}(t)})$ , instrumenting for prices, and recover  $\xi_{jdt}(\theta)$ . We then search for the nonlinear components of  $\theta$  that minimize a GMM objective including both standard instrument-exclusion moments and micro-moments (Berry et al., 2004).

Identification requires instruments for the mean price coefficient and nonlinear parameters. Our main class of instruments is the differentiation instruments formulated by Gandhi and Houde (2019). For each characteristic, we compute the quadratic distance between product  $j$ 's location in characteristic space and that of its competitors as  $z_{jdt}^x = \sum_{k \in J_{dt}} (x_j - x_k)^2$ , where  $J_{dt}$  is the set

of available products in market  $dt$ . Relevance comes from the notion that products that face fewer nearby competitors along a certain dimension of characteristics space tend to have higher prices, while exogeneity comes from the assumption that product characteristics are exogenous. These instruments are computed on the full set of characteristics before taking principal components. We then estimate a measure of predicted price given exogenous variables  $\hat{p}_{jdt} = \mathbb{E}[p_{jdt} | \mathbf{x}_j, \mathbf{z}_{jdt}^x]$  using a linear fixed effects regression, controlling for DMA-product fixed effects and all previously computed differentiation instruments, and construct a differentiation instrument from the fitted values  $z_{jdt}^{\hat{p}} = \sum_{k \in J_{dt}} (\hat{p}_{jdt} - \hat{p}_{kdt})^2$ .

We also consider two potential alternative classes of instruments. First, for many mergers, we are able to identify inputs that have producer price indices in FRED, and we use the indices as instruments that exogenously shift costs. We interact these with package size because input costs should be expected to shift the prices of larger products by proportionately more. For a subset of these mergers, only products sharing a certain characteristic are expected to use a particular input; in these cases, we also interact the input prices with an indicator for whether the product has that value of the characteristic. Second, as an easily generalizable measure of common cost shocks, we use Hausman instruments: the average price of product  $j$  in time  $t$  in DMAs bordering DMA  $d$ . Naturally, a concern with this strategy is that bordering DMAs may have correlated demand shocks. Nevertheless, we have decided to incorporate these instruments due to their ubiquity in the literature and the dearth of cost side instruments for some markets.

For the demand estimates that enter our simulations, we want to use the most relevant set of instruments while maintaining the plausibility of the exclusion restriction and avoiding weak instruments. To that end, we prefer the differentiation and input price instruments over Hausman instruments when possible, as Hausman instruments require relatively stronger assumptions to be valid, but must occasionally rely on the Hausman instruments when input prices and differentiation instruments are too weak. The following instrument selection procedure captures these goals. We start with differentiation instruments (which are used in all specifications), and add input prices if their partial  $F$ -statistic in the first-stage regression of prices on differentiation instruments, input price instruments, and fixed effects is at least 20.<sup>8</sup> If the input prices are insufficiently predictive, then we use only the differentiation instruments if their first-stage  $F$ -statistic is at least 100. Finally, if the differentiation instruments, by themselves or with input prices, are too weak ( $F < 100$ ), we rely on Hausman instruments and differentiation instruments. We run one-step GMM with the instruments resulting from this iterative procedure, then compute the feasible approximation to the optimal instruments (Chamberlain, 1987; Berry et al., 1995; Reynaert and Verboven, 2014) and run two-step GMM with these instruments. We take this set of results as successful if the objective

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<sup>8</sup>The input prices only vary temporally unless interacted with characteristics, and for some product markets we are unable to identify producer price indices for instruments that comprise a substantial share of input costs.

|                      | Mean  | St. Dev. | 10%   | 25%   | 50%   | 75%   | 90%   |
|----------------------|-------|----------|-------|-------|-------|-------|-------|
| Our Estimates        | -3.51 | 2.97     | -7.07 | -4.10 | -2.88 | -1.59 | -0.91 |
| Atalay et al. (2025) | -4.88 | 1.60     | -7.24 | -5.55 | -4.57 | -3.84 | -3.42 |
| Döpper et al. (2025) | -2.13 | 1.12     | -4.36 | -3.11 | -2.25 | -1.65 | -1.16 |

Table 3: Comparison of our estimated elasticities with Atalay et al. (2025) and Döpper et al. (2025). Elasticities from Atalay et al. (2025) are unweighted 2010 elasticities.

converges within our prespecified tolerance, the probability of an individual having a positive price sensitivity is less than 5%, the median own-price elasticity is between -0.5 and -20, and the absolute average predicted post-merger change in price is less than 25%. If estimation is unsuccessful with the initial set of instruments, we try the other possible sets of instruments in the following order: differentiation with input prices, differentiation only, and differentiation with Hausman.<sup>9</sup>

We also attempt to use the richest utility specification possible. We start by attempting to estimate the demand system where demographics enter all the random coefficients in (2). If estimation is unsuccessful, we consider a specification where demographics only interact with the constant and price coefficient. Finally, we consider a specification where demographics interact only with prices. If none of these specifications yield acceptable demand estimates, we set  $\sigma^0 = 0$  and go through this list again. In no case do we drop random coefficients on the characteristics or the price. To identify the demographic interactions, we use the Nielsen panel data to formulate micro-moments  $\mathbb{E}[x_j|D_i]$ , the average value of the characteristic conditional on the demographic (Berry et al., 2004; Conlon and Gortmaker, 2025). For demographic interactions with the constant, we use  $\mathbb{E}[D_i|j > 0]$ ; that is, the average value of each demographic level given purchase of an inside good.

To ensure we adhere to best practices, we use pyBLP (Conlon and Gortmaker, 2020) for estimation. Finally, to relieve concerns about local minima, we start estimation at two points: the result from nested logit using inside-outside nests, and the solution to the “FRAC” procedure of Salanié and Wolak (2022). If solutions from these initial points do not coincide, we restart estimation at a grid of starting points around these two values and pick the best solution. The instruments and specifications used for each merger in our estimation sample are listed in Table C.1.

### III.C. Estimates

Our main summary statistic for consumers’ price sensitivity in a given merger’s product market is the unweighted median own-price elasticity, taken across all observations (product-DMA-months) in the data. Figure 1 and Table 3 show the distribution of this statistic across mergers. The median

<sup>9</sup>Results using different sets of instruments are correlated, although not perfectly: the pairwise correlations in estimated median own-price elasticities are 0.59 with just differentiation instruments vs. differentiation and input prices, 0.36 for just differentiation vs. differentiation and Hausman, and 0.41 for differentiation and input prices vs. differentiation and Hausman.

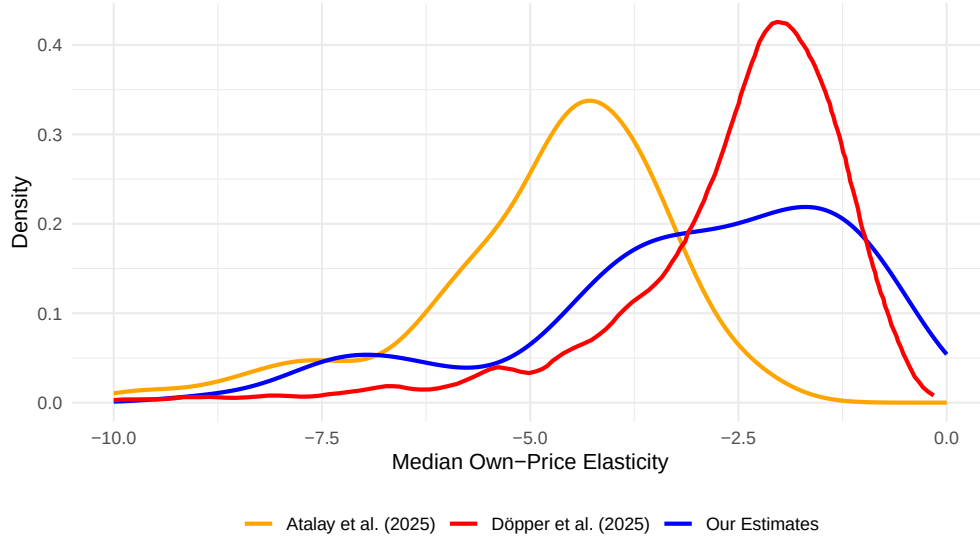


Figure 1: Comparison of our estimated elasticities with Atalay et al. (2025) and Döppler et al. (2025). Elasticities from Atalay et al. (2025) are unweighted 2010 elasticities.

of this distribution is -2.88. The bulk of the distribution of median elasticities is between -1 and -5, with an interquartile range of [-4.10, -1.59]. However, there is a relatively sizable tail of more elastic estimates (some of which are truncated from the plot for readability); the average across all mergers is -3.51, with a 10<sup>th</sup> percentile of -7.07.

As these demand elasticities are a key input into predictions from merger simulations, we perform a number of auxiliary checks on them. First, we compare these elasticities to others in the literature. Figure 1 and Table 3 report elasticity distributions estimated by Atalay et al. (2025) and Döppler et al. (2025). The main part of our distribution is more similar to that of Döppler et al. (2025), while Atalay et al. (2025)'s distribution is more elastic in the middle. However, the left tail of our distribution is more similar to that of Atalay et al. (2025). Table A.1 provides a point-by-point comparison of elasticities when possible. Of the 28 markets for which we have estimates from other papers in the literature, our estimates lie within the range of reported estimates for 8, are less elastic for 14, and more elastic for 6. Conditional on not being within the range, the median absolute deviation to the closest estimate is 0.95 elasticity points, a difference that is smaller than the range of estimates in the literature for many markets. These comparisons are not perfect given differences in the definition of products and markets across papers, but overall this exercise provides some evidence that our estimates are not out-of-line with the rest of the literature. Below, we perform a robustness exercise that allows for a broader range of demand elasticities as inputs to compute merger outcomes, and our takeaways are unchanged.

Second, we ask whether our demographic interactions are sensible. Work that estimates demand for consumer packaged goods typically finds that high-income consumers are generally less price

sensitive than low-income consumers (Backus et al., 2021; Döpper et al., 2025). Consistent with the literature, we find that this is the case for the majority of our demand estimates as well (see Figure A.1). The median median elasticity is -4.14 for low-income consumers, as compared to -2.17 for high-income consumers.

Finally, we ask whether the demand system picks up on the relative product similarity of the acquirer and target products. We find greater diversion between the merging parties: on average, the target to acquirer diversion ratio is 32% (1.8 pp) larger than the non-merging party to acquirer diversion ratio, while the acquirer to target diversion ratio is 36% (0.03 pp) larger than the non-merging party to target diversion ratio.<sup>10</sup> Figure A.2 provides a scatter. This suggests that the similarity in characteristics among the merging parties displayed in Table 2 are economically significant when analyzing potential merger effects. Moreover, we find that the mixed logit structure of the demand system does have an impact on these estimates: Figure A.3 plots the distribution of diversion ratios between merging parties and compares them to the share-based diversion we would expect from logit. We find that the diversion ratios are noticeably higher than logit in our estimates, especially in the upper tail, demonstrating that the demand system is estimating an important (and we think sensible) role for horizontal differentiation through the random coefficients.

In sum, our demand estimates appear reasonable. The own-price elasticity estimates, and the differences in price sensitivity between low- and high-income consumers, are in line with those from the literature. Finally, the demand systems pick up on the relative similarity in product characteristics between the merging parties, manifesting in closer substitution patterns between the merging parties than their competitors. These results give us some comfort that our demand estimates will be a reliable input into the merger simulations.

## IV. The Accuracy of Merger Simulations

### IV.A. Implementing the Merger Simulation

Under the Bertrand-Nash conduct assumption, prices  $\mathbf{p}$  in a market solve

$$D(\mathbf{p}; \boldsymbol{\theta}, \boldsymbol{\xi}) + (\boldsymbol{\Omega} \odot \boldsymbol{\Delta}(\mathbf{p}))(\mathbf{p} - \mathbf{c}) = 0, \quad (3)$$

where  $D(\mathbf{p}; \boldsymbol{\theta}, \boldsymbol{\xi})$  is the demand function (which we parameterize with the demand parameters  $\boldsymbol{\theta}$  and demand shocks  $\boldsymbol{\xi}$  to connect to our procedure),  $\boldsymbol{\Delta}(\mathbf{p})$  is the transpose of the Jacobian of demand with respect to  $\mathbf{p}$ ,  $\mathbf{c}$  is the vector of marginal costs, and  $\odot$  denotes elementwise multiplication. The  $jk$  entry of the ownership matrix  $\boldsymbol{\Omega}$  is 1 if products  $j$  and  $k$  are owned by the same firm, and 0

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<sup>10</sup>Note that the fact that dissimilar differences in levels map to similar differences in percentages is driven by acquirer products having smaller shares.

otherwise.

The typical merger simulation uses the equilibrium conditions in (3) in two ways. First, if the researcher is willing to maintain the assumption that conduct in the pre-merger period is Bertrand-Nash, then (3) can be inverted to solve for costs  $c$  in every market. Second, the researcher can change the ownership matrix from  $\Omega$  to  $\Omega^{\text{post}}$ , where  $\Omega_{jk}^{\text{post}}$  is now set to 1 if products  $j$  and  $k$  are both owned by the merging parties. Holding demand and these recovered costs fixed, the researcher then solves (3) for the new equilibrium prices using this new ownership matrix.

In this section, we compute a forecast for the price of product  $j$  in DMA  $d$  in month  $t$  in merger  $m$ . Doing so requires us to take a stance on (i) the assortment of products in the market in this future period, (ii) demand shocks  $\xi$  the firms would face, and (iii) the marginal costs in that market. For (i), we take the set of products in each DMA in the final period before the merger: we are not trying to test models of entry and exit of existing products, and predicting the introduction of new products is beyond the scope of most models. For (ii), we set  $\xi$  in each DMA as the average of the  $\xi$  in the three months before the merger. For (iii), for each merger  $m$ , we first recover marginal costs  $c_{jdtm}$  for each product-DMA-time combination in the pre-period. We then run a nonparametric regression of  $c_{jdtm}$  on product characteristics  $x_{jm}$ , cost shifters  $w_{jdtm}$ , and time. We implement this using B-splines and obtain a cost function  $h_m(x, w, t)$  and implied residuals  $\omega_{jdtm} \equiv c_{jdtm} - h_m(x_{jm}, w_{jdtm}, t)$ .<sup>11</sup> We then set  $c_{jdtm}$  in post-merger periods  $t$  as  $c_{jdtm}^{\text{Pred}} = h_m(x_{jm}, w_{jdtm}, t) + \bar{\omega}_{jdm}$ , where  $\bar{\omega}_{jdm}$  is the average of  $\omega_{jdtm}$  in the three periods before the merger. The rationale behind this procedure is that we do not want our measured prediction error to be due to large changes in input costs.

This procedure returns a prediction  $p_{jdtm}^{\text{Pred}}$  for the price of product  $i$  in DMA  $d$  in month  $t$  in merger  $m$ . We will compare these with the actual prices  $p_{jdtm}$  observed in the data. We typically average post-period prices over some time period, and so we drop the  $t$  subscript in the notation. We define prediction error as  $\text{Error}_{jdm} \equiv \log(p_{jdm}) - \log(p_{jdm}^{\text{Pred}})$ , so that a negative number corresponds to merger simulations overpredicting the price increase from a merger. We often aggregate prediction errors further. The product-level prediction error  $\text{Error}_{jm}$  is computed by taking the average across DMAs, the DMA-level error  $\text{Error}_{dm}$  is computed by taking the share-weighted average of prediction errors at the DMA level, and the merger-level error  $\text{Error}_m$  is computed by taking the mean of  $\text{Error}_{dm}$  across DMAs.

The distribution of prediction errors we recover is of course only representative for the sample of consummated mergers in CPG. This is a sample selection that would naturally apply to any research comparing predictions to reality. We note that since very few mergers are proposed but then withdrawn due to antitrust scrutiny in this market (Bhattacharya et al. (2026) document only four such cases, across two deals), and none are blocked by a court after a challenge from the agencies,

<sup>11</sup>We regularize the cost function by adding a ridge penalty term, and select hyperparameters by minimizing out-of-sample squared error using cross validation.

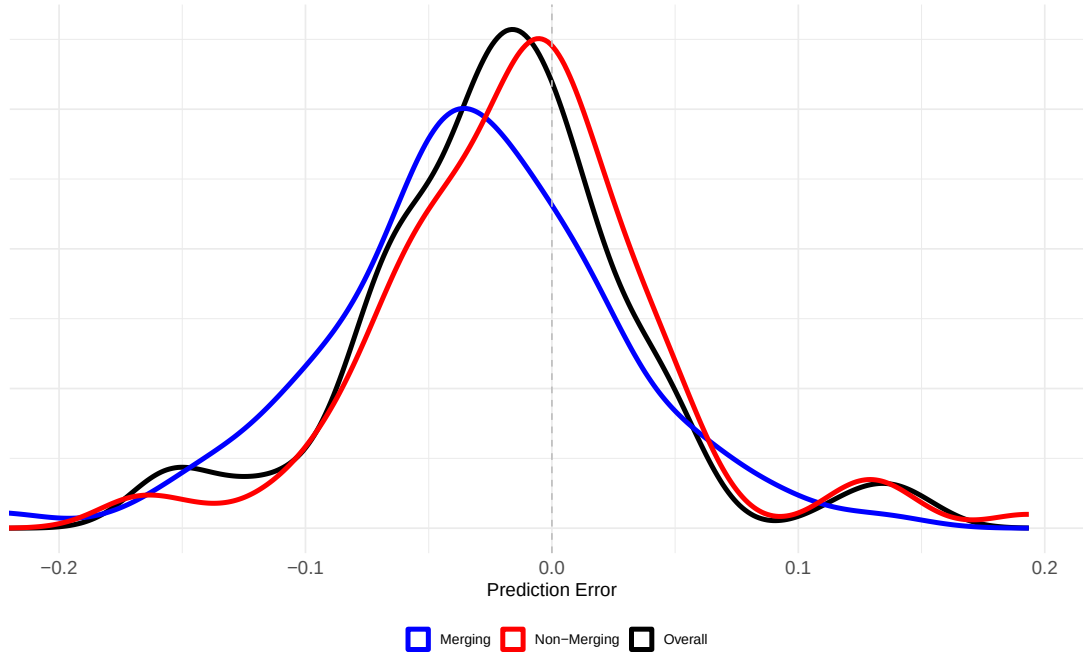


Figure 2: Distribution of prediction errors at the merger level

this sample is close to representative of the set of proposed mergers—and consequently the set of mergers an agency would have to evaluate. Thus, we believe it to be a useful class to consider, but one must be cautious about extrapolating these results to an arbitrary hypothetical merger between firms.

#### IV.B. The Distribution of Prediction Errors

Figure 2 plots the distribution of the merger-level prediction error  $\text{Error}_m$ , both overall and split by merging and non-merging parties. Panel A of Table 4 provides summary statistics of this distribution. Our main finding is that while there is a wide distribution of prediction errors, they are typically negative: merger simulations frequently overpredict price increases. The average prediction error for merging parties is -3.8%, and merger simulations overpredict in approximately three-quarters of mergers. The overprediction is smaller for overall price changes, with a mean of -2.1% and an error distribution that is shifted to the right.<sup>12</sup>

We check two potential sources of this prediction error that would be unrelated to merger simulations. The difference is not due to an initial prediction error that fades as firms adjust to the new equilibrium: the median prediction error across mergers grows in magnitude over time

<sup>12</sup>In one case in our sample, prior literature has documented underprediction from merger simulations. (Due to data restrictions, we cannot reveal the parties involved in this merger.) Our method replicates the literature’s finding that merger simulation assuming Bertrand-Nash conduct underpredicts price changes, especially for non-merging parties, for this particular merger: we find a prediction error of 2.1% for non-merging parties and 1.5% overall.



|                                                 |        |          | Quantiles |        |        |       |       |
|-------------------------------------------------|--------|----------|-----------|--------|--------|-------|-------|
|                                                 | Mean   | St. Dev. | 10%       | 25%    | 50%    | 75%   | 90%   |
| A. Baseline                                     |        |          |           |        |        |       |       |
| Merging                                         | -0.038 | 0.071    | -0.111    | -0.067 | -0.038 | 0.003 | 0.031 |
| Overall                                         | -0.021 | 0.056    | -0.081    | -0.054 | -0.020 | 0.004 | 0.039 |
| Non-Merging                                     | -0.012 | 0.058    | -0.077    | -0.043 | -0.010 | 0.017 | 0.045 |
| B. At Optimal Scaling of Price Coefficients     |        |          |           |        |        |       |       |
| Merging                                         | -0.031 | 0.052    | -0.088    | -0.057 | -0.025 | 0.002 | 0.025 |
| Overall                                         | -0.011 | 0.038    | -0.054    | -0.023 | -0.000 | 0.000 | 0.025 |
| Non-Merging                                     | -0.004 | 0.042    | -0.049    | -0.021 | -0.001 | 0.010 | 0.035 |
| C. At Optimal Scaling of Non-Price Coefficients |        |          |           |        |        |       |       |
| Merging                                         | -0.036 | 0.065    | -0.098    | -0.061 | -0.036 | 0.001 | 0.031 |
| Overall                                         | -0.016 | 0.040    | -0.062    | -0.033 | -0.008 | 0.000 | 0.022 |
| Non-Merging                                     | -0.007 | 0.045    | -0.058    | -0.025 | -0.002 | 0.013 | 0.035 |
| D. Using Post-Merger Demand and Assortment      |        |          |           |        |        |       |       |
| Merging                                         | -0.048 | 0.096    | -0.141    | -0.070 | -0.036 | 0.002 | 0.036 |
| Overall                                         | -0.023 | 0.095    | -0.123    | -0.049 | -0.017 | 0.011 | 0.045 |
| Non-Merging                                     | -0.014 | 0.094    | -0.092    | -0.049 | -0.009 | 0.014 | 0.051 |
| E. Pre-Merger Conduct                           |        |          |           |        |        |       |       |
| Merging                                         | -0.029 | 0.065    | -0.093    | -0.051 | -0.025 | 0.009 | 0.028 |
| Overall                                         | -0.018 | 0.052    | -0.075    | -0.042 | -0.015 | 0.004 | 0.039 |
| Non-Merging                                     | -0.013 | 0.056    | -0.076    | -0.042 | -0.009 | 0.013 | 0.045 |
| F. Post-Merger Conduct                          |        |          |           |        |        |       |       |
| Merging                                         | -0.028 | 0.064    | -0.091    | -0.049 | -0.024 | 0.009 | 0.031 |
| Overall                                         | -0.016 | 0.053    | -0.075    | -0.044 | -0.014 | 0.004 | 0.039 |
| Non-Merging                                     | -0.012 | 0.058    | -0.077    | -0.042 | -0.009 | 0.016 | 0.045 |
| G. Retail Pricing ( $\lambda = 0.2$ )           |        |          |           |        |        |       |       |
| Merging                                         | -0.053 | 0.138    | -0.180    | -0.093 | -0.038 | 0.010 | 0.082 |
| Overall                                         | -0.044 | 0.161    | -0.175    | -0.094 | -0.023 | 0.020 | 0.069 |
| Non-Merging                                     | -0.038 | 0.168    | -0.175    | -0.073 | -0.018 | 0.027 | 0.082 |

Table 4: Summary statistics of prediction error (log actual price minus log predicted price) on market characteristics, at different levels of observation.

(Figure A.4). Further, to validate that secular trends in costs or demand that are not captured by our forecasting models do not drive this overprediction, we conduct a placebo “merger simulation” with no ownership changes at a date one year before the merger and predict prices for one year after the placebo date. We find that the prediction error for all parties is centered around zero (Figure A.5), providing evidence that the prediction error we observe is not due to pre-merger trends.

#### IV.C. Potential Explanations for Prediction Errors

We first evaluate whether alternative models or estimates can eliminate the prediction error. We consider four potential explanations: (a) errors in demand estimation, (b) changes to demand and product assortment after the merger, (c) departures from Bertrand-Nash conduct, and (d) the vertical structure of the industry. The first addresses demand-side misspecification, but (b)–(d) are explanations that connect to supply-side changes, albeit without appealing to changes in marginal costs. Our main finding is that none of these individually eliminates the overprediction.

*Errors in Demand Estimation.* Merger simulation relies on accurately predicting how sensitive consumers are to changes in price (i.e., own-price elasticities) and how differentiated consumers perceive the product market to be (i.e., cross-price elasticities). As such, one potential explanation for overprediction is that our demand estimates themselves could be biased. This bias could arise from the possibility that our instruments, although standard in the industrial organization literature, do not eliminate the endogeneity issue between prices and unobserved characteristics—which typically would yield elasticities that are downward biased in magnitude. Alternatively, it could be that instruments are tailored towards estimating short-run elasticities, which could be lower in magnitude than long-run elasticities that may influence pricing decisions after mergers. Given our scale, it is infeasible for us to reestimate demand to evaluate robustness to each of the decisions we made, and such an exercise would also not be able to address the critique that standard IO procedures would not even capture these elasticities properly. We instead investigate whether our headline result of overprediction is robust to demand mismeasurement by perturbing our demand system in a neighborhood of the baseline estimates and picking the estimates that minimize prediction error.

We consider two types of perturbations. First, we scale all price-related coefficients, including the mean coefficient, random coefficient, and demographic interactions, by a common factor.<sup>13</sup> This changes both average price sensitivity as well as the intensity of competition among products with different price levels. Second, we scale all non-price-related coefficients, including random coefficients and demographic interactions on the constant and product characteristics. These affect the intensity of differentiation and competition in characteristics space.<sup>14</sup> For each set of coefficients, we recover pre-period costs and follow the same procedure to simulate the merger for scaling factors in a range from 0.5 to 3, and then select the scaling factors minimizing absolute overall prediction error separately for each merger. We believe this is a fairly wide range: Figure A.7 shows that the

<sup>13</sup>We scale these together because the mean price coefficient needs to be sufficiently negative relative to the random coefficient and demographic interaction to ensure that consumers almost surely have downward-sloping demand, a requirement for equilibrium existence. Scaling these coefficients together does not change the relative magnitudes.

<sup>14</sup>Given our finding that merging party products tend to be relatively close in characteristics space (Table 2), perturbing random coefficients on characteristics effectively perturbs diversion between the merging parties, an important component of the first-order unilateral effects of a merger (Jaffe and Weyl, 2013).

distribution of minimum and maximum elasticities are quite far apart, and of the 28 markets for which we have external estimates from the literature, our range intersects the elasticities reported from the literature in 25 of them.

Panels B and C of Table 4 show the distribution of prediction errors at the merger level at the optimal values of the scaling coefficients. By construction, the distributions are significantly tighter than the baseline. However, we observe that the overprediction still persists: scaling the price coefficient leads to a median overprediction for merging party prices of 2.5% (vs. 3.8% at baseline), while scaling the non-price coefficient leads to a median overprediction of 3.6%. Achieving these reductions in prediction error often requires us to go to the boundaries of our range (Figure A.6), far from our estimated demand systems. In the case of scaling price coefficients, this typically leads to significantly more elastic demand than estimated at baseline (Figure A.7): the left tail is especially long, as the distribution has a 10<sup>th</sup> percentile of -10.0 and a first quartile of -6.7. Meanwhile, scaling non-price coefficients often leads to attenuation, making the substitution patterns more similar to those in a simple logit; while this is unsurprising given the modal finding of overprediction along with the relative similarity of merging party products, it is notable given the common criticism that simple logit models produce unrealistic substitution patterns.

This exercise is naturally still “local” to mixed logit demand systems of the form we estimate. Some recent work has suggested that these demand systems may be too restrictive for experimental elasticities (Goldszmidt et al., 2025; Bray et al., 2026), or that they may have restrictive curvature (Miravete et al., 2026). Given that BLP is the workhorse demand system in industrial organization, we leave evaluating alternate classes of demand systems to future work.

*Changes to Demand and Assortment.* Even if demand is estimated accurately, the prediction error could be due to changes in demand or assortment itself, and the merger simulations in Section IV.B are conducted assuming that neither demand nor product assortment changes after the merger. Since we have post-merger data, we can evaluate how much smaller the prediction error would be if we had access to an oracle that predicts these changes to the environment. To implement this test, we take the observed post-merger assortment in each market as fixed. We also take the estimated demand parameters  $\theta$  as fixed as well to avoid the (rather expensive) process of re-estimating demand. We then use the post-merger market shares and invert the demand system to recover the post-merger demand shocks  $\xi$ , which differ from the forecasted shocks used to simulate the merger initially. We use the same marginal cost prediction as in Section IV.A and simulate post-merger prices as if demand were also known. We still use pre-merger shares to aggregate the prices so that we are comparing the same set of products.

Panel D of Table 4 shows the distribution of these prediction errors. Our interpretation is that changes in demand are not a main factor in explaining the prediction error. The mean prediction

error increases in magnitude to 4.8% for merging parties, and remains essentially unchanged for non-merging parties. The distribution of prediction error becomes wider in all cases.

The limited effect of demand changes on prediction error mirrors the findings in Peters (2006), where this procedure was first proposed, and Weinberg and Hosken (2013). Thus, we next turn to the explanations that would fall under the “supply-side residual,” as termed by Peters (2006). However, we should be clear that while Peters (2006) uses this term to encompass all changes to the environment not explained by observable changes to demand (or product positioning), some of the changes to demand or assortment may well be supply-side decisions that are not captured by our standard model of merger simulations. In fact, there is some evidence (Atalay et al., 2024; Bhattacharya et al., 2026) that mergers causally lead to changes in assortment, and mergers may also lead to other changes that affect demand (e.g., advertising, product restocking, etc.). However, for the purposes of our exercise, we can remain agnostic about the source of these changes—since we observe their true realizations. To quantify the role of other supply-side deviations from the standard exercise, we will need an explicit model.

*Partial Internalization of Merging Party Profits.* Since merger simulations rely on assuming a model of firm conduct in both the pre-merger and post-merger periods, deviations from Bertrand-Nash conduct will lead to incorrect predictions. Given that overprediction is the modal finding, we focus on alternative models that would deliver a lower price increase due to the merger than Bertrand-Nash. We consider a model of conduct in which merging firms internalize a fraction  $\tau_m^\times \in [0, 1]$ , for  $\times \in \{\text{pre}, \text{post}\}$  of their merging partners’ profits in either the pre- or post-merger periods. The standard model corresponds to  $\tau_m^{\text{pre}} = 0$  and  $\tau_m^{\text{post}} = 1$ . Decreases in  $\tau_m^{\text{post}}$  from 1 capture a trade-off between a division’s own profits and its new partner’s profits that is less than one-for-one: similar models are commonly used in the literature (Crawford et al., 2018; Backus et al., 2021), and in our setting this may come from, for instance, managerial incentives that depend both on firm and division profits. Increases in  $\tau_m^{\text{pre}}$  from 0 correspond to cases where the entities are pricing higher than would be expected in Bertrand-Nash; while such conduct parameters are often used to capture tacit collusion (Sullivan, 2020; Duarte et al., 2024), we remain agnostic about their source.

We ask whether models in this class could possibly explain the observed overprediction quantitatively. To do so, we calibrate  $\tau^{\text{pre}} \in [0, 1]$  (holding  $\tau_m^{\text{post}}$  fixed at 1) by choosing it to minimize (absolute) overall prediction error at the merger level. We then calibrate  $\tau^{\text{post}} \in [0, 1]$  (holding  $\tau^{\text{pre}}$  fixed at 0) in the same way. In Appendix A.2, we show that this procedure typically leads to corner solutions: to minimize prediction error, we often need to either assume that firms are maximizing joint profits in the pre-period or that they are acting separately in the post-period. However, even with these extreme assumptions, Panels E and F of Table 4 show that we continue to systematically

overpredict merging party price changes. The distribution of prediction errors is tighter than in the baseline, again by construction, but the median prediction error remains negative: -2.9% when considering pre-merger coordination and -2.8% for post-merger imperfect internalization.

*Retail Markups.* Our procedure assumes that retail prices are set by the manufacturers of products. This assumption is common in work in consumer packaged goods, dating back to Nevo (2000), and underlies the above cited papers studying markups at scale as well. However, in a model with retailer price setting, the retail markup would be a function of customer demand and the wholesale prices set by upstream firms, and the retail passthrough would be a function of the upstream prices. If passthrough is less than one-for-one, then a large predicted change in wholesale prices would lead to a smaller change in retail prices. Of course, in this environment, the implied prices in the pre-period would also lead to different implied costs in the pre-merger period.

Estimating a model of retailer competition is beyond the scope of this paper.<sup>15</sup> We instead follow an approach, proposed in Appendix E of Miller and Weinberg (2017), of calibrating a one-parameter model of competition in the retail sector. Given a set of upstream prices  $\mathbf{p}_{dtm}^U$ , a representative retailer will set a retail markup

$$\mu_{dtm}^R(\lambda, \mathbf{p}_{dtm}^U) \equiv -\lambda \cdot \left[ \left( \frac{\partial \mathbf{D}_{dtm}(\mathbf{p}_{dtm}^R(\mathbf{p}_{dtm}^U))}{\partial \mathbf{p}_{dtm}^R} \right)' \right]^{-1} \mathbf{D}_{dtm}(\mathbf{p}_{dtm}^R(\mathbf{p}_{dtm}^U)), \quad (4)$$

where the retail price is defined by

$$\mathbf{p}_{dtm}^R \equiv \mathbf{p}_{dtm}^U + \mu_{dtm}^R(\lambda, \mathbf{p}_{dtm}^U). \quad (5)$$

In (4),  $\lambda$  parameterizes the competitiveness of the retail sector, with  $\lambda = 0$  corresponding to a perfectly competitive retail sector, and  $\lambda = 1$  a monopolistic one; the formulation also assumes away additional marginal costs from the retailer. When setting wholesale prices, the upstream manufacturers know that the the retail prices will follow (5).

Fixing a  $\lambda$ , we implement the merger simulation by recovering costs from the first-order condition corresponding to the upstream manufacturer's problem in this model. We use the same forecasting procedure for costs as in Section IV.A, and we solve this two-stage model in the post-merger period using a modification of the algorithm from Morrow and Skerlos (2011) similar to the one proposed in Duarte (2025).

We report results with  $\lambda = 0.2$ , which is the upper bound of the range of robustness checks

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<sup>15</sup>In particular, by restricting attention to one product market at a time, we inherently rule out the possibility of cross-category spillovers (Thomassen et al., 2017). Separability of different product markets is a standard assumption in merger analysis, evaluation of which we leave to future work.

analyzed by Miller and Weinberg (2017). Panel G of Table 4 shows that this does not correct for the misprediction on average: the average prediction error becomes more negative for all parties, as does the median. It is important to note, however, that this is not a uniform shift to the left: the distribution of prediction errors does grow wider, and the upper quantiles increase.

#### IV.D. Implied Synergies

Across the models considered in Section IV.C, we find consistent evidence of overprediction. However, in all cases, we assumed that merging parties realized no cost synergies. Another potential explanation for the observed patterns of prediction errors is that merging parties realize marginal cost savings, on average, following mergers—which is another explanation grouped into the “supply-side residual” from Peters (2006). Prior work has documented such efficiencies using direct evidence in both consumer packaged goods (Ashenfelter et al., 2015) and other settings (Demirer and Karaduman, 2025; Braguinsky et al., 2015; Schmitt, 2017).

We report distributions of and analyze patterns of “implied synergies”: marginal cost changes that rationalize the observed price changes, given our estimates of demand and conduct assumptions. With estimates of post-merger demand, we invert (3) (or the equivalent condition in alternative models) using the realized prices at the realized product assortment and demand shocks  $\xi$  and obtain the implied marginal cost  $c_{jdtm}$ . Then, we define  $\text{Synergy}_{jdtm} \equiv (c_{jdtm}^{\text{Pred}} - c_{jdtm})/p_{jdm}^0$ , a quantity that is positive if marginal costs are below our prediction. We normalize this quantity by the original price of the good to bring units into the same scale and our synergies are thus measured as a proportion of price. As before, we aggregate this quantity over different units.

Table 5 shows the distribution of implied synergies in each of the settings and models considered in Section IV.C. The average implied synergy overall is essentially zero. Merging parties realize an average implied synergy of 2.1%, with them realizing positive implied synergies on average in slightly fewer than three-quarters of mergers. Non-merging parties realize negative implied synergies on average of about 1%, which correspond to cost increases. The remaining panels show similar results: merging parties enjoy implied synergies between 1.9% and 2.8%, while nonmerging party costs typically continue to increase slightly across models.

We are deliberate in using the term “implied” synergies. These synergies are model residuals; any price changes can be rationalized with the appropriate implied synergies. The true test of whether these correspond to actual synergies would be to compare them with direct measures of, or proxies for, costs. In this paper, we take an indirect approach and study patterns in these implied synergies that one may expect if they were capturing real changes in marginal costs. These patterns are summarized in Table 6 for each of our approaches, and we discuss them in turn.

First, prior work documents that synergies take time to materialize, sometimes even beyond

|                                                 |        |          | Quantiles |        |        |       |       |
|-------------------------------------------------|--------|----------|-----------|--------|--------|-------|-------|
|                                                 | Mean   | St. Dev. | 10%       | 25%    | 50%    | 75%   | 90%   |
| A. Baseline                                     |        |          |           |        |        |       |       |
| Merging                                         | 0.021  | 0.077    | -0.056    | -0.016 | 0.024  | 0.054 | 0.103 |
| Overall                                         | -0.002 | 0.123    | -0.050    | -0.030 | 0.001  | 0.029 | 0.068 |
| Non-Merging                                     | -0.012 | 0.124    | -0.068    | -0.032 | -0.006 | 0.024 | 0.061 |
| B. At Optimal Scaling of Price Coefficients     |        |          |           |        |        |       |       |
| Merging                                         | 0.013  | 0.060    | -0.056    | -0.020 | 0.020  | 0.048 | 0.079 |
| Overall                                         | -0.015 | 0.095    | -0.057    | -0.033 | -0.007 | 0.019 | 0.040 |
| Non-Merging                                     | -0.022 | 0.099    | -0.083    | -0.037 | -0.010 | 0.015 | 0.040 |
| C. At Optimal Scaling of Non-Price Coefficients |        |          |           |        |        |       |       |
| Merging                                         | 0.017  | 0.077    | -0.059    | -0.020 | 0.026  | 0.057 | 0.087 |
| Overall                                         | -0.012 | 0.122    | -0.058    | -0.031 | -0.007 | 0.022 | 0.054 |
| Non-Merging                                     | -0.023 | 0.119    | -0.070    | -0.037 | -0.011 | 0.014 | 0.050 |
| D. Pre-Merger Conduct                           |        |          |           |        |        |       |       |
| Merging                                         | 0.010  | 0.066    | -0.056    | -0.023 | 0.019  | 0.046 | 0.073 |
| Overall                                         | -0.007 | 0.118    | -0.050    | -0.030 | -0.004 | 0.023 | 0.065 |
| Non-Merging                                     | -0.011 | 0.122    | -0.067    | -0.032 | -0.007 | 0.023 | 0.059 |
| E. Post-Merger Conduct                          |        |          |           |        |        |       |       |
| Merging                                         | 0.019  | 0.076    | -0.056    | -0.016 | 0.021  | 0.053 | 0.092 |
| Overall                                         | -0.004 | 0.123    | -0.050    | -0.030 | -0.001 | 0.027 | 0.068 |
| Non-Merging                                     | -0.012 | 0.124    | -0.068    | -0.032 | -0.006 | 0.026 | 0.061 |
| F. Retail Pricing ( $\lambda = 0.2$ )           |        |          |           |        |        |       |       |
| Merging                                         | 0.028  | 0.149    | -0.096    | -0.018 | 0.028  | 0.066 | 0.131 |
| Overall                                         | -0.001 | 0.228    | -0.147    | -0.042 | 0.003  | 0.069 | 0.216 |
| Non-Merging                                     | -0.011 | 0.233    | -0.153    | -0.045 | 0.003  | 0.051 | 0.157 |

Table 5: Summary statistics for implied synergies

the analysis period of this paper.<sup>16</sup> This observation is consistent with the fact that many of the plausible economic mechanisms for the source of true marginal cost reductions—e.g., changes in distribution networks or production processes—are not immediate. Accordingly, we would expect synergies for merging parties to (i) increase over time and (ii) likely not take place immediately

<sup>16</sup>In the case of beer, Ashenfelter et al. (2015) uses distance to market as a measure of efficiencies and finds reductions about two years after the merger, stemming from the time it takes to reorganize production and distribution. Schmitt (2017) uses explicit measures of costs after hospital mergers and documents that about half the measured cost reduction happens 1–2 years after the merger but continues up to 4 years after the merger. Focarelli and Panetta (2003) and Sheen (2014) both use prices as a proxy for synergies: the former find price decreases in bank mergers starting approximately two years after consummation, and the latter finds a decrease in prices in consumer goods that speeds up after a year from the merger.

|                             | Baseline      | Optimal Demand<br>Scaling (Price) | Optimal Demand<br>Scaling (Non-Price) | Pre-Merger<br>Conduct | Post-Merger<br>Conduct | Retail Pricing<br>( $\lambda = 0.2$ ) |
|-----------------------------|---------------|-----------------------------------|---------------------------------------|-----------------------|------------------------|---------------------------------------|
| A. Synergies over Time      |               |                                   |                                       |                       |                        |                                       |
| Merging Party Synergies     |               |                                   |                                       |                       |                        |                                       |
| All                         | 2.5%          | 2.0%                              | 2.6%                                  | 1.9%                  | 2.1%                   | 2.8%                                  |
|                             | [-1.6%, 5.4%] | [-2.0%, 4.8%]                     | [-2.0%, 5.7%]                         | [-2.3%, 4.6%]         | [-1.6%, 5.3%]          | [-1.8%, 6.6%]                         |
| Months 1–6                  | 1.4%          | 1.5%                              | 1.7%                                  | 0.9%                  | 1.3%                   | 1.5%                                  |
|                             | [-1.8%, 6.2%] | [-1.2%, 4.4%]                     | [-1.8%, 5.5%]                         | [-2.0%, 4.1%]         | [-1.8%, 5.0%]          | [-2.9%, 6.2%]                         |
| Months 19–24                | 3.3%          | 2.8%                              | 3.7%                                  | 2.1%                  | 2.6%                   | 2.7%                                  |
|                             | [-2.6%, 7.8%] | [-2.4%, 6.9%]                     | [-1.8%, 7.7%]                         | [-2.3%, 6.3%]         | [-2.3%, 7.6%]          | [-2.0%, 9.1%]                         |
| Non-Merging Party Synergies |               |                                   |                                       |                       |                        |                                       |
| All                         | -0.6%         | -1.0%                             | -1.1%                                 | -0.7%                 | -0.6%                  | 0.3%                                  |
|                             | [-3.2%, 2.4%] | [-3.8%, 1.5%]                     | [-3.7%, 1.4%]                         | [-3.2%, 2.3%]         | [-3.2%, 2.6%]          | [-4.5%, 5.1%]                         |
| Months 1–6                  | -0.2%         | -0.4%                             | -0.6%                                 | -0.2%                 | -0.2%                  | -0.2%                                 |
|                             | [-2.1%, 2.4%] | [-2.8%, 1.5%]                     | [-2.5%, 1.7%]                         | [-2.1%, 2.3%]         | [-2.1%, 2.4%]          | [-4.3%, 4.5%]                         |
| Months 19–24                | -1.1%         | -1.3%                             | -1.5%                                 | -1.3%                 | -1.1%                  | -0.2%                                 |
|                             | [-4.4%, 2.4%] | [-4.7%, 1.2%]                     | [-5.0%, 2.0%]                         | [-4.3%, 2.6%]         | [-4.4%, 2.4%]          | [-7.0%, 5.8%]                         |
| B. Correlation Structure    |               |                                   |                                       |                       |                        |                                       |
| Adjusted $R^2$              |               |                                   |                                       |                       |                        |                                       |
| DMA                         | 3%            | 4%                                | 5%                                    | 3%                    | 2%                     | 7%                                    |
|                             | [-1%, 15%]    | [-1%, 14%]                        | [-1%, 15%]                            | [-1%, 15%]            | [-1%, 15%]             | [-1%, 19%]                            |
| Product                     | 36%           | 37%                               | 36%                                   | 34%                   | 35%                    | 29%                                   |
|                             | [16%, 49%]    | [16%, 49%]                        | [14%, 49%]                            | [15%, 50%]            | [16%, 49%]             | [12%, 49%]                            |
| Slope with Unilateral       |               |                                   |                                       |                       |                        |                                       |
| DMA                         | 0.35 (0.16)   | 0.41 (0.12)                       | 0.16 (0.12)                           | 0.11 (0.06)           | 0.07 (0.02)            | 0.49 (0.15)                           |
| Product                     | 0.74 (0.34)   | 0.50 (0.30)                       | 0.49 (0.10)                           | -0.51 (0.29)          | -0.23 (0.17)           | 0.10 (0.15)                           |

Table 6: Properties of synergies. Panel A shows median synergies at the merger level (along with first and third quartile in brackets underneath) for various time period after the merger. Panel B shows the share of variation at the product-DMA level (as measured by adjusted  $R^2$ ) explained by DMA fixed effects or product fixed effects, reporting the median (and first and third quartiles) across mergers. The second section of Panel B shows the slope of a regression of synergies at the merger-DMA and merger-product levels on unilateral effects, with merger fixed effects; standard errors are in parentheses.

after the merger. The top of Panel A of Table 6 shows that (i) typically holds across specifications: the median synergy 19–24 months after the merger is about 1–2 pp larger than that 1–6 months after the merger. The third quartile across mergers frequently increases by slightly more, while the change in the first quarter is typically less in magnitude. However, there is somewhat less support for (ii), as the median implied synergy across mergers is slightly above 1% within the first 6 months, and implied synergies are above 6.2% for a quarter of mergers. This observation may capture renegotiation of contracts with upstream suppliers (which could plausibly be done more quickly), but it could also suggest a different systematic departure from the modeling assumptions, particularly for a subset of mergers.

Second, we may not expect non-merging parties to realize synergies. As discussed above, we find that at the median merger, non-merging firms realize lower implied synergies in magnitude—although they still amount to dissynergies of approximately 0.5–1% over the course of the post-merger period we study. These implied dissynergies are not realized on impact, with the median being close to 0 in the initial six months. The magnitude does grow over time: this is not indefensible, as we could expect non-merging firms to suffer from spillovers in bargaining between merging



parties and upstream suppliers.

Third, the mechanisms hypothesized above imply a structure for the observed synergies for merging parties. If synergies are due to efficiencies in distribution, we would expect them to be correlated within DMA across products. If they are due to improvements in production costs, perhaps because of unifying production lines, we would expect them to be correlated across DMAs within products. To investigate this, we regress synergies at the DMA-product level for each merger on DMA fixed effects and then (separately) on product fixed effects. We do so separately for each merger, and the first two parts of Panel B report the median adjusted  $R^2$  from this regression, along with the first and third quartiles across mergers. For DMA fixed effects, we find that the  $R^2$  for merging parties is fairly low, with a median of 3% and a third quartile of 15% in the baseline specification. The  $R^2$  for product fixed effects is 36% at the median, with a third quartile of 49%. To benchmark these numbers, we conduct the same exercise on our placebo mergers, described in Section IV.B. The distribution of  $R^2$  for DMA fixed effects is very similar to the placebos, so we do not find widespread evidence of merger-induced cost synergies at the DMA level in our sample. However, the placebos show lower  $R^2$  at the product level, with a median of 21% and quartiles from 12–38%. We therefore find that synergies induced by mergers tend to be more concentrated at the product level than we would expect through random chance, consistent with production efficiencies.

Finally, we ask whether the mergers, DMAs, and products in which we recover the largest synergies are the ones where we also expect large effects due to the merger. If the markets or products in which our model predicts large price changes are the ones where we estimate large synergies, we may worry that the source of misprediction is model misspecification. However, there are also reasons to believe that these quantities should be related for consummated mergers at the merger level. If a merger is likely to generate large price effects in the absence of synergies, then the fact the enforcement agencies did not block it may suggest that they expected cognizable synergies to emerge. Alternatively, if firms go through the trouble of proposing a merger despite no incentives to raise price, then they must have expected synergies to materialize. These explanations, however, are more difficult to rationalize across DMAs or across products within-merger.

To investigate this question, we first compute  $\text{Unilateral}_{jdtm}$  as the unilateral effect of the merger for product  $j$  in DMA  $d$  in post-merger period  $t$  and merger  $m$ ; we define the specifics of this quantity in Section V.A, but it quantifies the incentives of the parties to raise prices due to the merger under the maintained conduct assumption. We aggregate as usual and regress  $\text{Synergy}_x$  on  $\text{Unilateral}_x$  at various levels, with merger fixed effects. The final section of Panel B reports the coefficients at the merger-DMA and the merger-product levels. Here we do find some heterogeneity across specifications. We find significant slopes at both the DMA and product levels in the baseline framework, and at the DMA level when we scale the price coefficients and introduce a retail markup. However, this is the one test that gives qualitatively different results across specifications. In

particular, the slope at the DMA level decreases when we consider optimal scaling of the non-price coefficients in the demand system. Furthermore, both slopes are much smaller in the specifications that change conduct. To the extent that one could view these exercises as a validation of the underlying model, this results could provide some suggestive evidence in favor of departures from Bertrand-Nash conduct.<sup>17</sup>

Together, we find consistent evidence of implied marginal cost synergies across specifications: demand scaling, misspecification of pre- or post-merger conduct, and modeling the vertical structure do not by themselves explain away the cost reductions. The median implied synergy for merging parties ranges from 1.9–2.8% of pre-merger prices, although there is substantial variation across mergers. Furthermore, to the extent that we view the slope with the unilateral effect as a way to falsify models of conduct, our tests suggest a scope for both marginal cost synergies and departures from Bertrand-Nash conduct in explaining the overprediction of merger effects.

We also caution that, even if a reader interprets these results as generally in favor of true merger-induced marginal cost reductions, they should not extrapolate this conclusion to a randomly-selected merger. Our sample consists of consummated mergers. Mergers that were expected *ex ante* to generate little in the way of efficiencies may have been less likely to be proposed, more likely to be abandoned, or more likely to be challenged or restructured before consummation. Conversely, mergers with more plausible efficiency justifications may be disproportionately represented among the transactions that ultimately close. As a result, the fact that many consummated mergers in our sample exhibit positive implied synergies should not be read as evidence that other hypothetical mergers generally generate synergies. Further, our comparisons are necessarily local to the antitrust enforcement environment firms faced during this time period; if antitrust enforcers became more likely to credit efficiencies, then one might expect firms to respond by proposing more mergers with fewer efficiencies.

## **V. The Predictive Power of Merger Simulations**

A practitioner of merger simulations typically does not view the output of merger simulations as an exact forecast but rather an indication of whether the merger is likely to cause consumer harm (Shapiro, 2010). By that token, a practitioner would worry about whether the unilateral effect under standard assumptions—the number that merger simulations are designed to target—is predictive of the true price change. This is of course a separate question from whether the simulations are

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<sup>17</sup>Another potential departure from Bertrand-Nash conduct that could resolve this correlation is bargaining between retailers and manufacturers. It is possible, although not guaranteed, that increases in manufacturers' bargaining power, or decreases in the retailers' disagreement points, could lead to lower retail prices that increase quantity (and manufacturers' profits) at the expense of retailers'. To the extent that these situations are more likely in markets with large DHHI, we would interpret this as a correlation between DHHI and the synergies implied by Bertrand-Nash conduct.

accurate (or unbiased) on average.

In this section, we consider the question of whether merger simulations are informative signals of the effect of the merger. As discussed above, our results will only apply to the set of consummated mergers in CPG and should not be extrapolated to any potential merger. However, since almost all proposed mergers are eventually consummated in our setting, the sample under consideration is relevant for antitrust agencies. Section V.A describes the prediction procedure. Sections V.B and V.C discuss different approaches to evaluating predictive power, and Section V.D comments on implications for antitrust enforcement.

### V.A. Defining the Predictive Exercise

In this section, we assess whether the unilateral effect is predictive for a “causal” effect of the merger, which is arguably the object of interest for an antitrust agency. Unlike in the exercise in Section IV, this requires us to take a stance on a counterfactual price that would have occurred but-for the merger. The observed post-merger price is due to not just due to changes in the pricing behavior of the parties in the market but also due to potential changes in demand, product assortment, and costs—both due to observable input costs and unobservable shifts to trends. Most commonly used methods to predict merger effects (merger simulations, pricing pressure indices, and presumptions) are not designed to predict changes in demand, assortment, or market-wide trends in costs, so we do not want to attribute limited predictive ability to their inherent inability to predict these factors.

In defining the causal effect of a merger, we take the approach of keeping as many of these factors close to the values that would be implied by the observed post-merger prices as possible. We take the post-merger product assortment as given. We invert the demand system at the observed post-merger shares and prices to determine the post-merger demand shocks  $\xi$  in each market. Finally, we assume that the post-merger costs for non-merging parties are those that would be implied by the Bertrand-Nash equilibrium at the observed prices, and we invert (3) to recover them. We are, however, unwilling to make this assumption for merging parties, as synergies are an often-cited explanation for merging. We instead assume that the pre-period cost function remains fixed for merging parties, and any shocks to costs for merging parties follow the same trend as the non-merging cost shocks do.<sup>18</sup> The counterfactual price  $p^{\text{cf}}$  is then the solution to (3) at the post-merger  $\xi$ , counterfactual cost vector  $c^{\text{cf}}$ , and pre-merger ownership matrix  $\Omega$ . Suppressing the dependence on  $m$ , the causal effect  $\text{Causal}_{jdt}$  is  $\log(p_{jdt}) - \log(p_{jdt}^{\text{cf}})$ . We define  $\text{Unilateral}_{jdt}$  as the log difference in prices with the post-merger and pre-merger ownership matrix at these demand

<sup>18</sup>Formally, we assume that  $h_m(x_{jm}, w_{jdtm}, t)$  stays the same after the merger, and that for a post-merger time period  $t'$  and a product owned by the merged firm  $j$ ,  $\omega_{jdt'} = \bar{\omega}_{jd,\text{pre}} + \bar{\Delta\omega}_{nm,dt'}$ , where  $\bar{\omega}_{jd,\text{pre}}$  is the average  $\omega$  for product-DMA  $jd$  in the last three months before the merger and  $\bar{\Delta\omega}_{nm,dt'}$  is the average within-DMA difference in  $\omega$  for non-merging parties from the last three months before the merger to period  $t'$ .

and cost schedules. We aggregate across time periods and DMAs in the same manner we aggregate  $\text{Error}_{jdm}$ .

All comparisons of simulated effects to actual effects must control for changes to the environment that are auxiliary to the merger. The main strength of our definition of the causal effect is that it allows us to control for these changes in a manner that is internally consistent with the model. Because we have estimated a demand system and cost function, we are able to use them to define the counterfactual prices directly, rather than having to rely on parametric functions of cost shifters and time—an approach some of us have used in previous work (Bhattacharya et al., 2026), but which may be difficult to microfound from primitives.<sup>19</sup> However, since the literature evaluating merger simulations has typically compared simulations to effects estimated from an event study or a difference-in-difference, we follow this approach in Appendix B. In particular, we compare the total change in the prices before and after the merger to the predictions using pre-merger data, controlling for the realized changes to observable cost shifters. Our takeaways are similar: predicted price changes are typically correlated with realized price changes, although we have to be careful about interpreting how much of this predictive power is coming from observable changes to marginal costs, which can be substantial in our setting. The caveats to this exercise are that the overall change from before to after the merger is likely not the object of interest for the enforcement agencies, and that much of the before and after variation in our setting comes from changes to unobserved input costs, which introduces noise to the exercise that is unrelated to the question of interest.

## **V.B. Comparing Causal and Unilateral Effects of Mergers**

A first question is whether unilateral effects are “calibrated” to the true causal effect of the merger: is an increase in the unilateral effect associated with a increase in the causal effect of the merger in expectation? While a one-for-one relation is strong, a positive association in expectation is in principle a fairly undemanding requirement of merger simulations. However, because it requires a sizable sample of mergers to test, it is not something that has been established before in the literature. Moreover, there are many reasons this could fail even if demand is well-estimated. For instance, mergers that would have large price effects might only be approved (or even proposed) if parties expect large cognizable synergies.<sup>20</sup> A more extreme reason is that the theory that underlies merger simulations—that of unilateral price effects tied to substitution patterns—could be a poor representation of firms’ pricing models in reality. Moreover, checking for directional accuracy at different levels is informative: even if parties change prices in a way that is on average consistent with predictions from merger simulations, effects in different geographies and products may be

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<sup>19</sup>See Aryal et al. (2025) for one microfoundation under parametric assumptions on demand and cost functions.

<sup>20</sup>The opposite correlation is also possible: firms may only be willing to merge in competitive markets if they expect to realize synergies.

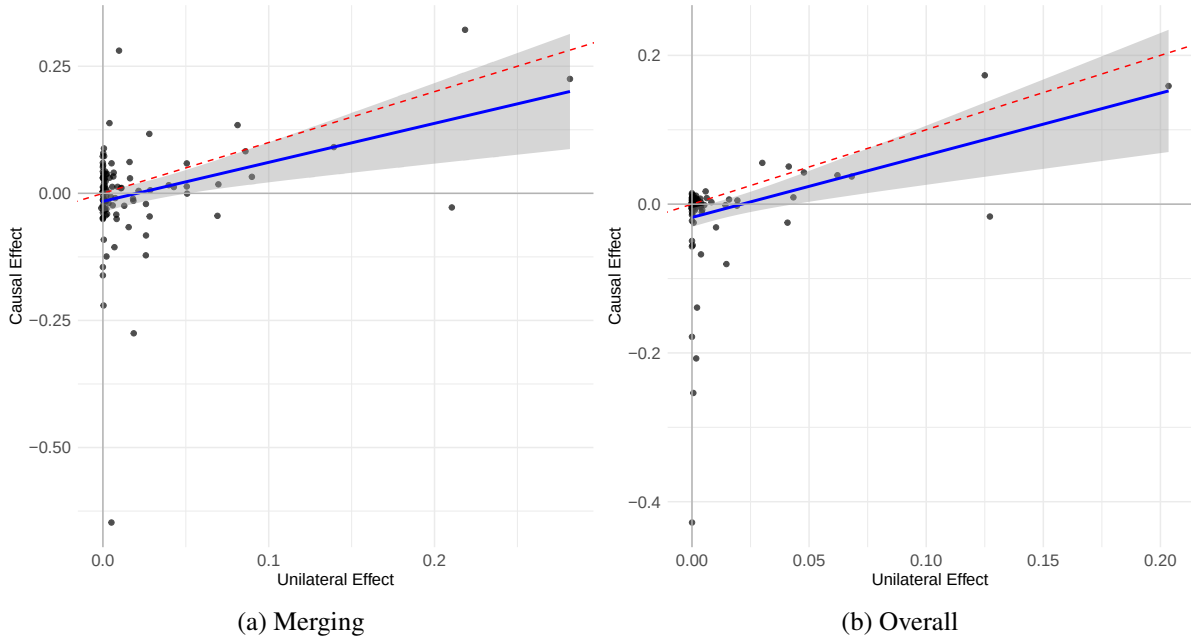


Figure 3: Scatter of the causal effect versus the unilateral effect, at the merger level, for (a) merging parties, and (b) all parties. The dotted red line indicates the 45° line, and the blue line is the line of best fit.

based on rules-of-thumb or uniform pricing rules (Escudero, 2018; DellaVigna and Gentzkow, 2019; Hitsch et al., 2021; Adams and Williams, 2019), mitigating the relation between predictions and realizations.

Figure 3 scatters  $\text{Causal}_m$  against  $\text{Unilateral}_m$  both for merging parties and overall. Due to the presence of divestitures, unilateral effects need not have been positive, although we find that they are in all mergers. Causal effects often are negative, an observation consistent with merger retrospectives in this industry (Bhattacharya et al., 2026). Both panels show that the predictions are correlated with actual price changes: the slope is 0.77 (s.e. 0.40) for merging price changes and 0.84 (s.e. 0.55) overall. The confidence intervals include the perfect calibration level of 1, although they are especially wide: the  $p$ -value for whether the slope is positive is 0.06 for merging party price changes and 0.13 overall.

A complementary exercise asks whether merger simulations are calibrated within a given merger: do DMAs and products predicted to have large price increases have large price increases on average? This within-merger test is especially useful because it differences out merger-specific shocks that shift prices broadly (for example, integration-wide cost shocks, aggregate demand shifts, or merger-level implementation disruptions), and it allays concerns about selection into merger proposal. Such results are also relevant for policy, as antitrust enforcement occurs not just at the merger level: agencies often require product divestitures, and they sometimes (but rarely) enforce regional divestitures as well. Relative to the cross-merger analysis, two additional, conflicting forces

|                   | Cross-Merger    | Within-Merger     |                  |                   |
|-------------------|-----------------|-------------------|------------------|-------------------|
|                   | (1)             | (2)               | (3)              | (4)               |
| A. Merging        |                 |                   |                  |                   |
| Constant          | -0.02<br>(0.03) |                   |                  |                   |
| Unilateral Effect | 0.77*<br>(0.40) | 0.88***<br>(0.10) | 0.32**<br>(0.13) | 0.43***<br>(0.09) |
| B. Overall        |                 |                   |                  |                   |
| Constant          | -0.02<br>(0.03) |                   |                  |                   |
| Unilateral Effect | 0.84<br>(0.55)  | 0.79***<br>(0.11) | 0.32**<br>(0.15) | 0.45***<br>(0.09) |
| C. Non-Merging    |                 |                   |                  |                   |
| Constant          | -0.01<br>(0.02) |                   |                  |                   |
| Unilateral Effect | 1.61<br>(2.37)  | 1.05***<br>(0.29) | 0.95**<br>(0.42) | 1.29***<br>(0.19) |
| Level             | Merger          | DMA               | Product          | P-DMA             |
| Merger FE         | No              | Yes               | Yes              | Yes               |

Table 7: Regressions of causal effects on unilateral effects. Each column runs the regression at a different level of observation (with “P-DMA” standing for product-DMA). Panels A and C focus on merging and non-merging price changes only, respectively. Panel B reports results for all price changes. Standard errors are clustered at the merger level when appropriate and account for estimation error in the dependent variables and regressors via bootstrapping the demand estimates and propagating this error through the regressions reported here via Rubin’s rules (Rubin, 1987).

affect our interpretation of the within-merger results. First, given geography-specific remedies are uncommon, it is unlikely that synergies at the DMA level are negatively correlated with effects from the merger due to selection into merger approval. This force would act to increase the correlation between predicted and actual price changes. On the other hand, there is evidence that price changes are not sensitive to local market structure due to uniform pricing rules at the retailer level (DellaVigna and Gentzkow, 2019), although national firms still seem sensitive to local cost shocks (Butters et al., 2022).

We next regress causal effects on unilateral effects at the DMA, product, and product–DMA levels and include merger fixed effects, so identification comes from deviations from each merger’s own mean predicted and realized price change. The results are reported in Columns (2)–(4) of Table 7. We find that these slopes are generally positive, suggesting that even after absorbing merger-level unobservables, calibration still remains a feature of these predictions.<sup>21</sup>

<sup>21</sup>Since elements of the regression in Table 7 are estimated quantities, we also verify in Table A.2 that a split-sample IV procedure to account for the sampling error of the regressors returns economically similar quantities.

|                   | Cross-Merger | Within-Merger |         |       |
|-------------------|--------------|---------------|---------|-------|
|                   | (1)          | (2)           | (3)     | (4)   |
| A. Merging        |              |               |         |       |
| $R^2$             | 0.115        | 0.529         | 0.176   | 0.175 |
| Spearman's $\rho$ | 0.120        | 0.206         | 0.138   | 0.120 |
| Within- $R^2$     | —            | 0.200         | 0.016   | 0.036 |
| B. Overall        |              |               |         |       |
| $R^2$             | 0.137        | 0.389         | 0.186   | 0.201 |
| Spearman's $\rho$ | 0.197        | 0.251         | 0.195   | 0.195 |
| Within- $R^2$     | —            | 0.107         | 0.016   | 0.032 |
| C. Non-Merging    |              |               |         |       |
| $R^2$             | 0.082        | 0.339         | 0.377   | 0.345 |
| Spearman's $\rho$ | 0.305        | 0.234         | 0.240   | 0.241 |
| Within- $R^2$     | —            | 0.051         | 0.014   | 0.048 |
| Level             | Merger       | DMA           | Product | P-DMA |
| Merger FE         | No           | Yes           | Yes     | Yes   |

Table 8: Measures of fit for the causal effect on the unilateral effect. In Columns (2)–(4), Within- $R^2$  and Spearman's  $\rho$  are computed after partialling out merger fixed effects.

Despite the positive slope, the scatters in Figure 3 suggest that the predictive power of simulations for the true price change is low, as the points are not tightly organized around the best-fit line. Table 8 verifies this observation. At the merger level, we find an  $R^2$  around 8%–14%, depending on whether we look at merging parties, non-merging parties, or all parties. This is not due to extreme outliers driving a low Pearson correlation: Spearman's  $\rho$  also indicates a fairly weak correlation. The within-merger analysis in Columns (2)–(4) illustrates that merger fixed effects do have substantial explanatory power, consistent with unmodeled merger-level changes being a source of this low correlation (e.g., nationwide synergies), but the within-merger analysis either yields similar results in the case of Spearman's  $\rho$ , or generally weaker correlations in the case of within- $R^2$ —except for merging parties across DMAs.

For context, we also compute the  $R^2$  of a regression of the causal effect on HHI, DHHI, and its interaction. We find similar levels of predictive power for merging parties (12%) but slightly larger ones for non-merging parties (15%). Thus, the presumptions and unilateral effects have similar predictive power, although the absolute amount is still low.

### V.C. Merger Simulations as a Screening Tool

The low predictive power documented in Table 8 does not necessarily rule out merger simulations being a useful tool for enforcement. Agencies are concerned with their ability to identify especially

harmful mergers, which could be distinct from the ability of a tool to predict the price change of all mergers. We assess this aspect of merger simulations in a statistical decision theory framework. We define a large price change to be situation where the causal effect is larger than a cutoff. We then use the unilateral effect as a predictor for this event: for any  $x$ , we consider the predicted event to be that the unilateral effect is larger than  $x$ . For this predicted event, we compute the so-called confusion matrix of true and false positives and negatives, and functions thereof. We then vary  $x$ .

We focus on precision-recall (PR) curves here: takeaways are similar with receiver operating characteristic (ROC) curves (Figure A.9). Precision is the share of mergers with unilateral effects larger than  $x$  that actually have large causal effects. Recall answers the complementary question of what share of mergers with large causal effects have unilateral effects larger than  $x$ . The curve itself traces out the relationship between the two quantities as  $x$  varies. Of course, “unilateral effect” can be replaced by any predictor, and we consider a variety. Points that are northeast dominate those in the southwest. In the extreme case, the oracle predictor has a PR curve that starts at  $(0, 1)$ , is horizontal initially, and drops vertically to  $(1, p)$ , where  $p$  is the share of positives in the data, while a completely random predictor has a PR curve that is flat at  $p$ .

Figure 4 plots the associated precision-recall (PR) curves for merging party price changes, where the level of the observation is the DMA-merger. We report merger-level PR and ROC curves, as well as those for overall price changes, in Appendix A. The PR curve for unilateral effects is in blue, and we mark points that correspond to marking 5%, 10%, or 20% of merger-DMA as high price changes using a diamond, circle, and square, respectively. We first find that the curves are far from the benchmark of a random predictor. Further, the classification is better for high-DHHI DMAs (those with  $\text{DHHI} \geq 100$ ) than for all DMAs: this is sensible, since for markets with especially small DHHI, the merger simulation would predict a negligible price effect. Finally, we do not see as large a difference in discrimination ability across the two cutoffs for high-DHHI DMAs.

We benchmark these curves against two classes of predictors. The first class involves approximations to a full merger simulation from a random coefficients logit model. In red, we show the PR curve corresponding to a merger simulation with a nested logit demand specification, using only inside-outside nests. We calibrate the nesting parameter so that the median diversion ratio to the outside option matches that of the random coefficients model; since this diversion ratio does not depend on the price coefficient, this can be matched exactly. We then scale the price coefficient to match the median own-price elasticity from the random coefficients specification. The PR curve corresponding to the merger simulation using a nested logit demand system is generally dominated by that using the random coefficients model: even though the aggregate diversion to the outside option and the own-price elasticity are matched exactly, the richer substitution patterns in mixed logit improve predictive power for high price changes.<sup>22</sup> The differences we observe have one-sided

<sup>22</sup>We have also tried estimating the nested logit demand system directly. Doing so yields some mergers with demand



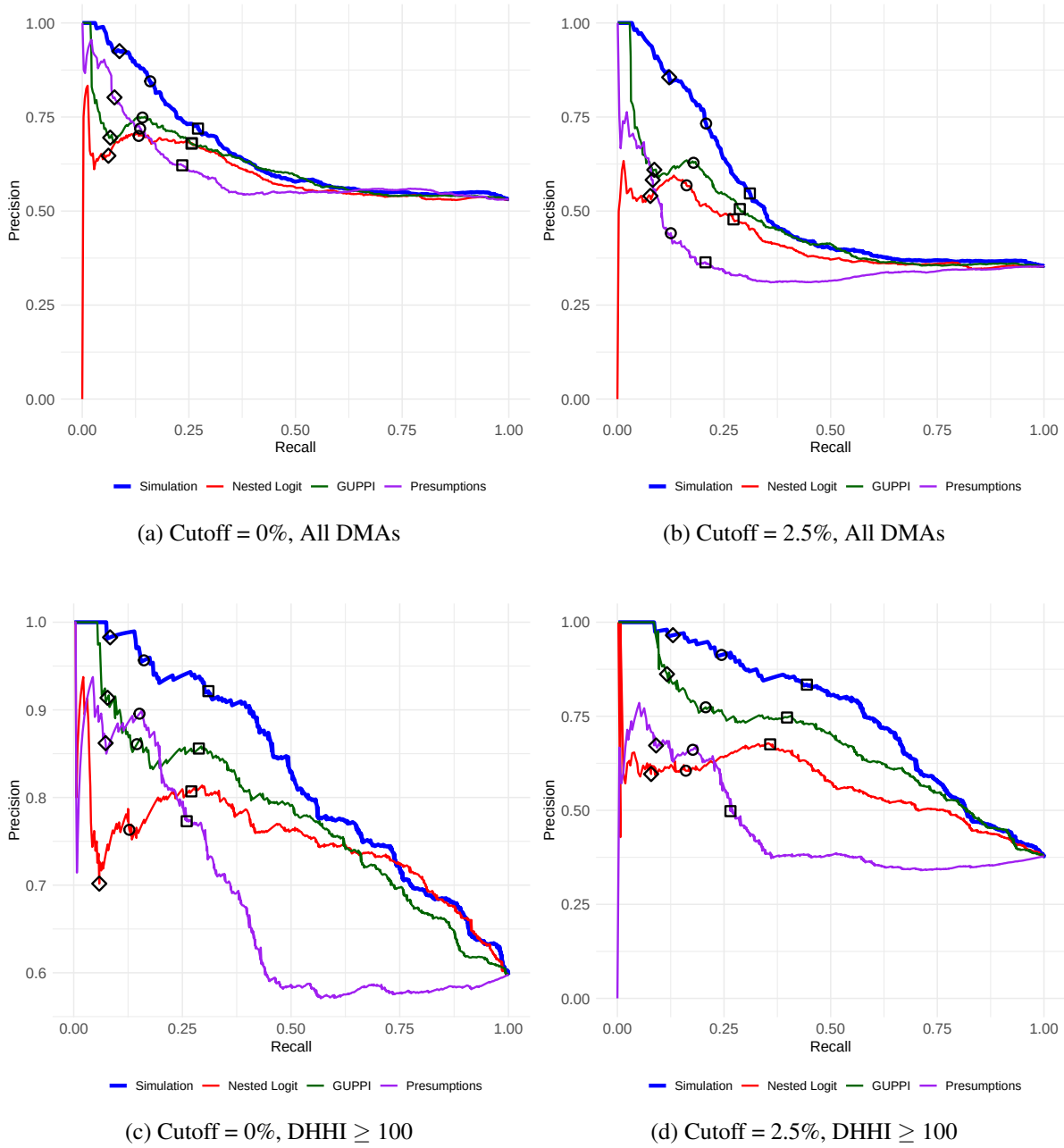


Figure 4: Precision-recall curves for merging-party price changes, using a variety of predictors. The diamond, circle, and square mark the points that correspond to marking 5%, 10%, and 20% of the merger-DMAs as high price changes.

$p$ -values (computed via a bootstrap clustered at the merger level, as described in Appendix A.3), as measured by areas under the PR curve, less than 0.15 at cutoffs of 0% and less than 0.10 at larger cutoffs (Table A.3).

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estimates that are not sensible for merger simulations. We have checked that using these estimates when possible and calibrations when not yield a worse PR curve.

Another approximation to a merger simulation is the gross upward pricing pressure index (GUPPI) proposed in Salop and Moresi (2009), the value of the diverted sales to the partner. This is closely tied to a first-order approximation to the price effect of a merger (Jaffe and Weyl, 2013). Although GUPPI abstracts away from curvature of the demand function and from competitor responses, prior work (Miller et al., 2017) has found that it serves as a good approximation to merger price effects when these considerations are limited.<sup>23</sup> Since the GUPPI involves the product of the diversion ratio and the margin, we need to calibrate both these parameters: we use margins and diversion ratios implied by the random coefficients model, and we take the share-weighted average of GUPPI from the target to the acquirer and the acquirer to the target. The PR curve for this predictor (green) generally falls short of that from the full simulation. The differences are significant (as measured by the area under the curve) at levels less than 0.10 for cutoffs less than 5%, and at levels less than 0.15 at a cutoff of 5% when restricting to DMAs with DHHI larger than 100 (Table A.3).

Of course, this exercise is predicated on already having estimated a random coefficients demand system. In practice, agencies that use GUPPI have alternate sources of margins and diversion to construct this approximation. We have experimented with using share-based diversion ratios, as well as diversion ratios computed from customer overlap following Einav et al. (2026). Since we do not have auxiliary sources of margins, we keep those implied by the random coefficients demand system in all our exercises. Both alternatives perform worse than the versions reported here (Figure A.9(b)).

The second class of benchmarks is based on measures of market structure cited as structural presumptions in the Merger Guidelines. For every cutoff we consider, we regress an indicator for the causal effect exceeding the cutoff on HHI, DHHI, and their interaction using a logistic specification. To avoid in-sample overfitting, we estimate this model using a leave-one-out procedure: for each merger, we fit the model on DMA-level observations from all other mergers and use the resulting coefficients to form predictions for the held-out merger. The PR curve for this approach, shown in purple, is also dominated by the full merger simulation.<sup>24</sup> These differences are significant at any conventional level. We make a few comments about these results. First, we try using DHHI and HHI separately as predictors, and we find that most of the predictive power of the structural presumptions come from DHHI: the PR curve using DHHI lies close to the PR curve using the combined predictor, while HHI shows limited predictive power (Figure A.9(c)), consistent with the findings in Nocke and Whinston (2022). Second, we construct a predictor using both the structural presumptions and the merger simulations themselves, and the PR curves do not improve markedly over the simulations themselves (see Figure A.9(c) again): the structural presumptions do not add

<sup>23</sup>We use the share-weighted GUPPI to aggregate across products.

<sup>24</sup>The in-sample fit of the presumptions model is, unsurprisingly, better, but it is still slightly worse than the GUPPI and nested logit benchmarks.

much information not captured by the simulations already. Finally, we also experimented with more flexible specifications, including generalized additive models (GAMs) of merging-party share, HHI, DHHI, and their interactions. These typically yield even worse out-of-sample predictions.

Another benchmark is how merger simulations compare to other methods used by practitioners. By modeling enforcement decisions, Bhattacharya et al. (2026) provide an estimate of the accuracy (share of classifications that coincide with the truth) of the agencies’ diligence process in determining whether mergers would raise or lower overall average price. They report an accuracy of 83%. The corresponding accuracy of merger simulations for predicting the event that the merger raises overall average price is 60%. This comparison must be viewed with caution, partly since the estimate in Bhattacharya et al. (2026) relies on a specific model of agency enforcement, but it underscores that neither the presumptions nor the simulations by themselves are a substitute for the work done by the agencies. With this caveat, we move to analyzing the implications of these results for enforcement in the next subsection.

## V.D. Implications for Enforcement

Another lens through which to interpret the results in Section V.C is that of an agency deciding on the level of enforcement. Agencies of course weigh a number of factors when deciding whether to challenge a case or request a remedy: the strength of auxiliary evidence, the probability of winning the case, the possibility of influencing precedent, non-price effects like innovation, and internal capacity constraints. However, to the extent that an agency focuses on price effects, an often-discussed concern in choosing the level of enforcement—even conditional on defining the “cutoff” price change the agency would want to target—is that between “Type I” and “Type II” errors (Hovenkamp, 2021). A Type I error is a false positive (challenging a pro-competitive merger), and a Type II error is a false negative (failing to challenge an anti-competitive merger).

We consider an agency endowed with a predictor of a merger’s causal effect on prices in each DMA. Its goal is to challenge mergers (or require remedies) in markets whose causal effects are greater than a cutoff  $C$ . The agency sets a threshold challenge rule: a value  $q \in [0, 1]$  such that a proposed merger is challenged in a given DMA if and only if the quantile of its predicted value among our sample of merger-DMAs is greater than  $q$ . The agency minimizes a loss function

$$L(q) \equiv \overbrace{\Pr(\text{Percentile of Predictor} > q | \text{Causal} \leq C)}^{\text{Type I Error Rate}} \cdot \Pr(\text{Causal} \leq C) + \kappa \cdot \underbrace{\Pr(\text{Percentile of Predictor} \leq q | \text{Causal} > C)}_{\text{Type II Error Rate}} \cdot \Pr(\text{Causal} > C), \quad (6)$$

which balances Type I and Type II error rates, with  $\kappa$  governing the trade-off. Regardless of the

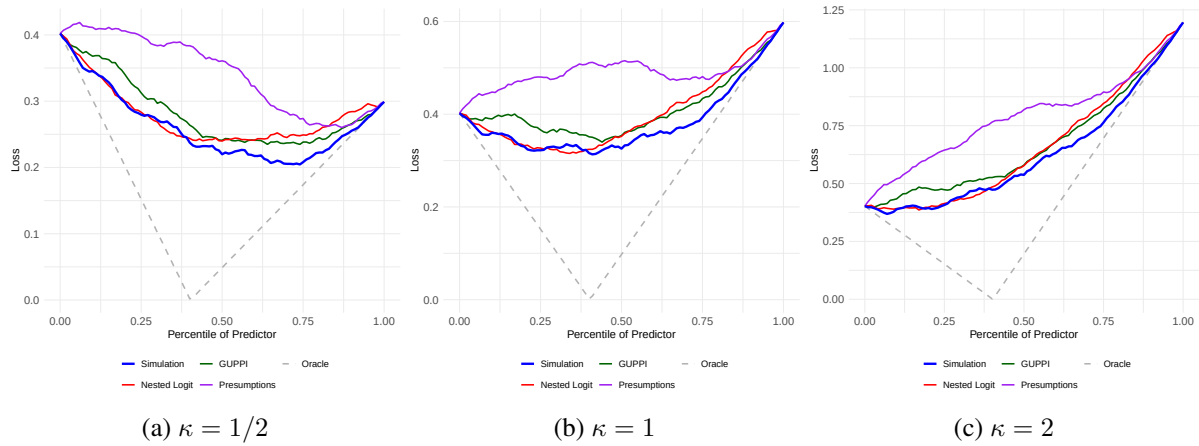


Figure 5: Loss functions at the DMA-merger level, restricting to markets with  $\text{DHHI} \geq 100$  at a cutoff of 0%

predictor,  $q = 0$  corresponds to challenging all mergers and  $q = 1$  corresponds to challenging no mergers.

Figure 5 shows this loss function for the unilateral effects from the merger simulation, as well as the three alternatives considered in Figure 4, for three different values of  $\kappa$ , at a cutoff of 0%, and restricting to DMAs with  $\text{DHHI}$  larger than 100. We also overlay a dashed gray line to illustrate the loss from the oracle. The horizontal axis depicts the percentile of the predicted value for the given predictor. Thus, antitrust agencies would begin by challenging the merger-DMAs on the right of the horizontal axis: an agency that wishes to challenge 10% of merger-DMAs would challenge ones whose percentile exceeds 0.9, and the loss they face under various predictors would correspond to the value of the function at a percentile of 0.9. Two observations about these lines mirror that from the PR curves in Section V.C. First, we find that the loss function from merger simulations typically dominates that from the other three—although the loss function from nested logit (calibrated to random coefficient elasticities) is comparable when challenge rates are larger than 50%, which we find to be implausibly high. Second, the loss functions in all three cases are significantly larger than the oracle loss, just as the PR curves were far from the oracle PR curve.

However, viewing the classification exercise through the lens of a loss function also allows us to analyze the optimal behavior of an agency. Consider an agency deciding whether to compute merger simulations or to just follow a predictor based entirely on the structural presumptions. If the agency were planning to challenge mergers in a fixed number of markets, it would only be concerned with the loss functions at one value of  $q$ . Under this scenario, simulations improve the decisions of the agency regardless of  $q$ . On the other hand, an unconstrained agency would choose  $q$  to minimize this loss, and thus the signal in the prediction could affect the challenge rate itself. Such an agency, if it were following the presumptions, would optimally challenge all mergers when  $\kappa$  equals 1 or 2, and would challenge in roughly 10% of markets when  $\kappa$  equals 1/2. In contrast,

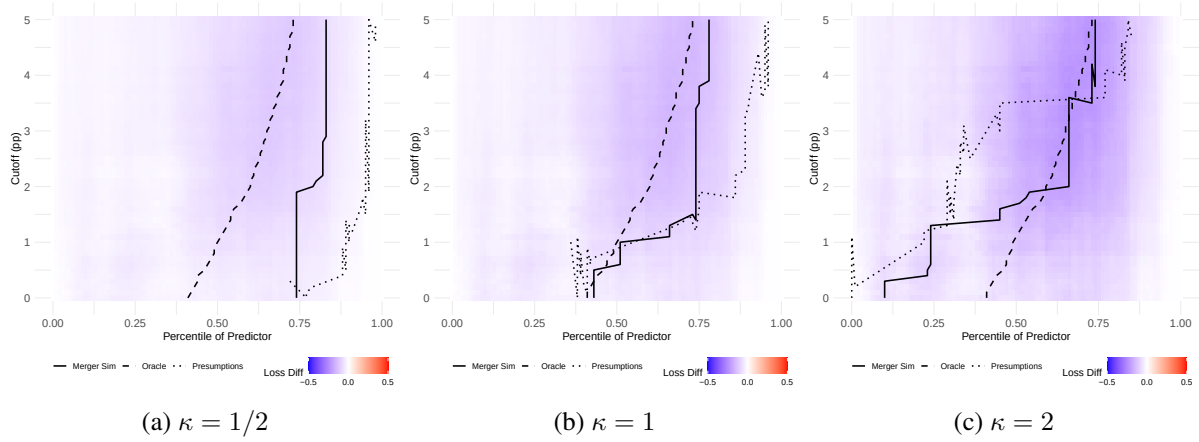


Figure 6: Difference in loss functions from simulations and presumptions, plotted against the percentile of the predictor and the cutoff, for DMAs with DHHI above 100. The solid line is the optimal challenge threshold using simulations, the dotted using presumptions, and the dashed the oracle-optimum.

such an agency would challenge in around 25%, 40%, and 95% of markets when  $\kappa$  equals 0.5, 1, and 2, respectively, if it used merger simulations.

Figure 6 plots the difference between the loss functions using the simulations and using the presumptions, for cutoffs ranging from 0% to 5%. The difference is always non-positive, showing that simulations dominate presumptions for a large set of parameter values. The figure also plots the optimal challenge rate using simulations or presumptions, and compares them to the oracle. The main observation is that simulations typically lead to an optimal challenge rate that is closer to the oracle-optimal. For  $\kappa = 1/2$  (i.e., the agency puts more weight on Type I errors), merger simulations lead to higher challenge rates than the presumptions, regardless of the cutoff. For  $\kappa = 1$ , simulations lead to higher challenge rates for cutoffs above 2%, and similar challenge rates for cutoffs below. Loss is still lower when following merger simulations because the selection of which mergers to challenge is more accurate. Finally, for  $\kappa = 2$  we find that simulations yield higher challenge rates for cutoffs above 4%, and lower challenge rates for cutoffs below.

To interpret this finding, consider an agency's optimal selection of  $q$  given a prespecified cutoff  $C$ . Under a truly random or uninformative predictor, the predictor and the causal effect are independent, and the loss function is linear in  $q$  with slope equal to  $\kappa \cdot \Pr(\text{Causal} > C) - \Pr(\text{Causal} \leq C)$ ; as such, the agency's will either block or approve all mergers depending on the sign of this quantity. Conversely, under the oracle predictor, the loss function is equal to zero at  $q = \Pr(\text{Causal} > C)$ , so regardless of  $\kappa$  the agency optimally challenges in a fraction of markets equal to the proportion with causal effects greater than  $C$ . Thus, the stronger the signal of the true causal effect, the more moderate optimal challenge rates become, regardless of  $\kappa$ . Since merger simulations are a more informative signal than structural presumptions, they push the optimal challenge rate more towards intermediate values, making challenge rates more uniform across agencies with different weights

on Type I versus Type II errors.<sup>25</sup>

## VI. Conclusion

This paper conducts the largest analysis of merger simulations thus far in the literature. Doing so allows us to find systematic patterns in how simulations depart from realizations of price changes. Merger simulations typically overpredict price changes by about 4% for merging parties. This overprediction could be rationalized with an average synergy of about 2% of initial price. We explore the possibility of demand misspecification, pre-merger coordination, imperfect internalization of the newly merged entity's profits, and retailer markups in explaining these results. While we find some evidence of imperfect internalization, we still find a role for synergies on average.

Our finding that merger simulation frequently overpredicts price increases cautions against naively taking the output of merger simulation as unbiased predictions. However, merger simulations could still be predictive even if they are not exactly calibrated to true price changes. While we find limited evidence in favor of their ability to predict price changes of a merger, we find that they are able to screen mergers with high price changes more effectively than measures of market structure by themselves, or than approximations to the full merger simulation. This has implications for antitrust enforcement: an unconstrained agency would have a challenge rate closer to the optimal rate, and enforcers with different preferences over the cost of Type I vs. Type II errors will tend to have more similar behavior if they base enforcement decisions on merger simulation.

Given these results, how should antitrust enforcers think about using merger simulation in practice? There are two broad takeaways from our analysis: merger simulation (a) is more informative than other methods at predicting price increases, even without incorporating merger-specific foreknowledge of marginal cost changes, and yet (b) is still relatively far away from being oracle-optimal. This suggests that while other less costly methods may credibly be used as initial screens, merger simulation is still more reliable as a screen than GUPPI or predicted price changes using only the structural presumptions. Merger simulation may also be made even more accurate with case-specific knowledge of post-merger marginal cost changes, which seemingly account for a substantial fraction of post-merger prediction error; unlike some other methods, merger simulation can incorporate assumptions about cost synergies to refine predictions in an internally consistent manner. However, the assumptions underpinning merger simulation—in particular those relating to firm conduct—should be critically evaluated on a case-by-case basis, as we find some suggestive evidence for violations of the standard Bertrand-Nash assumption. Ultimately, therefore, merger simulation should be viewed as a valuable part of the merger screening toolkit, but not a substitute

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<sup>25</sup>We caveat that this is a partial equilibrium exercise: changes in agency stringency would likely cause responses in firms' merger proposal behavior, changing the underlying distribution of causal effects.

for a more holistic merger review process.

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## ONLINE APPENDIX

### A. Additional Tables and Figures

#### A.1. Additional Results Related to Demand Estimates

We first present auxiliary results related to demand estimation. Figure A.1 scatters the median elasticity for high-income customer against that for low-income customers, at the merger level. This scatter shows that in almost all mergers, we estimate high-income customers to be less price sensitive than low-income ones.

Figures A.2 and A.3 show measures of substitutability between products. Figure A.2(a) scatters the diversion to acquirer products, from target products on the vertical axis and from non-merging products on the horizontal, at the merger (averaging diversion ratios with share-weights of the “from” group). While many diversion ratios are small, we find that diversion ratios from the target are typically larger than those from non-merging parties. Panel (b) repeats this exercise but taking the target firm as the destination, and the takeaway is generally similar, although there are a few mergers where diversion from the non-merging parties to the target is larger than that from the acquirer to the target (although both tend to be large). Larger diversion ratios between merging products is not an expected property of demand systems, but it is consistent with closer substitutability between products being one potential reason for the merger.

This heterogeneity is not just due to differences in shares. Figure A.3 compare diversion ratios from the estimated mixed logit system (red) to the share-based diversion one would expect from logit (blue). While many mergers have low diversion ratios in both models, we find that the median,

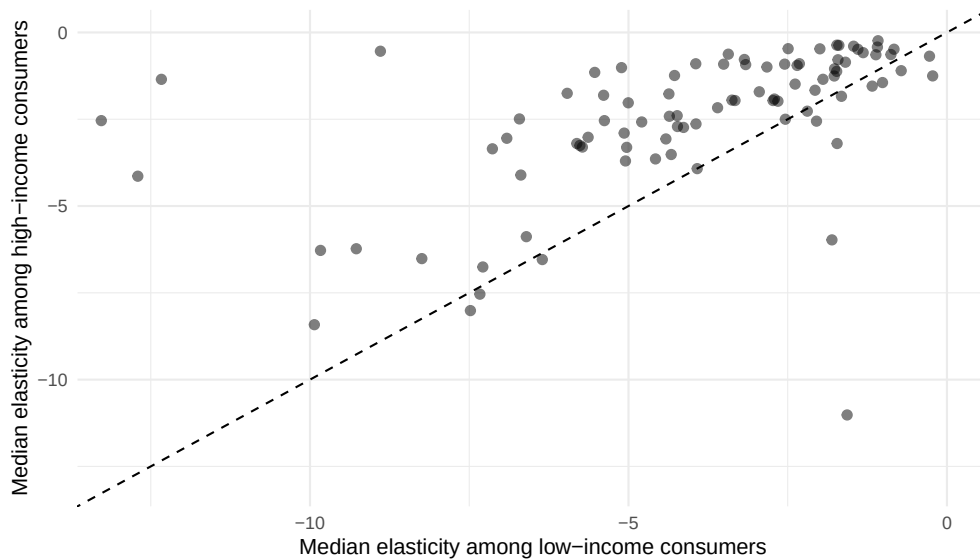


Figure A.1: Median elasticities for low- and high-income consumers

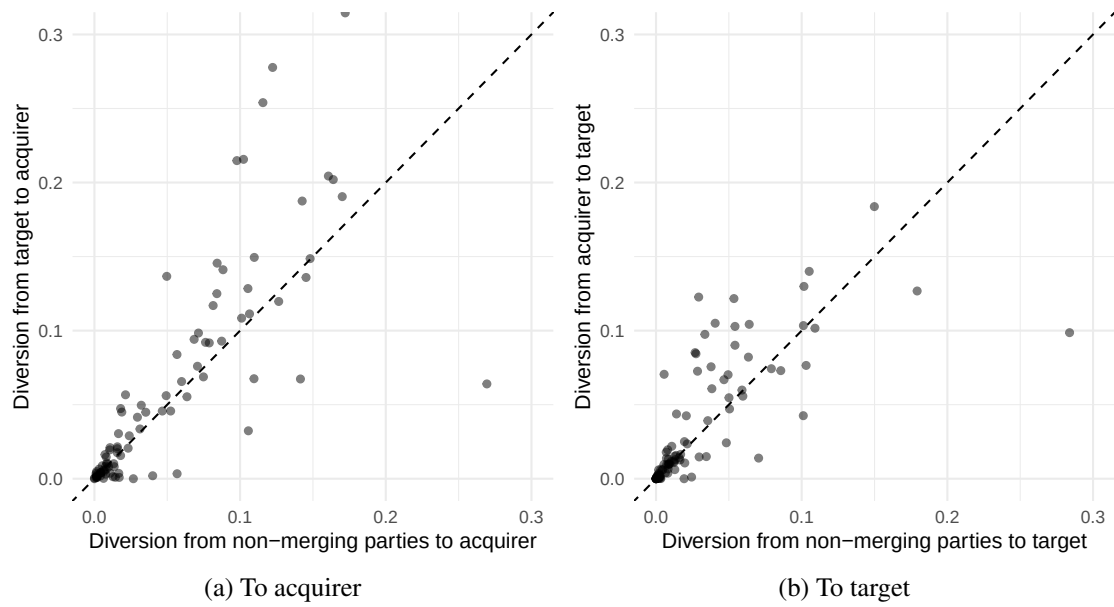


Figure A.2: Comparison of diversion ratios between merging parties to diversion ratios across merging and non-merging parties

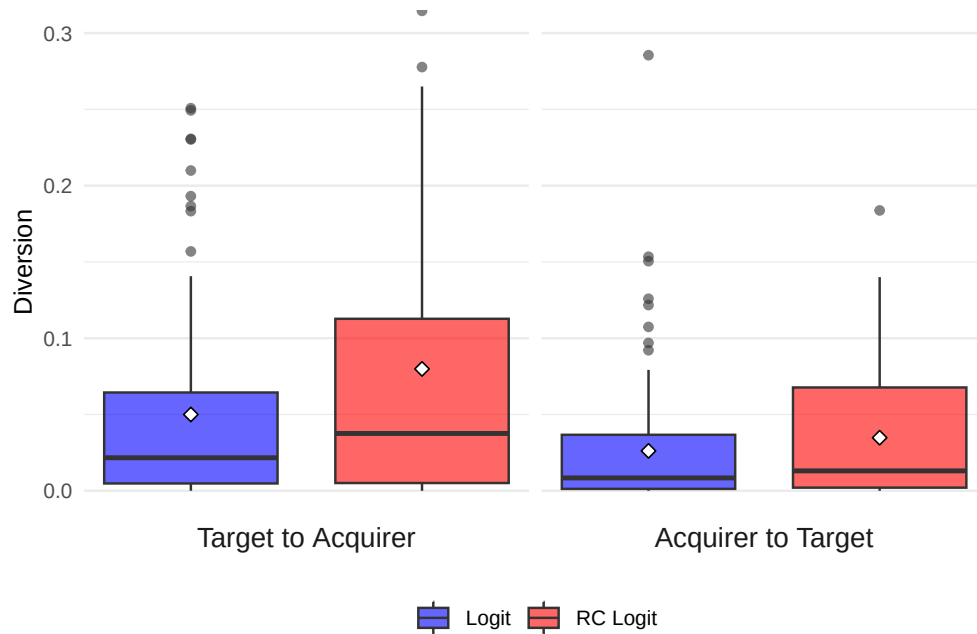


Figure A.3: Comparison of diversion ratios between merging parties under logit and random coefficients logit. The white diamonds are the mean across mergers, while the lower, middle, and upper lines of the boxes are the 25th, 50th, and 75th percentiles.

mean, and upper quartiles of the mixed logit system are all larger than from logit. This suggests that the random coefficients are influencing substitution patterns.

Finally, for additional validation of the demand estimates, Table A.1 compares estimated own-

price elasticities to those from other papers in the literature. When relevant, we include a range of estimates reported in the paper. We compare these elasticities to the closest market in our sample. We should note that the comparison is not perfect, as product definitions, selection of products in the market, market definitions, geographies, and time periods may differ between the papers and our decisions. The difference in time periods is especially relevant given the common finding (Atalay et al., 2024; Döpper et al., 2025) that demand has been getting less elastic over time. As discussed in the text, our estimates overlap with the literature’s in 8 out of 28 markets, are more elastic for 6, and are slightly less elastic on the others. However, the difference conditional on our range being different is typically small. On net, this provides some evidence that our procedure is not returning results entirely out of line with the literature’s.

## A.2. Additional Results Related to Prediction Errors

We next provide additional detail about the prediction errors themselves. Figure A.4 plots boxplots of prediction errors for all prices, merging party prices, and non-merging party prices. The first set of boxplots shows the prediction errors averaged over the entire post-period, and the other groups show them averaged in 6-month periods. We see that median prediction error for merging parties increases in magnitude over time, as do lower quantiles. A similar pattern holds for non-merging parties, although it is not as stark.

Figure A.5 repeats Figure 2 but overlays prediction errors from “placebo” mergers. We implement these mergers by using one year before the actual merger as a date for a “placebo” merger in each of our time periods. We then repeat our exercise and plot the prediction error in the year following this placebo date. While these distributions have some variance, the average placebo prediction error is very close to 0. Since we are using the same sample of markets, this exercise provides evidence that our observation of overprediction is not (on average) due to trends that our procedure does not cover.

Figure A.6 shows the factor by which (a) price coefficients and (b) non-price coefficients have to be multiplied to minimize overall prediction error. Both panels show that this value is often at the boundaries of the range we use. Figure A.7 shows the implications for elasticities. Scaling the price coefficients leads to a sizable left tail of median own price elasticities. Scaling non-price coefficients does not change the own-price elasticities appreciably (although it would change cross-price ones).

Figure A.8 shows the estimated  $\tau^{\text{pre}}$  and  $\tau^{\text{post}}$  that minimized optimal prediction error. We find that these are often at the boundary value of 0 and 1, suggesting that even extreme values do not lead to zero prediction error.

| Product Market                  | Estimates from Literature                                                                                                           | Our Estimates                |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| Beer                            | -3.8 to -6.1 (Miller and Weinberg, 2017, Table IV)<br>-3.4 (Asker, 2016, p. 398)                                                    | -3.9 (including ales)        |
| Bottled water                   | -5.0 to -13.0 (Bonnet and Dubois, 2010, Table 6)<br>-2.3 to -6.8 (Atalay et al., 2025, Table 16)                                    | -3.2                         |
| Breakfast cakes and sweet rolls | -3.2 to -4.0 (Atalay et al., 2025, Table 16)                                                                                        | -1.9                         |
| Breakfast sausages              | -4.6 to -6.5 (Atalay et al., 2025, Table 16)                                                                                        | -7.4                         |
| Coffee                          | -2.3 to -3.5 (Nakamura and Zerom, 2010, Table 5)<br>-4.6 to -5.7 (Atalay et al., 2025, Table 16)                                    | -1.2 (not cup)<br>-5.5 (cup) |
| Cookies                         | -3.8 to -4.8 (Atalay et al., 2025, Table 16)                                                                                        | -3.1                         |
| Doughnuts                       | -3.3 to -4.1 (Atalay et al., 2025, Table 16)                                                                                        | -1.1 to -1.8                 |
| Fresh bread                     | -2.5 to -3.5 (Atalay et al., 2025, Table 16)                                                                                        | -4.4 to -11.2                |
| Fresh buns                      | -4.4 to -6.5 (Atalay et al., 2025, Table 16)                                                                                        | -3.4                         |
| Frozen breakfasts               | -2.6 to -4.2 (Atalay et al., 2025, Table 16)                                                                                        | -4.2                         |
| Frozen pizza                    | -3.5 to -4.3 (Atalay et al., 2025, Table 16)                                                                                        | -3.5                         |
| Gum                             | -2.0 to -4.1 (Atalay et al., 2025, Table 16)                                                                                        | -17.2                        |
| Laundry detergent               | -2.5 to -4.3 (Hendel and Nevo, 2006, Table VII)<br>-5.3 to -10.0 (Atalay et al., 2025, Table 16)                                    | -3.4 (detergent boosters)    |
| Misc. snacks                    | -2.2 to -5.5 (Atalay et al., 2025, Table 16)                                                                                        | -3.0                         |
| Muffins                         | -2.3 to -4.2 (Atalay et al., 2025, Table 16)                                                                                        | -2.0                         |
| Potato chips                    | -1.5 to -5.4 (Dubois et al., 2018, Table D.5)<br>-4.5 to -8.4 (Atalay et al., 2025, Table 16)                                       | -4.0                         |
| Pretzels                        | -3.0 to -5.3 (Atalay et al., 2025, Table 16)                                                                                        | -4.0                         |
| Ready-to-eat cereal             | -2.3 to -4.3 (Nevo, 2001, Table VII)<br>-2.4 to -2.7 (Backus et al., 2021, Table 4)<br>-3.7 to -4.4 (Atalay et al., 2025, Table 16) | -1.9                         |
| Rolls                           | -3.4 to -4.4 (Atalay et al., 2025, Table 16)                                                                                        | -3.1                         |
| Sausages/frankfurters           | -3.2 to -3.6 (Atalay et al., 2025, Table 16)                                                                                        | -2.6                         |
| Sliced lunchmeat                | -2.8 (turkey) (Kreps, 2026, Table 4)<br>-1.5 to -3.9 (Atalay et al., 2025, Table 16)                                                | -4.9                         |
| Wine                            | -3.6 to -5.3 (Atalay et al., 2025, Table 16)                                                                                        | -1.4 (white)<br>-2.3 (red)   |

Table A.1: Comparisons of own-price elasticities from our procedure to those of similar markets in the literature. Ranges for our estimates indicate different mergers (with non-overlapping time periods) and ranges for published papers indicate different specifications, different time periods, or different products (if an average elasticity is not provided).

### A.3. Additional Results Related to Predictive Power

In the results in Table 7, we account for the estimation error in the regression variables when computing standard errors. However, since the variables themselves are estimates, an OLS regression could be biased. We correct for this bias using a split-sample IV procedure. For each merger, we first generate 50 draws of demand parameters from the asymptotic distribution of the BLP estimates. For each draw of the estimates, we redraw DMAs with replacement to re-estimate the cost function.

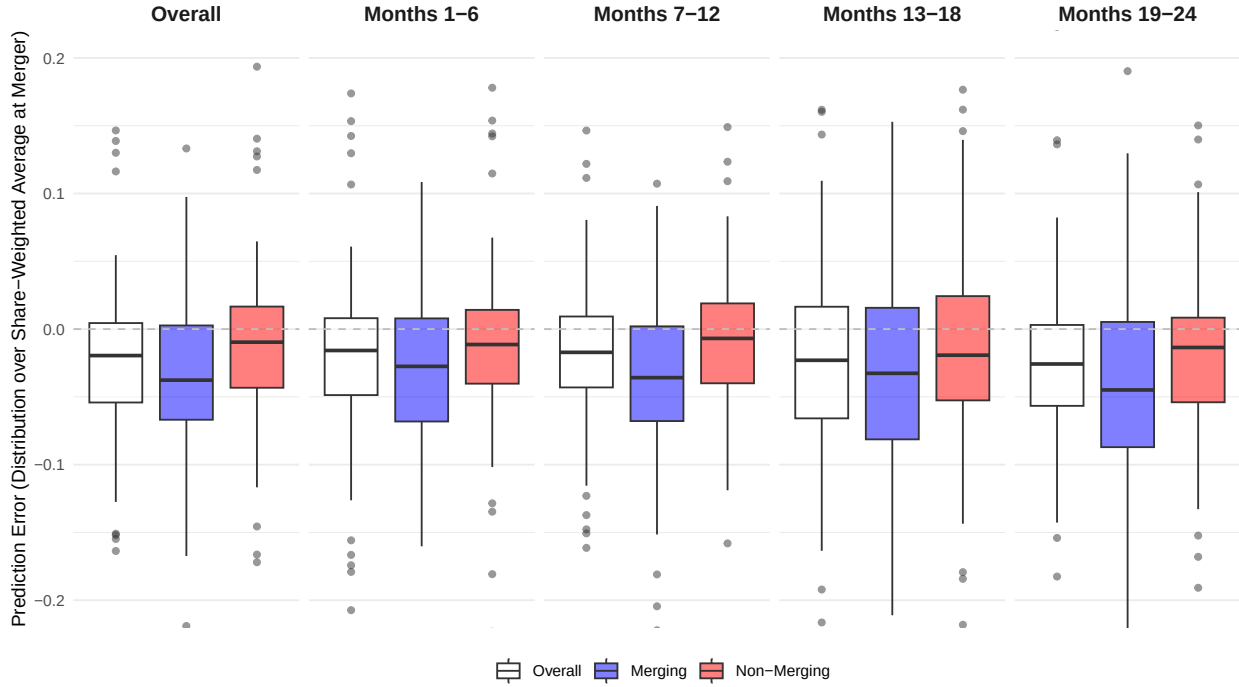


Figure A.4: Distribution of prediction error at the merger level, over time

This gives us bootstrap draws of unilateral and causal effects that reflect the estimation error in demand and the cost function. We then use a split sample IV procedure using the average of 25 bootstrap draws as instruments for the average of the other 25 bootstrap draws. We compute standard errors of this procedure via Rubin (1987). Table A.2 reports results of this procedure. Point estimates are sometimes larger than those of Table 7 (such as for Column (1)) and sometimes consistently smaller (such as for Column (4)), but the takeaways are generally similar.

Table A.3 provides statistical tests related to the discussions in Section V.C comparing the screening ability of merger simulations to other procedures. We compare the area under the PR curves (AUC-PR) for different predictors, with larger area under the curve indicating better discrimination ability. To get a distribution of the AUC-PR that reflects both sampling variation in DMAs and estimation error from demand and costs, we use the following bootstrap procedure. We first sample mergers with replacement.<sup>26</sup> We then pick one of the 50 bootstrap draws mentioned above and compute the relevant AUCs for this draw for all predictors (simulations, nested logit, GUPPI, etc.). We do the second step multiple times for each initial bootstrap draw. We then check whether the AUC for the full merger simulation exceeds that for the other predictors. Table A.3 reports the share of the “super-bootstrap” for which the AUC for simulations is less than that for

<sup>26</sup>We stratify merger sampling within bins of DHHI (0–50, 50–100, and more than 100) to ensure that there is a reasonable distribution of DHHI in our bootstrap samples.

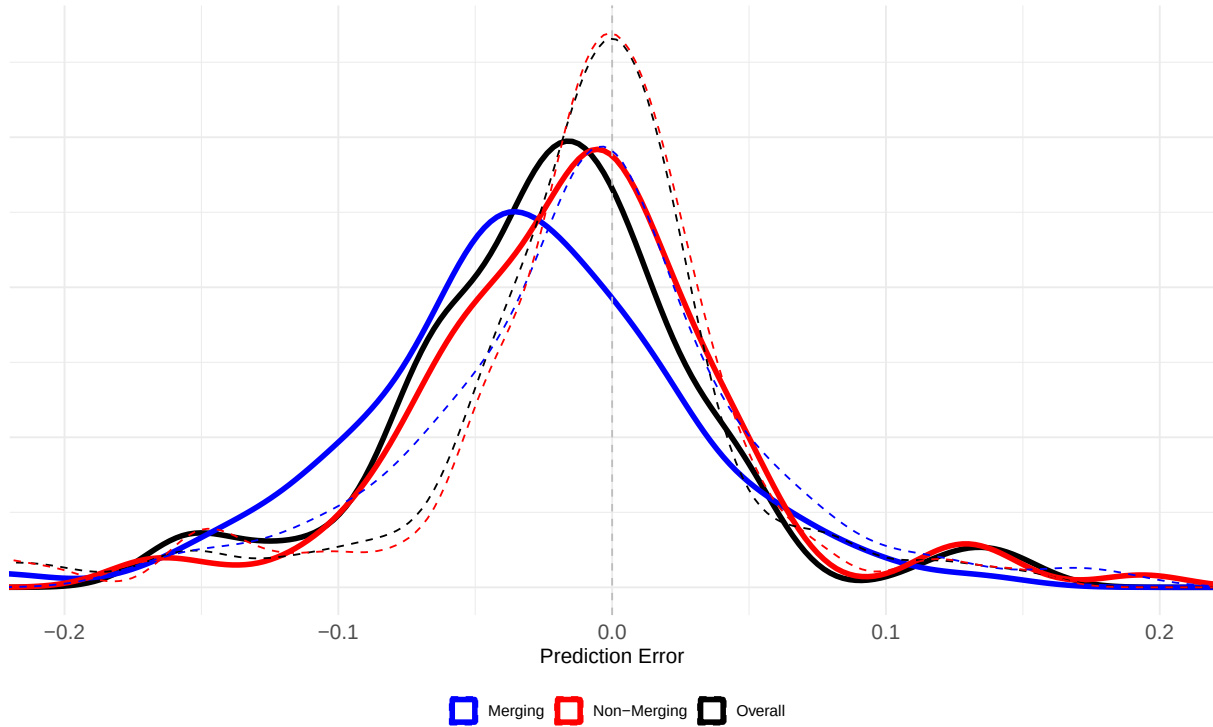


Figure A.5: Distribution of prediction error at the merger level, overlayed with distributions from placebo mergers (dashed)

other predictors. As discussed in the main text, these values are always small when comparing to structural presumptions. Comparisons to GUPPI and nested logit typically yield  $p$ -values less than 0.10, although low cutoffs with nested logit and high cutoffs using GUPPI are sometimes exceptions. Overall, these  $p$ -values give us confidence in the statistical statements made in Section V.C.

Finally, Figure A.9(a)–(c) provide other precision-recall curves. Panel (a) repeats Figure 4(c) but using overall price changes. Panel (b) shows PR curves for different methods for computing GUPPI: this panel shows that GUPPI calibrated to the random-coefficient logit parameters performs better than that with diversion ratios computed using share-based substitution (in green) or the consumer overlap metric from Einav et al. (2026) (in purple). We should note, however, that we use the panelist data to compute this overlap metric; this metric requires multiple purchases per panelist, and this could make it too demanding for the panelist data for smaller markets. Panel (c) shows results from using DHHI and HHI separately. This panel shows that the predictive power of the presumptions seems to come from DHHI itself: HHI (in yellow) has limited predictive power by itself. Moreover, using simulations together with the structural presumptions (green) does not materially add predictive power beyond the simulations themselves.

Panels (d)–(f) show receiver operating characteristic curves—which are another way to judge the predictive power of classifiers. These curves plot the true positive rate on the vertical axis: the



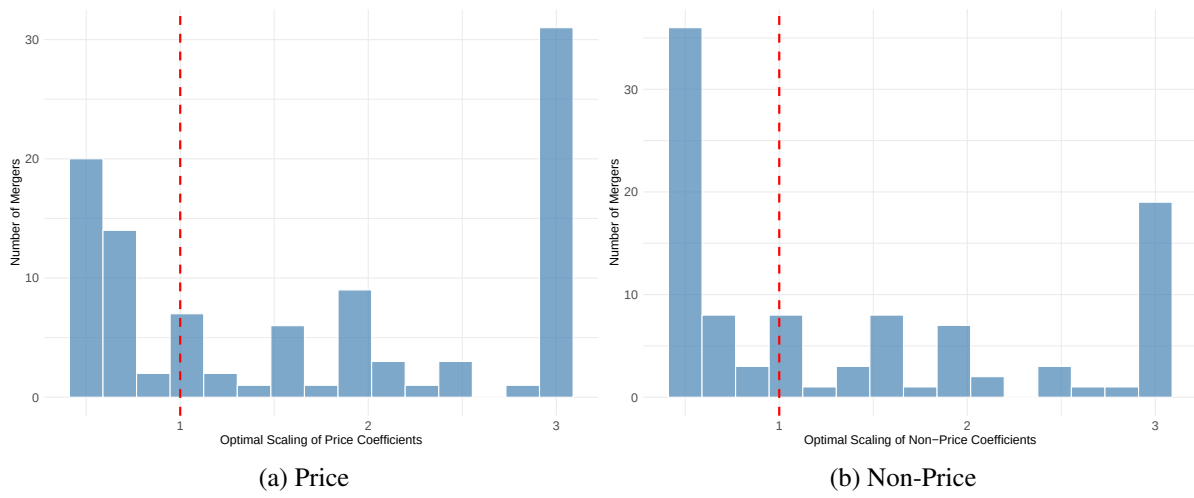


Figure A.6: Histograms of the optimal scaling of demand coefficients to minimize prediction error

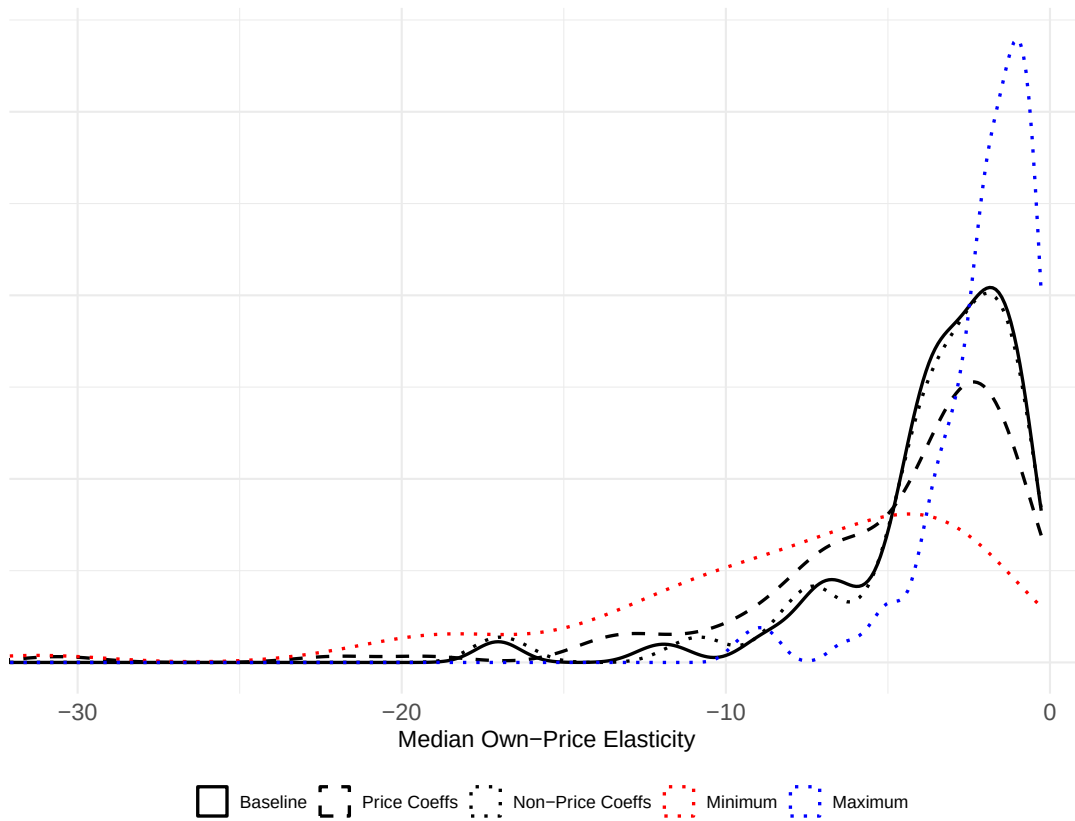


Figure A.7: Elasticities at optimally scaled demand systems

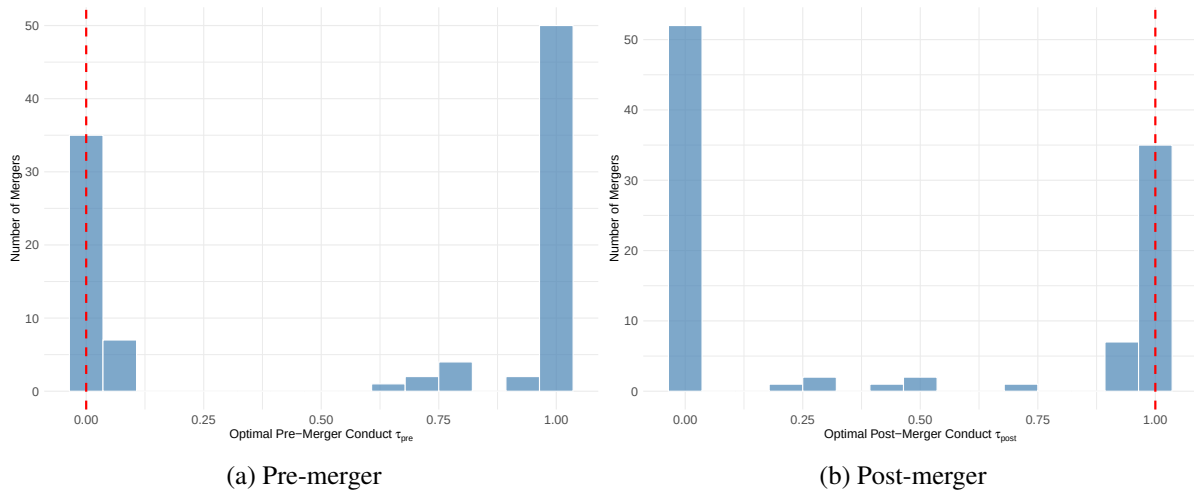


Figure A.8: Histograms of the pre-merger and post-merger profit internalization necessary to minimize overall prediction error

|                   | Cross-Merger       | Within-Merger     |                   |                   |
|-------------------|--------------------|-------------------|-------------------|-------------------|
|                   | (1)                | (2)               | (3)               | (4)               |
| A. Merging        |                    |                   |                   |                   |
| Constant          | -0.07***<br>(0.01) |                   |                   |                   |
| Unilateral Effect | 0.88***<br>(0.26)  | 0.67***<br>(0.06) | 0.36***<br>(0.06) | 0.23***<br>(0.04) |
| B. Overall        |                    |                   |                   |                   |
| Constant          | -0.06***<br>(0.01) |                   |                   |                   |
| Unilateral Effect | 0.91***<br>(0.22)  | 0.77***<br>(0.03) | 0.31***<br>(0.05) | 0.23***<br>(0.04) |
| C. Non-Merging    |                    |                   |                   |                   |
| Constant          | -0.06***<br>(0.01) |                   |                   |                   |
| Unilateral Effect | 3.31***<br>(1.22)  | 1.25***<br>(0.12) | 0.93***<br>(0.18) | 0.23***<br>(0.04) |
| Level             | Merger             | DMA               | Product           | P-DMA             |
| Merger FE         | No                 | Yes               | Yes               | Yes               |

Table A.2: Split-sample IV

proportion of actual positive cases the classifier identifies as positive. The horizontal axis plots the false positive rate: the proportion of actual negatives incorrectly flagged as positives (i.e., the Type I error rate). Curves bowed farther upwards from the 45° line are better predictors. Panels (d) and (e)

|                 | Overall |       |              | Merging |       |              |
|-----------------|---------|-------|--------------|---------|-------|--------------|
|                 | NL      | GUPPI | Presumptions | NL      | GUPPI | Presumptions |
| All Parties     |         |       |              |         |       |              |
| Cutoff = 0%     | 0.15    | 0.02  | 0.00         | 0.13    | 0.04  | 0.00         |
| Cutoff = 2.5%   | 0.03    | 0.03  | 0.00         | 0.06    | 0.02  | 0.00         |
| Cutoff = 5%     | 0.03    | 0.04  | 0.01         | 0.03    | 0.04  | 0.00         |
| DHHI $\geq 100$ |         |       |              |         |       |              |
| Cutoff = 0%     | 0.12    | 0.03  | 0.00         | 0.15    | 0.07  | 0.00         |
| Cutoff = 2.5%   | 0.06    | 0.05  | 0.00         | 0.05    | 0.07  | 0.00         |
| Cutoff = 5%     | 0.07    | 0.14  | 0.00         | 0.03    | 0.12  | 0.00         |

Table A.3:  $p$ -values of the hypothesis test that the AUC for the precision-recall curve for merger simulations is larger than the AUC for the precision-recall curve for other predictors. We report results for different DMAs and different cutoffs.

show that the merger simulations outperform on the ROC curve as well. Panel (f) shows the ROC curve at the merger level: estimates are generally rather noisy here.

## B. Comparing Predicted and Actual Changes

As discussed in Section V.A, an alternative method to evaluate the predictive power of merger simulations is to define the change as relative to the initial price of the product. Here, we define  $p_{jdtm}^{\text{Pred}}$  as in Section IV.B, and we define  $p_{jdm}^0$  as the price of the product in the period before the merger. Define  $\Delta\text{Actual}_{jdm} \equiv \log(p_{jdm}) - \log(p_{jdm}^0)$  and  $\Delta\text{Predicted}_{jdm} \equiv \log(p_{jdm}^{\text{Pred}}) - \log(p_{jdm}^0)$ . We aggregate these quantities across levels of analysis in the same way we aggregate  $\text{Causal}_{jdtm}$ .

Figure B.1 scatters  $\Delta\text{Actual}_m$  against  $\Delta\text{Predicted}_m$  both for merging parties and overall. Both panels show that the predictions are correlated with actual price changes, although the predicted price changes do not change one-for-one with actual changes. Since the predicted price changes account for observable changes in cost shifters, they can be negative. However, this also means that the positive correlation is partly due to shifts in costs rather than the structure of the merger simulation itself, unlike in our baseline procedure, which uses the same costs and demand shocks when computing the two prices that determine the change.

To control for such changes, we run an extended version of the Mincer and Zarnowitz (1969) regression:

$$\Delta\text{Actual}_\times = \alpha + \beta \cdot \Delta\text{Predicted}_\times + \text{Controls} + \epsilon_\times, \quad (7)$$

where the  $\times$  denotes that the level of the regression changes in different specifications. We include the predicted change in marginal cost, as a proportion of the original price, as a control.<sup>27</sup> When

<sup>27</sup>When computing this quantity, we winsorize extreme outliers: those where the marginal cost changes by more than 5

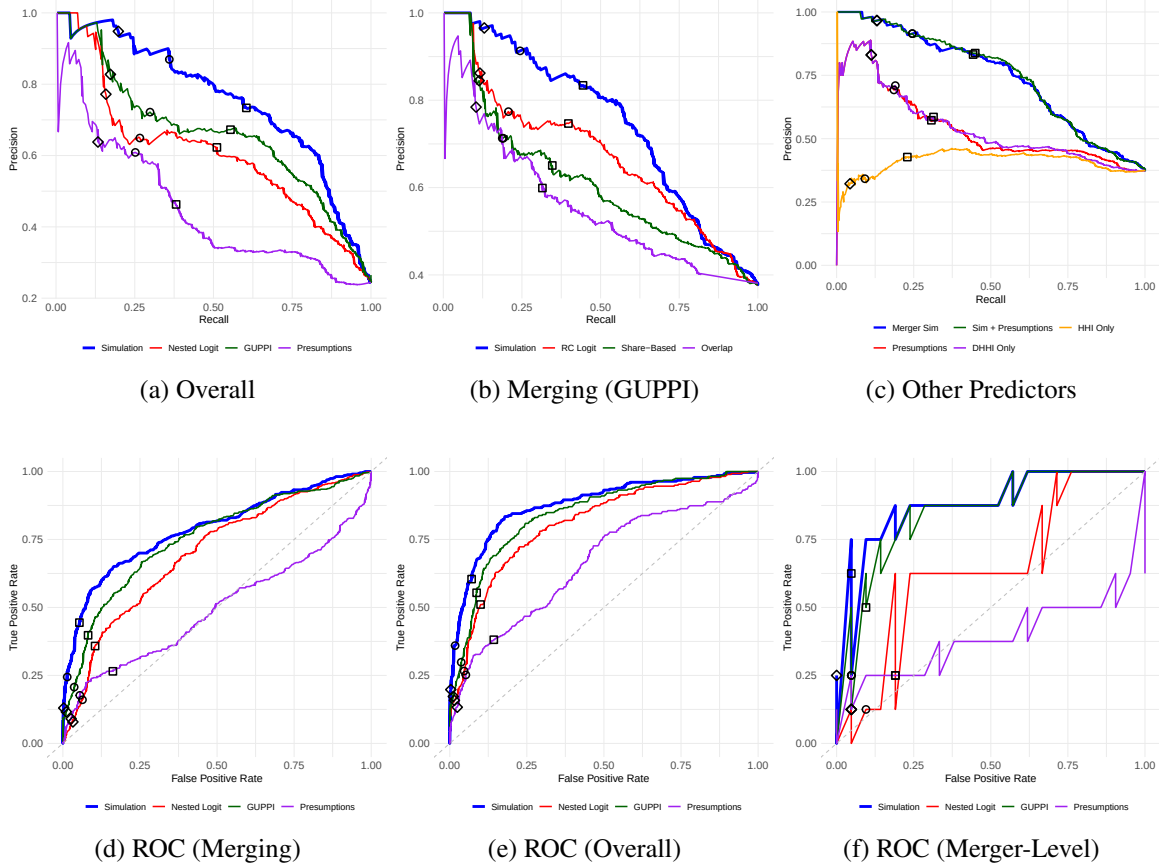


Figure A.9: Alternate illustrations of precision-recall (Panels (a)–(c)) and receiver operating characteristic curves (Panels (d)–(f)). All panels restrict to DMAs with  $\text{DHHI} \geq 100$ . Panel (a) shows overall price changes. Panel (b) shows merging party price changes, with alternative methods of estimating GUPPI. Panel (c) shows predictions using HHI and DHHI separately. Panels (d) and (e) show ROC curves using merging and overall price changes. Panel (f) shows an ROC curve at the merger level, using merging party price changes.

running regressions at the merger or the DMA level, we also control for HHI and DHHI—either averaged across DMAs in the first case, or at the DMA level in the second. Table B.1 reports results of this analysis, with each column corresponding to a different level of analysis and each panel corresponding to a different set of products.

Columns (1) shows that cross-merger, predicted changes in prices are correlated with actual changes. Unlike in our baseline approach, however, the coefficient does not include 1 in the confidence intervals. This result is not due to changes in marginal costs: we control for predicted changes in marginal costs, which are also typically statistically significant. We also control for market structure, suggesting that the prediction has some relation with the actual changes even controlling for (simple functions of) the market structure. Columns (2)–(4) add merger fixed effects,

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times the initial price.

|                                  | Cross-Merger      | Within-Merger      |                   |                   |
|----------------------------------|-------------------|--------------------|-------------------|-------------------|
|                                  | (1)               | (2)                | (3)               | (4)               |
| A. Merging                       |                   |                    |                   |                   |
| Constant                         | -0.01<br>(0.01)   |                    |                   |                   |
| Predicted $\Delta$ Price         | 0.27<br>(0.17)    | 0.36***<br>(0.06)  | 0.29<br>(0.23)    | 0.04***<br>(0.00) |
| Predicted $\Delta$ Marginal Cost | 0.23<br>(0.18)    | 0.34***<br>(0.09)  | 0.37<br>(0.45)    | 0.65***<br>(0.01) |
| HHI                              | -0.02<br>(0.04)   | -0.07***<br>(0.02) |                   |                   |
| DHHI                             | 0.10<br>(0.32)    | -0.26***<br>(0.09) |                   |                   |
| B. Overall                       |                   |                    |                   |                   |
| Constant                         | 0.01<br>(0.01)    |                    |                   |                   |
| Predicted $\Delta$ Price         | 0.27***<br>(0.07) | 0.27***<br>(0.04)  | 0.24***<br>(0.05) | 0.40***<br>(0.03) |
| Predicted $\Delta$ Marginal Cost | 0.02<br>(0.07)    | 0.10***<br>(0.03)  | 0.30***<br>(0.08) | 0.17***<br>(0.04) |
| HHI                              | -0.04<br>(0.03)   | 0.01<br>(0.01)     |                   |                   |
| DHHI                             | 0.08<br>(0.12)    | -0.09<br>(0.06)    |                   |                   |
| C. Non-Merging                   |                   |                    |                   |                   |
| Constant                         | 0.01<br>(0.01)    |                    |                   |                   |
| Predicted $\Delta$ Price         | 0.21***<br>(0.07) | 0.32***<br>(0.07)  | 0.24***<br>(0.07) | 0.21***<br>(0.00) |
| Predicted $\Delta$ Marginal Cost | 0.03<br>(0.07)    | 0.03<br>(0.03)     | 0.30***<br>(0.09) | 0.30***<br>(0.00) |
| HHI                              | -0.04*<br>(0.02)  | 0.02<br>(0.01)     |                   |                   |
| DHHI                             | 0.22<br>(0.18)    | -0.16***<br>(0.05) |                   |                   |
| Level                            | Merger            | DMA                | Product           | P-DMA             |
| Merger FE                        | No                | Yes                | Yes               | Yes               |

Table B.1: Regressions of actual change in (log) price on predicted change in (log) price, at various levels. Each column runs the regression at a different level of observation (with “P-DMA” standing for product-DMA). Panels A and C focus on merging and non-merging price changes only, respectively. Panel B reports results for all price changes. Measures of market structure are not defined at the product level and are thus omitted from Columns (3), (4), (6), and (7). Columns (2)–(4) drop changes above 0.25 log points. Standard errors are computed using Rubin’s rules to account for first-stage estimation error.

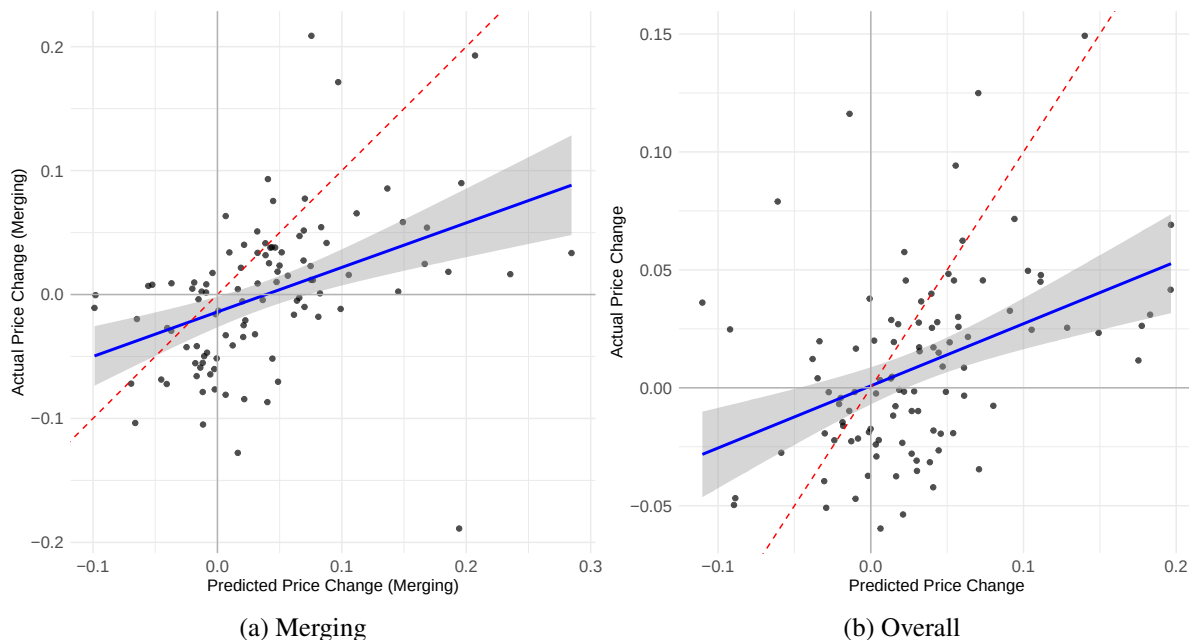


Figure B.1: Scatter of actual versus predicted changes in prices, at the merger level, for (a) merging parties, and (b) all parties. The dotted red line indicates the 45° line, and the blue line is the line of best fit.

and the takeaways are similar.<sup>28</sup>

Table B.2 lists partial  $R^2$  values for regressions of  $\Delta\text{Actual}_{jdm}$  on  $\Delta\text{Predicted}_{jdm}$  at various levels of analysis, controlling for costs and market structure. The first row in each panel shows the unconditional  $R^2$ , while the second and third rows show the partial  $R^2$  of predictions from the simulation above costs (second row) and costs and market structure (third row). Simulations explain approximately 17% of cross-merger variation in merging party price changes, although much of that explanatory power comes from the simulation incorporating observable changes to marginal costs. Simulations also have predictive power across geographies within-merger, although less so across products. For comparison, Table B.3 shows analogous quantities for the predictive power of HHI, DHHI, and their interaction. We find that simulations have substantially larger marginal predictive power than does market structure.

Figure B.2 shows ROC curves for actual price changes, both overall and for merging parties, using predicted effects. We restrict to high DHHI DMAs, but the takeaways we discuss in this paragraph are the same for all DMAs. We see that the ROC curves are far from the 45° line, suggesting that the prediction has strong predictive power. We also overlay two other ROC curves: one from a prediction using changes in marginal cost, HHI, DHHI, and an interaction; and one from just using marginal cost. The comparisons are somewhat more nuanced than those in the PR and

<sup>28</sup>The coefficient on DHHI has an unexpected sign in Column (2), although this could be explained by the fact that it is measuring the residual correlation in actual price changes conditional on the merger simulation itself.

|                                 | Cross-Merger | Within-Merger |         |       |
|---------------------------------|--------------|---------------|---------|-------|
|                                 | (1)          | (2)           | (3)     | (4)   |
| A. Merging                      |              |               |         |       |
| Individually                    | 0.173        | 0.155         | 0.064   | 0.083 |
| Over Costs                      | 0.086        | 0.041         | 0.013   | 0.012 |
| Over Costs and Market Structure | 0.063        | 0.046         | —       | —     |
| B. Overall                      |              |               |         |       |
| Individually                    | 0.163        | 0.077         | 0.038   | 0.054 |
| Over Costs                      | 0.112        | 0.052         | 0.007   | 0.019 |
| Over Costs and Market Structure | 0.111        | 0.053         | —       | —     |
| C. Non-Merging                  |              |               |         |       |
| Individually                    | 0.101        | 0.039         | 0.036   | 0.052 |
| Over Costs                      | 0.046        | 0.033         | 0.006   | 0.019 |
| Over Costs and Market Structure | 0.062        | 0.035         | —       | —     |
| Level                           | Merger       | DMA           | Product | P-DMA |
| Merger FE                       | No           | Yes           | Yes     | Yes   |

Table B.2: Partial  $R^2$  of predicted change in prices over different characteristics of the merger. Columns (2)–(4) report within- $R^2$  values and drop changes above 0.25 log points.

|                            | Merging |       |       | Overall |       |       | Non-Merging |       |       |
|----------------------------|---------|-------|-------|---------|-------|-------|-------------|-------|-------|
|                            | (1)     | (2)   | (3)   | (4)     | (5)   | (6)   | (7)         | (8)   | (9)   |
| Individually               | 0.022   | 0.006 | 0.005 | 0.018   | 0.007 | 0.003 | 0.012       | 0.002 | 0.007 |
| Over Costs                 | 0.037   | 0.007 | 0.005 | 0.031   | 0.008 | 0.002 | 0.019       | 0.001 | 0.007 |
| Over Costs and Predictions | 0.012   | 0.002 | 0.011 | 0.027   | 0.003 | 0.003 | 0.036       | 0.002 | 0.009 |
| Level                      | Merger  | DMA   | DMA   | Merger  | DMA   | DMA   | Merger      | DMA   | DMA   |
| Merger FE                  | No      | No    | Yes   | No      | No    | Yes   | No          | No    | Yes   |

Table B.3: Partial  $R^2$  of market structure in prices over different characteristics of the merger

ROC curves using causal and unilateral effects. First, we see that the marginal cost prediction by itself has lower AUC than the merger simulations—but it is not dominated by the merger simulations. In fact, for especially low challenge rates (or high cutoffs), the marginal cost prediction by itself dominates. Second, we see that marginal cost with the structural presumptions is comparable to the predictions themselves: the AUC is typically lower, but this dominates the predictions in certain regions. (Of course, one should remember that this model is estimated exactly on the sample on which it is being used to predict, which artificially inflates the ROC curve; however, estimating on a holdout sample does not change the results appreciably.) On net, our takeaway from this exercise is predictions do have significantly predictive power for high price changes, but the marginal contribution of the simulations when analyzing predictions is limited—unless the

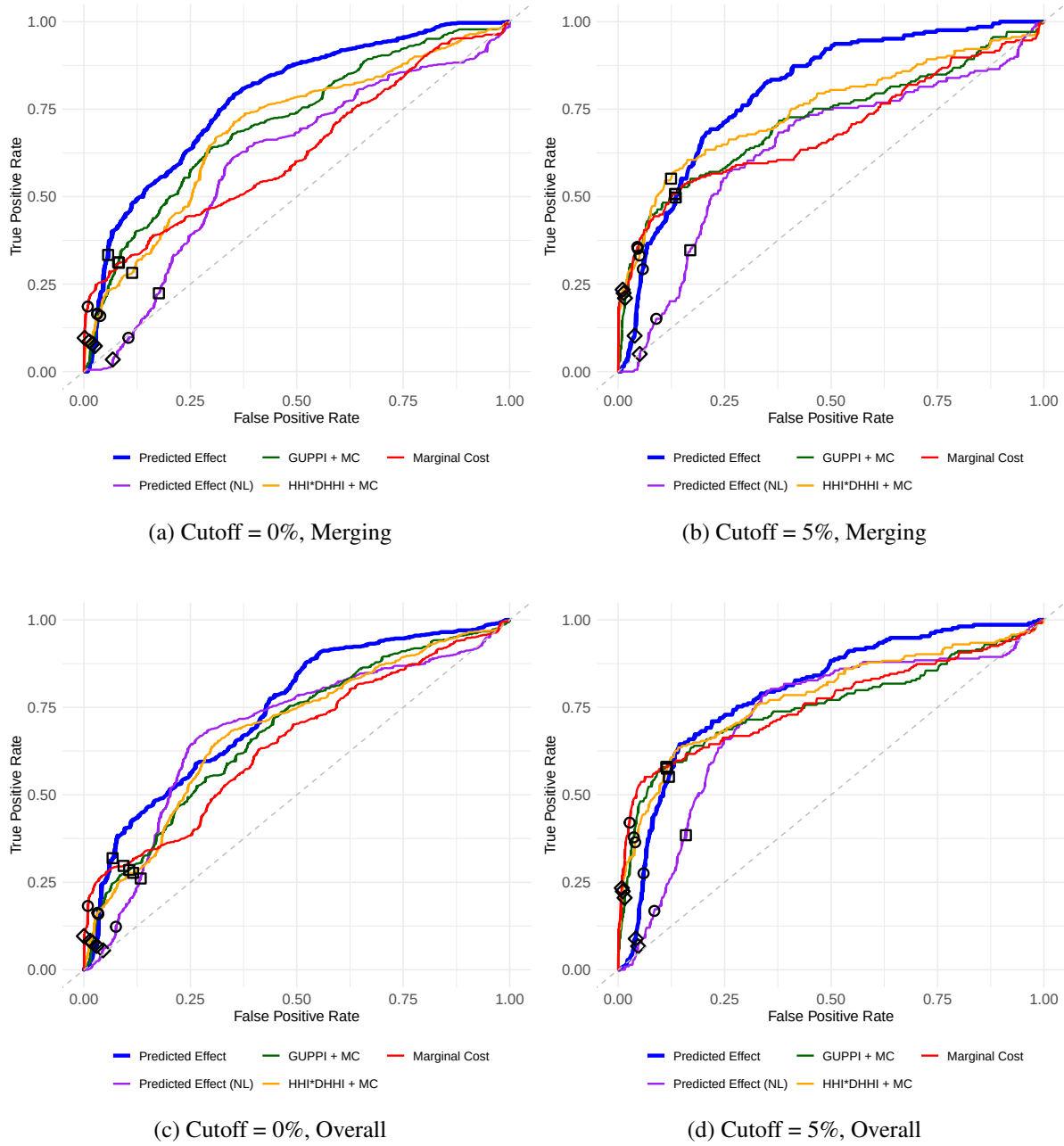


Figure B.2: ROC curves for actual price changes, using predicted price changes, marginal costs with market structure, and marginal cost by themselves as predictors. We restrict to DMAs with  $\text{DHHI} \geq 100$ .

agencies are challenging a large set of mergers. These results provide another justification for the unilateral/causal analysis done in the main text.



### C. List of All Mergers

Table C.1 summarizes all of the mergers included in our sample. For each merger, we list the NielsenIQ department and product group of the relevant product market, and the NielsenIQ product modules that comprise the relevant product market, along with criteria used for dividing the modules into separate markets (if any). We then define all characteristics that we extract from the scanner data, the share of variance of those characteristics (along with a private-label indicator and the package size, which are included in all mergers) that are explained by the first two principal components, the set of candidate cost shifter instruments, and whether the merger is included in our final estimation sample. For mergers that are included, the table also lists the set of instruments (out of Gandhi-Houde, cost shifters, and Hausman) and specification of demographic interactions (out of Price, 1+Price, and 1+PC+Price) used in the demand estimates included in our sample.

The “cost shifters” column can be read as follows. Each instrument is separated by a semicolon. Generally, these share a name with the FRED series from which they are taken. Sometimes, the instruments are multiplied by a boolean expression of a characteristic, in which case the instrument is interacted with an indicator for whether the boolean is true. If this column is empty then there are no cost shifter instruments defined for this merger and we rely on the other two types of instruments for estimation (differentiation IVs and Hausman instruments).

| Merger | NielsenIQ department | NielsenIQ product group | Product market definition                                                                            | Characteristics                                 | % variance explained by two PCs | Cost shifters                                                                                                                                                                                                                                                                                                                                                                                  | In estimation sample | Instruments used | RC on constant | Demographic interactions |
|--------|----------------------|-------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|------------------|----------------|--------------------------|
| 1      | Alcoholic beverages  | Beer                    | Ale; Beer; Light Beer (Low Calorie/Alcohol); Stout And Porter                                        | Container; ice; imported; module                | 41%                             | Aluminum x container == "can"; barley; barley x module != "light"; dollar strength x imported == 1; glass x container == "bottle"; wheat                                                                                                                                                                                                                                                       | Yes                  | GH               | Yes            | 1+PC+Price               |
| 2      | Alcoholic beverages  | Beer                    | Ale; Beer; Light Beer (Low Calorie/Alcohol); Stout And Porter                                        | Container; ice; imported; module                | 50%                             | Aluminum x container == "can"; barley; barley x module != "light"; dollar strength x imported == 1; glass x container == "bottle"; wheat                                                                                                                                                                                                                                                       | Yes                  | GH+CS            | Yes            | 1+PC+Price               |
| 3      | Alcoholic beverages  | Liquor                  | Alcoholic Cocktails                                                                                  | Flavor; liquor; party pack; quality             | 28%                             | Barley x liquor in ["whiskey", "gin"]; corn x liquor == "whiskey"; dollar strength; grapes x liquor == "other"; russet potatoes x liquor == "vodka"; sugar x liquor in ["tequila", "rum"]; wheat x liquor in ["gin", "vodka"]; wine grapes x liquor == "other"                                                                                                                                 | Yes                  | GH               | Yes            | 1+PC+Price               |
| 4      | Alcoholic beverages  | Liquor                  | Bourbon-Blended; Bourbon-Straight/Bonded; Canadian Whiskey; Irish Whiskey; Remaining Whiskey; Scotch | Imported; liquor type; proof; quality           | 49%                             | Barley; corn; dollar strength; dollar strength x imported == 1                                                                                                                                                                                                                                                                                                                                 | Yes                  | GH+CS            | Yes            | 1+PC+Price               |
| 5      | Alcoholic beverages  | Liquor                  | Bourbon-Blended; Bourbon-Straight/Bonded; Canadian Whiskey; Irish Whiskey; Remaining Whiskey; Scotch | Imported; liquor type; proof; quality           | 49%                             | Barley; corn; dollar strength; dollar strength x imported == 1                                                                                                                                                                                                                                                                                                                                 | Yes                  | GH               | Yes            | 1+PC+Price               |
| 6      | Alcoholic beverages  | Liquor                  | Cordials & Proprietary Liqueurs                                                                      | Alcohol content; flavor; imported; liquor       | 28%                             | Barley x liquor == "neutral spirit"; corn x liquor == "whiskey or rum"; dollar strength; dollar strength x imported == 1; grapes x liquor == "brandy or cognac"; oranges x liquor == "triple sec"; russet potatoes x liquor == "neutral spirit"; sugar x liquor in ["whiskey or rum", "triple sec"]; wheat x liquor in ["neutral spirit", "other"]; wine grapes x liquor == "brandy or cognac" | Yes                  | GH               | Yes            | 1+PC+Price               |
| 7      | Alcoholic beverages  | Liquor                  | Cordials & Proprietary Liqueurs                                                                      | Alcohol content; flavor; imported; liquor       | 28%                             | Barley x liquor == "neutral spirit"; corn x liquor == "whiskey or rum"; dollar strength; dollar strength x imported == 1; grapes x liquor == "brandy or cognac"; oranges x liquor == "triple sec"; russet potatoes x liquor == "neutral spirit"; sugar x liquor in ["whiskey or rum", "triple sec"]; wheat x liquor in ["neutral spirit", "other"]; wine grapes x liquor == "brandy or cognac" | Yes                  | GH               | Yes            | 1+PC+Price               |
| 8      | Alcoholic beverages  | Liquor                  | Cordials & Proprietary Liqueurs                                                                      | Alcohol content; flavor; imported; liquor       | 27%                             | Barley x liquor == "neutral spirit"; corn x liquor == "whiskey or rum"; dollar strength x imported == 1; grapes x liquor == "brandy or cognac"; oranges x liquor == "triple sec"; russet potatoes x liquor == "neutral spirit"; sugar x liquor in ["whiskey or rum", "triple sec"]; wheat x liquor in ["neutral spirit", "other"]; wine grapes x liquor == "brandy or cognac"                  | Yes                  | GH               | Yes            | 1+PC+Price               |
| 9      | Alcoholic beverages  | Liquor                  | Gin                                                                                                  | Flavored; gin type; imported; quality           | 46%                             | Barley; dollar strength; dollar strength x imported == 1; wheat                                                                                                                                                                                                                                                                                                                                | Yes                  | GH               | Yes            | 1+PC+Price               |
| 10     | Alcoholic beverages  | Liquor                  | Rum                                                                                                  | Flavored; imported; proof; quality; type of rum | 42%                             | Dollar strength; dollar strength x imported == 1; sugar                                                                                                                                                                                                                                                                                                                                        | Yes                  | GH               | Yes            | 1+PC+Price               |
| 11     | Alcoholic beverages  | Liquor                  | Tequila                                                                                              | Flavored; proof; quality; type of tequila       | 39%                             | Dollar strength; sugar                                                                                                                                                                                                                                                                                                                                                                         | Yes                  | GH+CS            | Yes            | 1+PC+Price               |
| 12     | Alcoholic beverages  | Liquor                  | Vodka                                                                                                | Flavored; imported; proof; quality              | 53%                             | Dollar strength; dollar strength x imported == 1; russet potatoes; wheat                                                                                                                                                                                                                                                                                                                       | Yes                  | GH               | Yes            | 1+PC+Price               |

|    |                     |                       |                                                                      |                                                                                                                                                 |     |                                                                                                                                                                                                                |     |       |     |            |
|----|---------------------|-----------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------|-----|------------|
| 13 | Alcoholic beverages | Liquor                | Vodka                                                                | Flavored; imported; proof; quality                                                                                                              | 53% | Dollar strength; dollar strength x imported == 1; russet potatoes; wheat                                                                                                                                       | Yes | GH+CS | Yes | 1+PC+Price |
| 14 | Alcoholic beverages | Wine                  | Wine-Domestic Dry Table; Wine-Imported Dry Table; is red = 0         | Grape type; imported; is red; quality                                                                                                           | 37% | Dollar strength x imported == 1; wine grapes                                                                                                                                                                   | Yes | GH+H  | Yes | 1+PC+Price |
| 15 | Alcoholic beverages | Wine                  | Wine-Domestic Dry Table; Wine-Imported Dry Table; is red = 1         | Grape type; imported; is red; quality                                                                                                           | 32% | Dollar strength x imported == 1; wine grapes                                                                                                                                                                   | Yes | GH+H  | Yes | 1+PC+Price |
| 16 | Dry grocery         | Baby Food             | Baby Milk And Milk Flavoring; infant and toddler = 0                 | Form; infant and toddler; organic                                                                                                               | 47% | Corn sweeteners; starch vegetable fats oils; vitamin nutrient hematonic human                                                                                                                                  | Yes | GH    | No  | Price      |
| 17 | Dry grocery         | Baby Food             | Baby Milk And Milk Flavoring; infant and toddler = 1                 | Form; infant and toddler; organic                                                                                                               | 77% | Corn sweeteners; starch vegetable fats oils; vitamin nutrient hematonic human                                                                                                                                  | Yes | GH+H  | Yes | 1+Price    |
| 18 | Dry grocery         | Bread And Baked Goods | Bakery - Bread - Fresh                                               | Bread type; bread usage; is herb garlic; is potato; is sourdough; is sweet; is wheat; low cholesterol; low fat; regular health                  | 32% | Russet potatoes x is potato == 1; wheat                                                                                                                                                                        | Yes | GH    | No  | Price      |
| 19 | Dry grocery         | Bread And Baked Goods | Bakery - Bread - Fresh                                               | Bread type; bread usage; is herb garlic; is potato; is sourdough; is sweet; is wheat; low cholesterol; low fat; regular health                  | 33% | Russet potatoes x is potato == 1; wheat                                                                                                                                                                        | Yes | GH    | Yes | 1+PC+Price |
| 20 | Dry grocery         | Bread And Baked Goods | Bakery-Bagels-Fresh                                                  | Bagel size; is low fat; primary flavor; quality tier; secondary flavor                                                                          | 8%  | Wheat; wheat flour                                                                                                                                                                                             | Yes | GH    | Yes | 1+PC+Price |
| 21 | Dry grocery         | Bread And Baked Goods | Bakery-Bagels-Fresh                                                  | Bagel size; has no salt; is low fat; is sliced; is wheat; primary flavor; quality tier; secondary flavor                                        | 9%  | Sugar x primary flavor == "sweet"; wheat; wheat flour                                                                                                                                                          | Yes | GH    | Yes | Price      |
| 22 | Dry grocery         | Bread And Baked Goods | Bakery-Breakfast Cakes/Sweet Rolls-Fresh                             | Has nuts; is filled; primary flavor; primary type; quality tier; secondary flavor; secondary type                                               | 6%  | Cocoa beans x primary flavor == "chocolate"; corn sweeteners; sugar; wheat; wheat flour                                                                                                                        | Yes | GH    | Yes | 1+Price    |
| 23 | Dry grocery         | Bread And Baked Goods | Bakery-Buns-Fresh                                                    | Ingredient; is low cholesterol; is low fat; is potato; is specialty; is wheat; primary flavor; primary type; secondary flavor; secondary type   | 10% | Russet potatoes x is potato == 1; wheat                                                                                                                                                                        | Yes | GH+CS | Yes | 1+PC+Price |
| 24 | Dry grocery         | Bread And Baked Goods | Bakery-Buns-Fresh                                                    | Ingredient; is low cholesterol; is low fat; is potato; is specialty; is wheat; primary flavor; primary type; secondary flavor; secondary type   | 9%  | Russet potatoes x is potato == 1; wheat                                                                                                                                                                        | Yes | GH+H  | Yes | 1+PC+Price |
| 25 | Dry grocery         | Bread And Baked Goods | Bakery-Doughnuts-Fresh                                               | Donut style; has chocolate; is frosted; is glazed; primary flavor; secondary flavor                                                             | 11% | Cocoa beans x has chocolate == 1; corn sweeteners; sugar; vegetable oil; wheat                                                                                                                                 | Yes | GH    | Yes | 1+PC+Price |
| 26 | Dry grocery         | Bread And Baked Goods | Bakery-Doughnuts-Fresh                                               | Donut style; has chocolate; is frosted; is glazed; primary flavor; secondary flavor                                                             | 10% | Cocoa beans x has chocolate == 1; corn sweeteners; sugar; vegetable oil; wheat                                                                                                                                 | Yes | GH    | Yes | 1+PC+Price |
| 27 | Dry grocery         | Bread And Baked Goods | Bakery-Muffins-Fresh                                                 | Is english muffin; is mini; is wheat; low cholesterol; low fat; primary flavor; quality tier; secondary flavor                                  | 8%  | Cocoa beans x primary flavor == "chocolate dessert"; other grains x primary flavor == "breakfast grains"; sugar x is english muffin == 0; wheat; wheat flour                                                   | Yes | GH+H  | Yes | 1+PC+Price |
| 28 | Dry grocery         | Bread And Baked Goods | Bakery-Pies-Fresh                                                    | Is tiny; primary flavor; secondary flavor                                                                                                       | 21% |                                                                                                                                                                                                                | Yes | GH    | Yes | 1+Price    |
| 29 | Dry grocery         | Bread And Baked Goods | Bakery-Rolls-Fresh                                                   | Bread region; is low cholesterol; is low fat; is potato; is sourdough; is wheat; primary flavor; primary type; secondary flavor; secondary type | 3%  | Russet potatoes x is potato == 1; wheat                                                                                                                                                                        | Yes | GH+H  | Yes | 1+PC+Price |
| 30 | Dry grocery         | Bread And Baked Goods | Bakery-Rolls-Fresh                                                   | Bread region; is low cholesterol; is low fat; is potato; is sourdough; is wheat; primary flavor; primary type; secondary flavor; secondary type | 4%  | Russet potatoes x is potato == 1; wheat                                                                                                                                                                        | Yes | GH+H  | Yes | 1+PC+Price |
| 31 | Dry grocery         | Candy                 | Candy-Chocolate; Candy-Chocolate-Miniatures; Candy-Chocolate-Special | Chocolate shape; chocolate type; flavor; high quality; holiday; module info                                                                     | 21% | Almonds x flavor == "almond"; cocoa beans; cocoa beans x chocolate type != "white"; corn sweeteners x flavor == "caramel"; peanuts x flavor == "peanut butter"; sugar; sugar x chocolate type != "bittersweet" | No  |       |     |            |

|    |             |                                 |                                                                                         |                                                                                                |     |                                                                                                                                                                                                                                                       |     |       |     |            |  |
|----|-------------|---------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------|-----|------------|--|
| 32 | Dry grocery | Candy                           | Candy-Hard Rolled; Candy-Lollipops; Candy-Non-Chocolate; Candy-Non-Chocolate-Miniatures | Candy type                                                                                     | 52% | Corn sweeteners; sugar                                                                                                                                                                                                                                | No  |       |     |            |  |
| 33 | Dry grocery | Candy                           | Candy-Hard Rolled; Candy-Lollipops; Candy-Non-Chocolate; Candy-Non-Chocolate-Miniatures | Candy type; flavor; holiday                                                                    | 28% | Corn sweeteners; sugar                                                                                                                                                                                                                                | No  |       |     |            |  |
| 34 | Dry grocery | Cereal                          | Cereal - Granola & Natural Types                                                        | Has chocolate; high fiber; high protein; low sugar; nut type; primary flavor; secondary flavor | 11% | Almonds x nut type == "almond"; other grains; peanuts x nut type == "peanut"; pecans x nut type == "pecan"; sugar x low sugar == 0                                                                                                                    | Yes | GH    | Yes | 1+PC+Price |  |
| 35 | Dry grocery | Cereal                          | Cereal - Ready To Eat                                                                   | Additions; audience; high fiber; ingredient; is sweet; low cholesterol; low sugar              | 38% | Corn; corn sweeteners; corn sweeteners x is sweet; corn x ingredient == "corn"; oats; oats x ingredient == "oats"; rough rice; rough rice x ingredient == "rice"; sugar; sugar x is sweet; wheat; wheat x ingredient in ["wheat", "multigrain/other"] | Yes | GH+H  | Yes | 1+PC+Price |  |
| 36 | Dry grocery | Coffee                          | Coffee - Soluble; Coffee - Soluble Flavored                                             | Decaf; drying method; flavor; is flavored; milk beverage; roast                                | 37% | Cocoa beans x flavor == "chocolate"; coffee beans; dairy x milk beverage == 1                                                                                                                                                                         | Yes | GH+H  | Yes | 1+PC+Price |  |
| 37 | Dry grocery | Coffee                          | Ground And Whole Bean Coffee; cup = 0                                                   | Cup; decaf; flavored                                                                           | 59% | Coffee beans                                                                                                                                                                                                                                          | Yes | GH+H  | Yes | Price      |  |
| 38 | Dry grocery | Coffee                          | Ground And Whole Bean Coffee; cup = 1                                                   | Cup; decaf; flavored                                                                           | 58% | Coffee beans                                                                                                                                                                                                                                          | Yes | GH+H  | Yes | Price      |  |
| 39 | Dry grocery | Condiments, Gravies, And Sauces | Fish & Seafood & Cocktail Sauce                                                         | Sauce type                                                                                     | 46% | Mayonnaise and dressing x sauce type == "tartar"; pickles and horseradish x sauce type == "tartar"; shrimp; tomatos x sauce type == "cocktail"; unprocessed finfish                                                                                   | Yes | GH+CS | Yes | 1+PC+Price |  |
| 40 | Dry grocery | Condiments, Gravies, And Sauces | Meat Sauce; Worcestershire Sauce                                                        | Flavor; sauce type                                                                             | 41% | Salt pepper spices; tomatos x sauce type == "steak"; vinegar                                                                                                                                                                                          | No  |       |     |            |  |
| 41 | Dry grocery | Condiments, Gravies, And Sauces | Sauce & Seasoning Mix-Remaining                                                         | Cuisine; flavor; preparation                                                                   | 31% | Dry onions; salt pepper spices; spices; tomatos; vinegar                                                                                                                                                                                              | Yes | GH    | Yes | 1+PC+Price |  |
| 42 | Dry grocery | Condiments, Gravies, And Sauces | Sauce & Seasoning Mix-Remaining                                                         | Bbq; beef; burger; curry; pork; sausage; turkey; wing                                          | 32% | Dry onions; salt pepper spices; spices; tomatos; vinegar                                                                                                                                                                                              | Yes | GH    | No  | Price      |  |
| 43 | Dry grocery | Condiments, Gravies, And Sauces | Sauce & Seasoning Mix-Remaining Mexican; Sauce Mix - Taco                               | Hot; other flavor; type of sauce                                                               | 28% | Dry onions; salt pepper spices; spices; tomatos x type of sauce == "salsa"; vinegar                                                                                                                                                                   | Yes | GH    | Yes | 1+Price    |  |
| 44 | Dry grocery | Condiments, Gravies, And Sauces | Sauce Mix - Spaghetti                                                                   | Mushroom; pesto or alfredo                                                                     | 61% | Salt pepper spices; spices; tomatos                                                                                                                                                                                                                   | Yes | GH    | Yes | 1+PC+Price |  |
| 45 | Dry grocery | Condiments, Gravies, And Sauces | Seasoning Mix - Chili                                                                   | Hot; regular; texas                                                                            | 65% | Dry onions; salt pepper spices; spices; tomatos                                                                                                                                                                                                       | Yes | GH+CS | Yes | 1+PC+Price |  |
| 46 | Dry grocery | Condiments, Gravies, And Sauces | Seasoning Mix - Chili                                                                   | Hot; regular; texas                                                                            | 63% | Dry onions; salt pepper spices; spices; tomatos; vinegar                                                                                                                                                                                              | Yes | GH    | Yes | 1+PC+Price |  |
| 47 | Dry grocery | Condiments, Gravies, And Sauces | Seasoning Mix - Sloppy Joe                                                              | No msg                                                                                         | 87% | Dry onions; salt pepper spices; spices; tomatos                                                                                                                                                                                                       | Yes | GH+H  | Yes | 1+PC+Price |  |
| 48 | Dry grocery | Cookies                         | Cookies                                                                                 | Bite size; coated; cookie type; flavor; holiday; low fat                                       | 22% |                                                                                                                                                                                                                                                       | Yes | GH    | Yes | 1+PC+Price |  |
| 49 | Dry grocery | Gum                             | Gum-Bubble; Gum-Bubble-Sugarfree; Gum-Chewing; Gum-Chewing-Sugarfree                    | Bubblegum; flavor; presentation; sugarfree                                                     | 38% | Corn sweeteners x sugarfree == 0; resin and synthetic rubber; sugar x sugarfree == 0                                                                                                                                                                  | Yes | GH    | Yes | Price      |  |
| 50 | Dry grocery | Gum                             | Gum-Bubble; Gum-Bubble-Sugarfree; Gum-Chewing; Gum-Chewing-Sugarfree                    | Flavor; sugarfree                                                                              | 42% | Corn sweeteners x sugarfree == 0; resin and synthetic rubber; sugar x sugarfree == 0                                                                                                                                                                  | No  |       |     |            |  |
| 51 | Dry grocery | Jams, Jellies, Spreads          | Garlic Spreads                                                                          | Herbs                                                                                          | 93% | Soybean oil; vegetable oil                                                                                                                                                                                                                            | No  |       |     |            |  |
| 52 | Dry grocery | Pet Food                        | Cat Food - Dry Type                                                                     | Bulk; flavor; kitten; quality                                                                  | 47% | Corn; other grains; rough rice; slaughter cattle x flavor == "beef"; slaughter poultry x flavor in ["chicken", "turkey"]; unprocessed finfish x flavor == "fish"; wheat                                                                               | Yes | GH    | Yes | 1+Price    |  |
| 53 | Dry grocery | Pet Food                        | Dog & Cat Treats; animal = cat                                                          | Animal; flavor                                                                                 | 53% | Corn; other grains; rough rice; slaughter cattle x flavor == "meat poultry"; slaughter poultry x flavor == "meat poultry"; unprocessed finfish x flavor == "seafood"                                                                                  | Yes | GH+H  | Yes | 1+PC+Price |  |

|    |             |                              |                                                                                            |                                                                      |     |                                                                                                                                                                                                                                       |     |       |     |            |
|----|-------------|------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------|-----|------------|
| 54 | Dry grocery | Pet Food                     | Dog & Cat Treats; animal = dog                                                             | Animal; flavor                                                       | 44% | Corn; other grains; rough rice; slaughter cattle x flavor == "meat poultry"; slaughter poultry x flavor == "meat poultry"; unprocessed finfish x flavor == "seafood"                                                                  | Yes | GH+CS | Yes | 1+PC+Price |
| 55 | Dry grocery | Pet Food                     | Dog Food - Dry Type                                                                        | Bulk; flavor; puppy; quality                                         | 36% | Corn; other grains; rough rice; slaughter cattle x flavor == "beef"; slaughter poultry x flavor in ["chicken", "turkey"]; unprocessed finfish x flavor == "fish"; wheat                                                               | Yes | GH    | Yes | 1+Price    |
| 56 | Dry grocery | Pet Food                     | Dog Food - Moist Type; Dog Food - Wet Type; puppy = 0                                      | Flavor; puppy                                                        | 46% | Corn; other grains; rough rice; slaughter cattle x flavor == "beef"; slaughter poultry x flavor in ["chicken", "turkey"]; unprocessed finfish x flavor == "fish"; wheat                                                               | No  |       |     |            |
| 57 | Dry grocery | Pet Food                     | Dog Food - Moist Type; Dog Food - Wet Type; puppy = 1                                      | Flavor; puppy                                                        | 75% | Corn; other grains; rough rice; slaughter cattle x flavor == "beef"; slaughter poultry x flavor in ["chicken", "turkey"]; unprocessed finfish x flavor == "fish"; wheat                                                               | Yes | GH+H  | Yes | Price      |
| 58 | Dry grocery | Pickles, Olives, And Relish  | Pickles - Sweet                                                                            | Cut; spicy; style                                                    | 28% | Cucumbers; sugar; vinegar                                                                                                                                                                                                             | Yes | GH    | Yes | 1+PC+Price |
| 59 | Dry grocery | Pickles, Olives, And Relish  | Relishes                                                                                   | Condiment type; dill; fruity; hot; sweet                             | 52% | Cucumbers x condiment type == "pickle relish"; sugar x sweet == 1; vinegar                                                                                                                                                            | Yes | GH+CS | Yes | 1+PC+Price |
| 60 | Dry grocery | Pickles, Olives, And Relish  | Relishes                                                                                   | Condiment type; dill; fruity; hot; sweet                             | 49% | Cucumbers x condiment type == "pickle relish"; sugar x sweet == 1; vinegar                                                                                                                                                            | Yes | GH    | Yes | 1+PC+Price |
| 61 | Dry grocery | Prepared Food-Ready-To-Serve | Chicken - Shelf Stable                                                                     | Chicken type; has broth; white only                                  | 40% | Aluminum; poultry processing; slaughter poultry; wheat x chicken type in ["noodle", "pot pie", "dumplings"]                                                                                                                           | Yes | GH+CS | Yes | 1+PC+Price |
| 62 | Dry grocery | Prepared Food-Ready-To-Serve | Chili-Shelf Stable                                                                         | Con carne; has beans; hot; meat                                      | 32% | Aluminum; beans city average x has beans == 1; poultry processing x meat in ["chicken", "turkey"]; slaughter cattle x meat == "regular"; slaughter poultry x meat in ["chicken", "turkey"]                                            | Yes | GH    | Yes | 1+PC+Price |
| 63 | Dry grocery | Prepared Food-Ready-To-Serve | Stew - Beef - Shelf Stable; Stew - Chicken - Shelf Stable; Stew - Remaining - Shelf Stable | Brunswick; contains beef; contains chicken; microwave; regular       | 57% | Beans city average x brunswick == 1; poultry processing x contains chicken == 1; sauces; slaughter cattle x contains beef == 1; slaughter poultry x contains chicken == 1; tomatos x brunswick == 1                                   | Yes | GH+H  | Yes | 1+PC+Price |
| 64 | Dry grocery | Shortening, Oil              | Cooking Sprays                                                                             | Butter; oil type; pump                                               | 44% | Dairy x butter == 1; olive oil x oil type == "olive"; soybean oil x oil type == "soybean"; vegetable oil x oil type == "vegetable"                                                                                                    | Yes | GH    | No  | Price      |
| 65 | Dry grocery | Snacks                       | Dip - Mixes                                                                                | Type of mix                                                          | 34% | Dry onions; salt pepper spices                                                                                                                                                                                                        | Yes | GH    | Yes | 1+PC+Price |
| 66 | Dry grocery | Snacks                       | Popcorn - Popped; Snacks - Caramel Corn                                                    | Flavor; is caramel corn; low fat; low salt                           | 31% | Cheese x flavor == "cheese"; corn; corn sweeteners x is caramel corn == 1; sugar x flavor == "sweet"                                                                                                                                  | Yes | GH    | Yes | 1+PC+Price |
| 67 | Dry grocery | Snacks                       | Popcorn - Popped; Snacks - Caramel Corn                                                    | Flavor; is caramel corn; low fat; low salt                           | 34% | Cheese x flavor == "cheese"; corn; corn sweeteners x is caramel corn == 1; sugar x flavor == "sweet"                                                                                                                                  | Yes | GH    | Yes | Price      |
| 68 | Dry grocery | Snacks                       | Snacks - Potato Chips; Snacks - Potato Sticks                                              | Low fat; low salt                                                    | 57% | Russet potatoes; salt pepper spices; vegetable oil; vegetable oil x low fat == 0                                                                                                                                                      | Yes | GH    | Yes | 1+PC+Price |
| 69 | Dry grocery | Snacks                       | Snacks - Potato Sticks                                                                     | Cut; low fat; low salt; style                                        | 31% | Russet potatoes; soybean oil x style != "baked"; vegetable oil x style != "baked"                                                                                                                                                     | Yes | GH    | Yes | 1+PC+Price |
| 70 | Dry grocery | Snacks                       | Snacks - Pretzel                                                                           | Flavor; is coated; is filled; low fat; low salt; pretzel size        | 32% | Peanuts x flavor == "nutty"; sugar x flavor == "sweet"; wheat; wheat flour                                                                                                                                                            | Yes | GH    | Yes | 1+PC+Price |
| 71 | Dry grocery | Snacks                       | Snacks - Pretzel                                                                           | Flavor; is coated; is filled; low fat; low salt; pretzel size        | 32% | Peanuts x flavor == "nutty"; sugar x flavor == "sweet"; wheat; wheat flour                                                                                                                                                            | Yes | GH    | Yes | Price      |
| 72 | Dry grocery | Snacks                       | Snacks - Remaining                                                                         | Flavor; ingredient; low fat; low salt; snack preparation; snack type | 19% | Beans city average x ingredient == "legume"; corn x ingredient == "corn"; rough rice x ingredient == "rice"; russet potatoes x ingredient == "potato"; wheat flour x ingredient == "wheat grain"; wheat x ingredient == "wheat grain" | No  |       |     |            |
| 73 | Dry grocery | Snacks                       | Snacks - Remaining                                                                         | Flavor; ingredient; low fat; low salt; snack preparation; snack type | 18% |                                                                                                                                                                                                                                       | Yes | GH    | Yes | Price      |
| 74 | Dry grocery | Soft Drinks-Non-Carbonated   | Water-Bottled                                                                              | Flavored; single bottle; type of water                               | 46% | Diesel; plastic; vitamins x type of water == "enhanced"                                                                                                                                                                               | Yes | GH    | Yes | 1+PC+Price |

|    |                     |                                  |                                                                                            |                                                                                                          |      |                                                                                                                                                                                                                                                                                                          |     |       |     |            |
|----|---------------------|----------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------|-----|------------|
| 75 | Dry grocery         | Spices, Seasoning, Extracts      | Meat Marinades & Tenderizers                                                               | Asian; citrus or mojo; herb and garlic; mesquite bbq; steakhouse; tenderizer; teriyaki                   | 32%  | Lemons x citrus or mojo == 1; salt pepper spices; spices                                                                                                                                                                                                                                                 | Yes | GH    | Yes | 1+PC+Price |
| 76 | Dry grocery         | Spices, Seasoning, Extracts      | Pepper                                                                                     | Black; garlic; ground; lemon; red; whole                                                                 | 42%  | Lemons x lemon == 1; salt pepper spices; spices                                                                                                                                                                                                                                                          | Yes | GH    | Yes | 1+PC+Price |
| 77 | Dry grocery         | Spices, Seasoning, Extracts      | Salt - Cooking/Edible/Seasoned                                                             | Garlic; kosher; seasoned                                                                                 | 63%  |                                                                                                                                                                                                                                                                                                          | Yes | GH    | Yes | 1+PC+Price |
| 78 | Dry grocery         | Spices, Seasoning, Extracts      | Seasoning-Dry; spice = All purpose seasoning or blend                                      | Spice                                                                                                    | 100% |                                                                                                                                                                                                                                                                                                          | Yes | GH    | No  | Price      |
| 79 | Dry grocery         | Spices, Seasoning, Extracts      | Seasoning-Dry; spice = Chili powder                                                        | Spice                                                                                                    | 100% |                                                                                                                                                                                                                                                                                                          | Yes | GH+H  | Yes | 1+Price    |
| 80 | Dry grocery         | Spices, Seasoning, Extracts      | Seasoning-Dry; spice = Cinnamon powder                                                     | Spice                                                                                                    | 100% |                                                                                                                                                                                                                                                                                                          | Yes | GH+H  | Yes | 1+Price    |
| 81 | Dry grocery         | Spices, Seasoning, Extracts      | Seasoning-Dry; spice = Garlic powder/granules                                              | Spice                                                                                                    | 100% |                                                                                                                                                                                                                                                                                                          | Yes | GH+H  | Yes | Price      |
| 82 | Dry grocery         | Spices, Seasoning, Extracts      | Seasoning-Dry; spice = Meat seasoning                                                      | Spice                                                                                                    | 100% |                                                                                                                                                                                                                                                                                                          | Yes | GH+H  | Yes | 1+Price    |
| 83 | Dry grocery         | Spices, Seasoning, Extracts      | Vegetables - Onions - Instant                                                              | Chopped; minced                                                                                          | 75%  | Dry onions                                                                                                                                                                                                                                                                                               | Yes | GH+CS | Yes | 1+PC+Price |
| 84 | Dry grocery         | Vegetables - Canned              | Vegetables-Mixed-Canned                                                                    | Brand tier; low salt; preparation style; vegetable type                                                  | 28%  | Beans city average x vegetable type == "bean"; tomatos x vegetable type contains "tomato"                                                                                                                                                                                                                | Yes | GH    | Yes | 1+PC+Price |
| 85 | Dry grocery         | Vegetables - Canned              | Vegetables-Peas & Carrots-Canned; Vegetables-Peas-Canned; Vegetables-Peas-Remaining-Canned | Has carrots; has meat; salt; type of peas                                                                | 49%  | Aluminum; carrots x has carrots; green peas x type of peas != "blackeye"; salt pepper spices x salt                                                                                                                                                                                                      | Yes | GH    | Yes | 1+PC+Price |
| 86 | Dry grocery         | Vegetables And Grains - Dried    | Rice - Instant                                                                             | Instant                                                                                                  | 79%  | Fertilizer; rough rice                                                                                                                                                                                                                                                                                   | Yes | GH    | Yes | Price      |
| 87 | Fresh produce       | Fresh Produce                    | Fresh Fruit-Remaining                                                                      | Baby; banana; gold; organic; pineapple                                                                   | 63%  |                                                                                                                                                                                                                                                                                                          | Yes | GH+H  | Yes | 1+Price    |
| 88 | Frozen foods        | Baked Goods-Frozen               | Bakery-Bagels-Frozen                                                                       | Bagel size; has no salt; is low fat; is sliced; is wheat; primary flavor; quality tier; secondary flavor | 32%  | Sugar x primary flavor == "sweet"; wheat; wheat flour                                                                                                                                                                                                                                                    | Yes | GH    | Yes | 1+PC+Price |
| 89 | Frozen foods        | Breakfast Foods-Frozen           | Frozen/Refrigerated Breakfasts                                                             | Breakfast type; carb; has cheese; healthy; is veg; protein                                               | 26%  | Eggs x protein == "egg"; poultry processing x protein in ["chicken", "turkey"]; slaughter cattle x protein == "beef"; slaughter hogs x protein in ["ham/bacon", "sausage"]; wheat x carb != "other"                                                                                                      | Yes | GH+CS | Yes | 1+PC+Price |
| 90 | Frozen foods        | Pizza/Snacks/Hors Doeurves-Frzn  | Pizza-Frozen                                                                               | Crust; healthy; microwave; quality; topping                                                              | 26%  | Cheese; slaughter cattle x topping in ["pepperoni", "other meats"]; slaughter hogs x topping in ["pepperoni", "other meats"]; tomatos; wheat; wheat flour                                                                                                                                                | Yes | GH    | Yes | 1+PC+Price |
| 91 | Frozen foods        | Prepared Foods-Frozen            | Entrees - Meat - 1 Food - Frozen                                                           | Cheesy; cuisine; has sides; protein; saucy                                                               | 24%  | Beef and veal x protein == "beef"; cheese x cheesy == 1; poultry processing x protein in ["turkey", "chicken"]; processed meat; salt pepper spices; slaughter cattle x protein == "beef"; slaughter hogs x protein in ["pork", "meatball sausage"]; slaughter poultry x protein in ["turkey", "chicken"] | Yes | GH    | Yes | 1+PC+Price |
| 92 | Frozen foods        | Unprep Meat/Poultry/Seafood-Frzn | Frozen Poultry                                                                             | Boneless; chicken; cut; meat                                                                             | 34%  | Poultry processing; processed foods and feeds; slaughter poultry                                                                                                                                                                                                                                         | Yes | GH    | Yes | 1+PC+Price |
| 93 | General merchandise | Kitchen Gadgets                  | Beverage Storage Container                                                                 | Bottle type; quality; size description; volume oz                                                        | 34%  | Plastic; stainless steel                                                                                                                                                                                                                                                                                 | Yes | GH    | Yes | 1+Price    |
| 94 | General merchandise | Stationery, School Supplies      | Dry Erase Bulletin Board And Accesory                                                      | Board type; has calendar; has cork; is magnetic                                                          | 47%  |                                                                                                                                                                                                                                                                                                          | Yes | GH    | Yes | 1+PC+Price |
| 95 | General merchandise | Stationery, School Supplies      | Personal Planners Binders And Folders                                                      | Product type                                                                                             | 35%  | Plastic; pulp paper                                                                                                                                                                                                                                                                                      | Yes | GH+CS | Yes | 1+Price    |

|     |                      |                                 |                                                                                |                                                                                        |     |                                                                                                                                                                                                                                                                              |     |       |     |            |
|-----|----------------------|---------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------|-----|------------|
| 96  | Health & beauty care | Cosmetics                       | Cosmetic Kits                                                                  | Kit type                                                                               | 42% |                                                                                                                                                                                                                                                                              | Yes | GH    | No  | 1+PC+Price |
| 97  | Health & beauty care | Cosmetics                       | Cosmetics-Eyebrow & Eye Liner                                                  | Product type                                                                           | 71% |                                                                                                                                                                                                                                                                              | Yes | GH    | No  | 1+PC+Price |
| 98  | Health & beauty care | Cosmetics                       | Cosmetics-Foundation-Cream And Powder; Cosmetics-Foundation-Liquid             | Foundation type                                                                        | 55% |                                                                                                                                                                                                                                                                              | No  |       |     |            |
| 99  | Health & beauty care | Cosmetics                       | Cosmetics-Lipsticks                                                            | Product purpose                                                                        | 45% | Fatty acids; starch vegetable fats oils                                                                                                                                                                                                                                      | Yes | GH    | No  | Price      |
| 100 | Health & beauty care | Cosmetics                       | Talcum & Dusting Powder                                                        | Scent                                                                                  | 66% | Corn starch; talc                                                                                                                                                                                                                                                            | Yes | GH+H  | Yes | 1+Price    |
| 101 | Health & beauty care | Fragrances - Women              | Cologne & Perfume-Women's                                                      | Common consumer name; value                                                            | 46% | Coal; coal x common consumer name == "body spray"; ethanol; ethanol x common consumer name == "body spray"; other grains; other grains x common consumer name == "body spray"; soybeans; soybeans x common consumer name == "body spray"                                     | No  |       |     |            |
| 102 | Health & beauty care | Grooming Aids                   | Cosmetic And Nail Grooming Accessory                                           | Primary type                                                                           | 53% |                                                                                                                                                                                                                                                                              | Yes | GH    | No  | Price      |
| 103 | Health & beauty care | Hair Care                       | Shampoo-Aerosol/ Liquid/ Lotion/ Powder; Shampoo-Combinations                  | Dandruff; liquid; regular or other; two in one                                         | 43% | Fatty acids; surfactants                                                                                                                                                                                                                                                     | No  |       |     |            |
| 104 | Health & beauty care | Medications/Remedies/First Aids | Foot Preparations-Athlete's Foot                                               | Cream; deodorant; extra strength; liquid; liquid spray; ointment; powder; powder spray | 41% | Basic organic compounds; basic organic compounds x extra strength == 1                                                                                                                                                                                                       | Yes | GH+CS | Yes | Price      |
| 105 | Health & beauty care | Men's Toiletries                | Cologne/Lotion-Men's                                                           | Common consumer name                                                                   | 53% | Coal; coal x common consumer name == "body spray"; ethanol; ethanol x common consumer name == "body spray"; other grains; other grains x common consumer name == "body spray"; soybeans; soybeans x common consumer name == "body spray"                                     | No  |       |     |            |
| 106 | Health & beauty care | Men's Toiletries                | Cologne/Lotion-Men's                                                           | Common consumer name; form; value                                                      | 41% | Coal; coal x common consumer name == "body spray"; ethanol; ethanol x common consumer name == "body spray"; other grains; other grains x common consumer name == "body spray"; soybeans; soybeans x common consumer name == "body spray"                                     | Yes | GH+H  | Yes | 1+Price    |
| 107 | Health & beauty care | Skin Care Preparations          | Skin Cream-All Purpose                                                         | High price tier; scent; target condition; vitamin e                                    | 44% | Fatty acids; starch vegetable fats oils; vitamins x vitamin e == 1                                                                                                                                                                                                           | Yes | GH+CS | Yes | 1+PC+Price |
| 108 | Non-food grocery     | Detergents                      | Detergents - Heavy Duty - Liquid; Detergents - Light Duty; Detergents-Packaged | Scent none                                                                             | 70% | Surfactants                                                                                                                                                                                                                                                                  | No  |       |     |            |
| 109 | Non-food grocery     | Laundry Supplies                | Detergent Boosters                                                             | Liquid; powder                                                                         | 67% | Surfactants                                                                                                                                                                                                                                                                  | Yes | GH+CS | Yes | 1+PC+Price |
| 110 | Non-food grocery     | Tobacco & Accessories           | Cigarettes                                                                     | Low tar; menthol                                                                       | 74% | Pulp paper; tobacco                                                                                                                                                                                                                                                          | Yes | GH    | Yes | 1+PC+Price |
| 111 | Packaged meat        | Packaged Meats-Deli             | Bratwurst & Knockwurst; Frankfurters-Refrigerated; Sausage-Dinner              | Form; frozen; hot; meat; sausage type; skinless                                        | 14% | Beef and veal x meat contains "(beef veal)"; poultry processing x meat contains "(chicken turkey)"; processed meat; salt pepper spices; slaughter cattle x meat contains "beef"; slaughter hogs x meat contains "pork"; slaughter poultry x meat contains "(chicken turkey)" | Yes | GH+CS | Yes | 1+PC+Price |
| 112 | Packaged meat        | Packaged Meats-Deli             | Bratwurst & Knockwurst; Frankfurters-Refrigerated; Sausage-Dinner              | Form; frozen; hot; meat; sausage type; skinless                                        | 16% | Beef and veal x meat contains "(beef veal)"; poultry processing x meat contains "(chicken turkey)"; processed meat; slaughter cattle x meat contains "beef"; slaughter hogs x meat contains "pork"; slaughter poultry x meat contains "(chicken turkey)"                     | Yes | GH    | Yes | 1+PC+Price |

|     |               |                     |                                     |                                            |     |                                                                                                                                                                                                                                                                                                                                |     |    |     |            |  |
|-----|---------------|---------------------|-------------------------------------|--------------------------------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|-----|------------|--|
| 113 | Packaged meat | Packaged Meats-Deli | Lunchmeat-Deli Pouches-Refrigerated | Flavor; form; low fat; low salt; meat type | 27% | Beef and veal x meat type contains “(beef veal)”; poultry processing x meat type in [“turkey”, “chicken”]; processed meat; salt pepper spices; slaughter cattle x meat type in [“beef”, “salami pepperoni”]; slaughter hogs x meat type in [“ham”, “salami pepperoni”]; slaughter poultry x meat type in [“turkey”, “chicken”] | No  |    |     |            |  |
| 114 | Packaged meat | Packaged Meats-Deli | Lunchmeat-Sliced-Refrigerated       | Flavor; form; low fat; low salt; meat type | 25% | Beef and veal x meat type contains “(beef veal)”; poultry processing x meat type in [“turkey”, “chicken”]; processed meat; salt pepper spices; slaughter cattle x meat type in [“beef”, “salami pepperoni”]; slaughter hogs x meat type in [“ham”, “salami pepperoni”]; slaughter poultry x meat type in [“turkey”, “chicken”] | Yes | GH | Yes | Price      |  |
| 115 | Packaged meat | Packaged Meats-Deli | Sausage-Breakfast                   | Flavor; form; low fat; meat; precooked     | 21% | Beef and veal x meat contains “(beef veal)”; poultry processing x meat contains “(chicken turkey)”; processed meat; salt pepper spices; slaughter cattle x meat contains “beef”; slaughter hogs x meat contains “pork”; slaughter poultry x meat contains “(chicken turkey)”                                                   | Yes | GH | Yes | 1+PC+Price |  |

Table C.1: All mergers in our sample. Characteristics explained by principal components always include an indicator for whether the product is a store brand, and the package size. The possible values for “Instruments used” are “GH” (only differentiation instruments), “GH+CS” (differentiation and cost shifter instruments), and “GH+H” (differentiation and Hausman instruments). The possible values for “Demographic interactions” are “Price” (high income indicator interacted with only price), “1+Price” (interacted with price and the constant), and “1+PC+Price” (interacted with price, the constant, and principal components of characteristics).